

# In-Kind Transfers and Early Childhood Development: Evidence from Housing Assistance Lotteries\*

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## Abstract

This paper examines how public housing and housing vouchers affect families' housing circumstances and their children's development. Using randomized housing assistance offers from lottery-ordered waiting lists and housing applications linked to data on kindergarten readiness assessments, test scores, and Medicaid claims, we examine the impact of both programs on housing consumption, neighborhood quality, and child development. Vouchers improve children's early literacy skills, school readiness, behavioral development, and subsequent test scores, while public housing has no detectable effects on average. The pattern of results is consistent with each program's effects on housing and neighborhood quality. The positive impacts of vouchers accrue largely to children in households that experienced the greatest improvement in housing and neighborhood quality, rather than those predicted to experience the largest increase in disposable income. For public housing, we find heterogeneous effects by development quality: children moving to higher-quality developments experience improved outcomes, while those in lower-quality developments see worse outcomes. Together, our results suggest that in-kind transfers *can* yield improvements in child development, but that the effects depend on how transfers impact consumption of the subsidized good.

**Keywords:** In-Kind Transfers, Child Development, Housing Policy

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# 1 Introduction

The past three decades have marked a significant shift in low-income housing policy, as the federal government moved from an emphasis on government-managed public housing toward government-subsidized vouchers redeemed on the private market. Since 1990, the number of households served by public housing in the U.S. has declined by nearly 40 percent, while the number of households served by the housing voucher program has more than doubled.<sup>1</sup> This decline in public housing investment was due, at least in part, to policymakers’ concerns about whether public housing was “bad for kids” (Currie and Yelowitz, 2000; Jacob, 2004).

Despite its importance for policy, there is relatively little direct evidence on the effects of moving into public housing on children, or comparable evidence on the effects of voucher receipt from the same setting.<sup>2</sup> Moreover, theory provides little guidance on how these two forms of housing assistance might differ in their effects. Like most in-kind transfers, these programs alleviate material hardship, which could lead to improvements in children’s subsequent short- and long-term outcomes (Currie and Gahavri, 2008; Aizer et al., 2022). Yet, while both programs have similar predicted effects on family resources, they may generate different impacts on child outcomes if their effects on neighborhood or housing quality differ. More generally, predicting whether and how in-kind transfers might impact children has proven difficult for researchers, in part because the effects likely depend on consumption responses, which vary with program design (Page, 2024).

In this paper, we provide new experimental evidence on the role of two large in-kind transfers — public housing and housing vouchers — in children’s development, tracing how the programs’ divergent effects on housing and neighborhood conditions correspond to differences in early childhood development in the context of U.S. low-income housing policy. We assemble novel data linking children on waiting list applications to outcomes data from several sources. We leverage an unusual housing assistance lottery randomly offering public housing *or* housing vouchers to a large population of low-income households in Miami-Dade County, Florida.<sup>3</sup> The lottery represents a unique setting in which we can *directly* observe children either receiving or not receiving assistance and estimate the impacts on housing

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<sup>1</sup>See Figure 1.

<sup>2</sup>Most existing studies of public housing use plausibly exogenous *exits* from public housing, such as through Moving To Opportunity (MTO) (Kling et al., 2007; Chetty et al., 2016) or as a result of housing demolitions Chyn (2018) to draw conclusions about the effects of public housing. However, nearly all of these studies focus on the effects of leaving some of the most distressed public housing developments, which may differ considerably from the effects of moving into average quality public housing.

<sup>3</sup>Miami-Dade Public Housing and Community Development (PHCD), the county’s lead housing authority, is the seventh largest housing authority in the United States in terms of total households assisted at approximately 25,000.

consumption and subsequent developmental outcomes drawn from several sources.

Using records of more than 10,000 young children listed on housing assistance applications linked to several administrative datasets, we estimate the causal effects of public housing and housing vouchers on families' housing circumstances and children's developmental outcomes. In particular, we link children to school-entry assessments, allowing us to examine impacts on children's early reading skills, kindergarten readiness, and cognitive and noncognitive skills, as demonstrated in a classroom setting. Linked data on elementary and middle school test scores allow us to investigate whether developmental gains accrued by formal school entry persist in later schooling years. To characterize effects on housing consumption, neighborhood quality, and housing instability, we construct hedonic quality measures using rent listings data, link address histories to neighborhood quality metrics, and use data on housing-unstable students derived from education records. Finally, we leverage Medicaid claims data to study whether housing assistance affects measures of children's cognitive and behavioral health.

To estimate the causal effects of public housing and housing vouchers, we instrument for assistance receipt with randomized initial waiting list positions for each program. To our knowledge, we provide the first large-scale experimental evidence on public housing and the first evidence of an existing safety net program's impact on early childhood development in the United States, a piece often missing from studies of the impact of in-kind transfers or household resources. To the extent resources or early environments impact children's early skill development, cognition, and behavioral health, these are precisely the measures on which we would expect to observe effects.

We show that housing vouchers and public housing have different impacts on the quality of housing and neighborhood in which low-income families reside. Using address data linked to hedonic housing quality estimates and Census tract measures like poverty and neighborhood crime, we estimate the effects of voucher receipt and public housing admission on the change in quality relative to baseline neighborhoods. Leasing-up with a voucher leads families to move to housing with a roughly 10-percent higher hedonic quality and neighborhoods with 3 percentage points lower poverty and about 5.4 fewer violent crimes per 1,000 residents (a 40-percent reduction). In contrast, leasing-up in public housing causes families to move, on average, to neighborhoods with 8 percentage points *higher* poverty and about 8 more violent crimes per 1,000 residents (a 65-percent increase), and lower estimated hedonic quality.

While we find contrasting effects on housing and neighborhood quality, we find strikingly similar results for housing *instability*. Both programs substantially reduce the chances that schools flag a child as having unstable housing. Vouchers and public housing reduce the probability that a child is homeless or doubled-up by roughly 7 percentage points relative to a control mean of 8 percentage points. Our results are consistent with earlier work exclusively

on housing vouchers (Wood et al., 2008; Gubits et al., 2015). Uniquely, we document effects for both public housing and housing vouchers on students’ experiences of housing instability using novel McKinney-Vento data from schools.

We find that voucher receipt substantially improves children’s early developmental outcomes. Children whose families lease-up with a voucher score 0.2 standard deviations higher on an assessment of pre-reading skills. This estimate grows to 0.29 standard deviations when we focus on families that were unlikely to have public housing as an alternative. Voucher receipt also boosts kindergarten readiness by 13 percentage points (a 22 percent increase relative to the control mean). Relative to control group children at the bottom of the waiting list, children in families receiving vouchers are considerably less likely to be identified as “not demonstrating” appropriate developmental milestones. We find treatment group children are 10 percentage points (67 percent) less likely to be categorized as “not emerging or demonstrating” key skills during observational classroom evaluations.

Public housing admission has no statistically significant effect on child development outcomes, on average. Our point estimates typically lie near zero, and we can rule out substantial positive effects, on average. We can also rule out large negative effects, a surprising result given our finding that public housing admission generally exposes children to low quality neighborhoods. The results, along with our housing stability findings, echo earlier non-experimental work by Currie and Yelowitz (2000), which shows that admission to conventional public housing may not be as harmful for children’s outcomes as suggested by public perception or implied by the *positive* effects of escaping public housing found in previous work (Chetty et al., 2016; Chyn, 2018). One possibility, which we explore in Section 6.2, is that the beneficial effects for some families of moving to better-quality public housing developments appear to be offset by the deleterious effects of moving to lower-quality for others.

The effects of voucher receipt on child development are driven by large positive effects for boys. Boys in families randomly receiving a voucher score 0.5 standard deviations higher in evaluations of pre-reading skills, and they are 28 percentage points (49 percent) more likely to be deemed kindergarten ready. Moreover, they are substantially less likely than those in the control group to receive the lowest developmental categorization in classroom-based observational assessments. Girls show no consistent positive or negative effects from housing voucher receipt. Our finding of concentrated effects for boys aligns with existing literature documenting gender differences in the returns to early-life investments, and boys’ heightened susceptibility to disadvantage, particularly as it affects their behavioral skill development Autor et al. (2019); Bertrand and Pan (2013).

The positive effects of housing vouchers persist into elementary and middle school. For a

portion of our early childhood sample, we can observe reading and math exam scores seven to eight years after the lottery. We find that voucher receipt increases reading scores by 0.34 standard deviations and combined math and reading scores by 0.23 standard deviations. Notably, we also find that vouchers increase test scores in both reading and math when we expand to the full sample of children that we observe in the relevant test years. The lack of fade-out suggests that the gains made before formal schooling could be especially durable. Still, we find little impact of public housing on scores in either sample, though our estimates for the early childhood sample are somewhat imprecise.

Housing assistance also appears to shape children’s cognitive and behavioral health. Using linkages of children to Medicaid claims data, we examine impacts on diagnoses for impulse control problems, oppositional defiance disorder, ADHD, and delays in speech, reading or math. We find evidence that vouchers contribute to lower incidence of impulse control problems and developmental delays. Combining diagnoses into a summary measure, we estimate that children in families that lease-up with a voucher have 0.14 standard deviations better cognitive and behavioral health. Consistent with our analysis of the kindergarten readiness assessments, we find that improvements in children’s behavioral development accrue primarily to boys.

To further explore the mechanisms behind our results, we investigate heterogeneity in housing assistance effects by estimated baseline housing consumption. Due to the structure of rent rules in federal rental assistance programs, a household spending a significant portion of their income on housing *before* rental assistance receipt should experience a relatively large increase in disposable income, while a household spending relatively little on housing at baseline should have a comparatively small change in disposable income but a large improvement in housing quality. Using hedonic quality as a proxy for housing spending, we first show that housing quality increases substantially with voucher receipt for those in low-quality housing at baseline and is essentially unchanged for households that were in higher-quality housing at baseline. Next, we explore what these consumption changes mean for child development. We find that vouchers’ positive effects are driven by the large improvements for children in low quality housing at baseline. We interpret this as further evidence that child development impacts of in-kind transfers are moderated through their particular impacts on consumption.

Lastly, we explore whether the impacts of public housing are heterogeneous depending on the type of public housing that families lease. Using the leasing “zones” created by the housing authority for the purpose of making public housing apartment offers, we categorize public housing developments as either “Higher Quality” or “Lower Quality.” We then instrument for leasing-up in either type of public housing using zone-specific estimated randomized waiting

times and interactions of these waiting times with families’ pre-lottery zone of residence. We find evidence of sharply different impacts of public housing based on whether it is higher or lower quality. Children who move into lower-quality public housing are exposed to worse housing and neighborhood conditions, and experience sizable reductions in kindergarten readiness and worsened cognitive health. While children moving into higher-quality public housing experience improvements in housing and neighborhood conditions, and appear to potentially benefit with higher rates of kindergarten readiness and fare better in terms of cognitive health, though these effects are imprecisely estimated. These differences further illustrate how improvements in child development appear to depend crucially on the degree of improvement in housing and neighborhood environment, rather than simply increases in family resources.

We make contributions to two literatures. The first is on the effects of the safety net on child development. To our knowledge, we provide the first randomized evidence of existing U.S. in-kind transfers, at scale, on measures of early skill development. We show that in-kind transfers *can* yield improvements in child development outcomes, but the effects are moderated by their particular impacts on consumption. Despite generating similar changes in family resources, vouchers and public housing impact housing and neighborhood quality differently — voucher recipients move to better quality housing in safer neighborhoods, while public housing recipients move to poorer, less safe neighborhoods — and consequently differentially affect child development. These cross-program differences align with our within-program analyses of mechanisms, which find that the positive effects of vouchers are driven by households getting the largest gains in housing and neighborhood quality, further underscoring the importance of housing and neighborhood changes for the impacts on child development. We think that our results are helpful for interpreting mixed findings in the literature on the effects of in-kind and cash transfers on children (Page, 2024), which has found both positive (Aizer et al., 2016; Bailey et al., 2023; Duncan et al., 2011; Pollakowski et al., 2022; Barr et al., 2022; East et al., 2023) and null effects (Hawkins et al., 2025; Jacob et al., 2015; Krause et al., 2025), and are particularly relevant given the behavioral economics literature that documents how consumption patterns differ with how money or transfers are labeled (Thaler, 1990; Hastings and Shapiro, 2018).

Second, we make important contributions to the literature on low-income housing policy in the U.S., providing direct experimental evidence of the effects of both housing vouchers and public housing on policy-relevant outcomes for a large sample of low-income children and families. Previously, public housing and voucher debates primarily used evidence from the Moving to Opportunity demonstration (Kling et al., 2007; Sanbonmatsu et al., 2011; Chetty et al., 2016) or public housing demolitions (Jacob, 2004; Chyn, 2018). These studies are

critical to understanding neighborhood effects, but they mostly examine the effects of escaping troubled public housing. Previous work on the effects of admission to public housing found a mix of positive and negative effects using a variety of research designs (Currie and Yelowitz, 2000; Pollakowski et al., 2022). We provide the first experimental evidence of the effects of U.S. public housing on housing consumption, neighborhood quality, child homelessness, and a broad set of child development outcomes. We find that public housing reduces housing instability for families, but worsens their neighborhood quality, and has no detectable effects on child development, on average. However, we do find evidence that the effects of public housing are heterogeneous based on the quality of public housing developments in which families lease, with moves to better-quality developments appearing to improve neighborhood conditions, producing improved child outcomes, and lower-quality developments having the opposite effects. This relates to work on the effects of public housing on adult outcomes outside the U.S., which also finds heterogeneity in effects by development quality (van Dijk, 2019).

For vouchers, we find evidence of improvements across a range of outcomes including early literacy skills, developmental benchmarks, test scores, and cognitive and behavioral health. These results contrast with the only other large-scale experimental evidence of the effects of vouchers on children. Using a Chicago-based voucher lottery, Jacob et al. (2015) find that vouchers have few if any meaningful long-run impacts on childhood outcomes. In Section 6.3 of this paper, we review this discrepancy and related evidence, and conclude that differences in (i) housing and neighborhood quality effects, and (ii) data and measurement likely drive the divergent findings.

The remainder of the paper is organized as follows: Section 2 provides background on the programs and waiting list lottery that we study, Section 3 describes the data, Section 4 outlines our empirical strategy, Section 5 reports results, Section 6 explores mechanisms and how the results align with related literature, and Section 7 concludes.

## 2 Background

### 2.1 Public Housing and Housing Choice Voucher Programs

The Housing Choice Voucher program, formerly Section 8, is the largest rental assistance program administered by the U.S. Department of Housing and Urban Development (HUD). Housing vouchers serve roughly 2.5 million households (5.3 million people) with an annual budget of roughly \$30 billion (Collinson et al., 2019). Public housing serves slightly fewer than 1 million households (1.7 million people) at annual budget expense of roughly \$8 billion.

Both programs are typically funded exclusively using federal funds, but they are administered by local quasi-government organizations called Public Housing Authorities (PHAs).

Families issued housing vouchers use them to lease units on the private rental housing market, while public housing families are offered apartments in public housing developments owned by a PHA. Other than this key difference, the public housing and housing vouchers programs have very similar rules. For both programs, tenants typically pay 30 percent of their adjusted income towards rent.<sup>4</sup> Income eligibility for each program are based on HUD's official income limits, which are set at based metropolitan Area Median Income (AMI), and are adjusted by household size. Families typically must re-certify their incomes each year in meetings with housing authority staff to remain in good standing. There are slightly different rules for remaining income eligible for those at the top of the income distribution: families in living in public housing can remain income eligible earning up to HUD's "Low-Income" limit (80 percent of AMI), while for vouchers this limit is usually the "Very Low-Income Limit" 50 percent of AMI. That being said, the vast majority of tenants in both programs fall below 50 percent of AMI. Nationally, 90 percent of public housing tenants and 95 percent of voucher tenants are below 50 percent of AMI, and in our setting of Miami Dade county, these rates are identical at 94 percent. These income limits are not binding for most households, as 83 percent of households in public housing (79 percent of voucher households) earn less than 30 percent of AMI, what HUD classifies as "Extremely Low-Income."

In Miami-Dade County, Florida, the local PHA is Miami-Dade Public Housing and Community Development (PHCD). PHCD is the seventh largest housing authority in the country in terms of total households assisted at approximately 25,000. It owns and operates 16 large public housing developments, and about 56 scattered-site properties which together total 9,000 units. The tenant-based voucher program operated by PHCD serves approximately 16,000 households per year. PHCD serves the entire county, with voucher families located in nearly all jurisdictions in Miami-Dade County.

## **2.2 2008 Waiting List**

As a result of a lawsuit filed against PHCD, the agency agreed to open their waiting lists for all their assisted housing: housing vouchers and public housing and separately randomize waitlist positions for vouchers and public housing. In July 2008, PHCD opened their waiting list for a four-week period to applicants seeking any type of federal housing assistance. Households that submitted an application were informed about eligibility for the

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<sup>4</sup>Adjusted income is calculated the same under both programs, and allows for the expenses associated with dependent care, or care for disabled or elderly members to be excluded from over annual income

programs and agreed to allow their submitted information to be used to determine eligibility. After four weeks, the application period was closed and in September every household that applied during the open period was randomly assigned a ranking on a housing voucher waiting list and a separate random ranking on a public housing waiting list.<sup>5</sup> Housing assistance was heavily oversubscribed: more than 70,000 households applied for roughly 3,500 vouchers and 4,500 public housing apartments to be allocated over the ensuing eight years. As a result, applicants randomly assigned to the top of either list were likely to receive an offer of housing assistance, while those randomly assigned to the bottom of the list were unlikely to receive assistance. Our research design compares the outcomes of children in households that were randomly-assigned to the top of an assistance waiting list to children in families outside the top of the list.

Households were notified of their initial position after randomization, and the initial static ranking on each list was published online. PHCD then began drawing down from each waiting list as vouchers and public housing units became available. When households approached the top of the waiting list, PHCD would attempt to contact the household to undergo a final eligibility determination. If a household wanted to learn about their current position, they needed to call the leasing department. Households deemed eligible in the final eligibility review received offers of assistance in the order they became available.

For public housing vacancies, PHCD would attempt to offer one unit with the appropriate number of bedrooms within each of three broad geographic zones that divide the county. If households rejected all the available units, they would be permanently removed from the public housing waiting list. A household that accepted the offer of a public housing unit could technically remain eligible for a voucher during the first 12 months of their public housing leases, but in practice no one was issued a voucher after leasing up in public housing. And after 12 months in their unit, families accepting public housing were removed from the voucher waiting list. For housing vouchers, households issued a voucher were typically given 90 days to find an acceptable unit. When a family was issued a housing voucher they were removed from both waiting lists. Finally, households that could not be reached at follow-up or who were found ineligible at follow-up screening were removed from the waiting list as PHCD worked through both lists. Recent work suggests that housing authorities often struggle to contact households on their waiting list [Waldinger \(2021\)](#).

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<sup>5</sup>The randomized orders were created from a random number generating function in a spreadsheet.

## 3 Data

### 3.1 Waiting List Data

The primary sample is drawn from the more than 70,000 applications entered in the 2008 housing assistance lottery. Importantly, these data contain information on both the head of the household and all members of the household at the time of application. For household heads, the data include race, ethnicity, need for an accessible unit, age, whether the household had an email address, and baseline address, as well as first name, last name, and date of birth. For members, it contains unique identifiers, first name, last name, date of birth, and gender. The waitlist sample includes more than 70,000 unique households with more than 158,000 unique members. The waitlist file also includes the initial randomized position for both public housing and housing vouchers.

We determine whether a household is at the “top” of a given waiting list by estimating a series of first-stage regressions for receipt of a voucher (public housing) on initial voucher (public housing) position searching over all plausible cutoffs values of position rank in bins of 200 spots, and selecting the threshold which maximizes the first-stage F-statistic. This is analogous to the procedure for identifying discontinuities in regression discontinuity designs with unknown cutoffs (and zero bias), and has been shown by [Porter and Yu \(2015\)](#) to yield a super-consistent estimator.<sup>6</sup> Encouragingly, this approach yields similar cutoffs to simply choosing the 90th percentile of position ranks among the “treated” households. We show in Section 5.4 that our results are robust to a variety of alternative cut-offs to denote “top” positions. Throughout, we define treatment as leasing-up with assistance *during* the period spanned by our outcome window.<sup>7</sup>

We make two sample restrictions for our analysis. First, we restrict the sample to applicants living in Miami-Dade County at the time of application. This is because applicants outside of Miami-Dade were typically ineligible to receive assistance. Second, we exclude the 4 percent of applicants living at public housing at baseline, since we are primarily interested in the effects of the transfer relative to counterfactual of no housing assistance.

### 3.2 Address History Data

We rely on several data sources to measure the address histories of waiting list families. Appendix A details how we construct these histories using the sources described here. We

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<sup>6</sup>This approach also avoids the bias described in ([de Chaisemartin and Behaghel, 2020](#)), as it is a version of their “initial offer” estimator.

<sup>7</sup>That is, some “control” households will be receive assistance years later. We explore whether this affects our results in Section 5.4, and find that it does not.

can observe *baseline* addresses for all families directly from the lottery applications. For households that lease-up in public housing or with a housing voucher, we can observe their assistance addresses using housing authority records and use these records to validate the other address data sources. For unassisted households, we use two data sources for address histories. The first is voter records from the state of Florida, for the years 2012 and 2015. The second is consumer reference data from Infutor Data Solutions, covering addresses from the early 2000’s to 2020.<sup>8</sup> Collectively, these sources allow us to assign a post-lottery address for nearly 80 percent of the families in our sample. As with most outcome data, a concern with these address data sources is that receiving housing assistance could affect whether a household appears in these data at all, potentially biasing our estimates. In Appendix A we also investigate differential attrition in the address data, and find no evidence that waiting list positions impact appearance in any of the address data.

### 3.3 Data on Child Outcomes

Our primary outcomes data consist of measures of children’s early literacy and observed skill development from the Florida Kindergarten Readiness Screener, accessed through the Florida Department of Education (FDOE). FDOE linked children listed on housing assistance applications to their individual-level readiness assessment data for the 2009–10 through 2013–14 academic years.<sup>9</sup> The state requires that public schools screen all kindergarten students within the first 30 days they are enrolled.<sup>10</sup> At the time, the school-entry screener consisted of two distinct components: an emerging literacy assessment, the Florida Assessment for Instruction in Reading (FAIR), and a classroom-embedded, observational assessment of school readiness, the Early Childhood Observation System (ECHOS). These assessments measure children’s early development and school readiness relative to a number of benchmarks, including broader measures of social and behavioral development—in addition to early literacy and numeracy skills—than are typically available in assessment data.

We use three measures from the readiness screener: (1) a standardized “probability of reading success” score from the FAIR literacy assessment, (2) an indicator of kindergarten readiness also derived from the FAIR literacy assessment, and (3) a categorical designation

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<sup>8</sup>However, we focus on address records during the 2009–2015 period to align as much as possible with our other data sources.

<sup>9</sup>As we show in Appendix Table A.2, about 74 percent of children 0–4 link to some type of screener data. In practice, most of our sample – more than three-quarters – are under the age of 3 at the time of the lottery. For these children, we link 87 percent to a FAIR record and 88 percent to an ECHOS assessment. FDOE linked children probabilistically using first name, last name, and date of birth.

<sup>10</sup>Children enrolled in the state’s voluntary, public pre-kindergarten (VPK) program or kindergarten in non-public schools may also participate in the screening, however, nearly all of our records are from public elementary schools, including charters.

of “Demonstrating,” “Emerging/Progressing,” or “Not Yet Demonstrating” readiness skills as measured across the developmental benchmarks captured on the observational ECHOS assessment. While assessments of young children likely measure noncognitive development (e.g., task persistence, attention) even when the target of the assessment is a cognitive skill, such as literacy, these data importantly contain direct measures of social-behavioral development. The ECHOS evaluates individual student progress towards 19 benchmarks in seven domains: language and literacy, mathematics, social/personal skills, science, social studies, physical development, and creative arts. The data description found in Appendix C provides more details on the contents of these data.

### **3.3.1 Standardized Test Scores**

In addition to the Kindergarten Readiness Screener data, the FDOE also linked children from lottery applications to the Florida Statewide Assessment (FSA) for academic years 2014-15 and 2015-16. The FSA is a mandatory statewide exam for grades 3-8, typically taken in the month of May.<sup>11</sup> Our extract of FSA data covers the English Language Arts and Mathematics components of the assessment. We examine whether vouchers and public housing impact standardized: reading scores (ELA), mathematics scores (Math), and an average of both scores (Combined).

### **3.3.2 Data on Cognitive and Behavioral Health**

We supplement data on early skill development with cognitive and behavioral health measures derived from Medicaid claims records. The Medicaid claim records are from the Florida Agency for Health Care Administration (AHCA), covering the years 2005-2016.<sup>12</sup> The data record every doctor visit, hospital or emergency department (ED) admission, behavioral health center or ambulatory surgery visit in the state of Florida covered by Medicaid. In addition, AHCA provided data on the coverage spells for each individual listed in lottery applications that could be linked to Medicaid records. Among children under 5 years of age at the time of the lottery, we estimate that 92.8 percent appear covered by Medicaid for one or more years in the four years leading up to the lottery.

Using International Classification of Diseases (ICD-9) codes tied to each visit, we create indicators for four common diagnosed conditions related to cognitive and behavioral health: impulse control problems, oppositional defiance disorders, attention deficit/hyper-activity, and developmental delays in reading, math, or speech. These measures take the value 1 if the child has any diagnosis for the listed condition in the post-lottery period, and zero otherwise.

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<sup>11</sup>ELA assessments also typically occur in grades 9 and 10

<sup>12</sup>For our main analysis, we restrict to years prior to 2015 when Florida transitioned to more significant use of managed care plans and our data is more limited.

Then we construct a summary index of cognitive and behavioral health by standardizing each measure, summing them, and normalizing by dividing by the standard deviation (Kling et al., 2007). A standard concern with using Medicaid claims data to measure health is that it may capture differences in utilization rather than health per se. Another related concern is that the interventions we study could, in principle, impact appearance in the Medicaid data, either by impacting geography, eligibility, or take-up. In Appendix F we investigate these possibilities in our sample and find little evidence that the differential appearance in Medicaid drives our results.

### 3.4 Housing Instability and Homelessness

We also draw on data from the Florida Department of Education to measure housing instability. In linking children listed on the housing assistance lottery files to their student records, we also received annual information collected from local school districts on students experiencing housing instability. Under the McKinney-Vento Act, as a condition of receiving funds, school districts are responsible for making every effort to identify homeless students, or students “who lack a fixed, regular, and adequate nighttime residence.”<sup>13</sup> The data records whether a student has experienced any of the following living situations during each academic year: Living in emergency or transitional shelters (i.e. “shelter”), awaiting foster care, sharing the housing of other persons due to loss of housing or economic hardship (“doubled-up”); living in hotels or motels, or living in a place not meant for human habitation (i.e. living in cars, parks, public spaces, abandoned buildings etc). For our linked sample, the most common housing homelessness states are living in emergency shelter and doubled-up, which comprise 34 and 58 percent of recorded instability in our sample, respectively. We focus on whether the child has been flagged as experiencing *any* of the categories of housing instability (“Homeless or Doubled-up”), and separately whether they appear as living in shelter or doubled-up.

### 3.5 Rent and Hedonic Quality Data

To measure housing consumption, we construct measures of hedonic quality (Rosen, 1974; Black, 1999) at baseline and follow-up addresses. The hedonic estimates are formed using address-level rents from two sources. The first is pre-lottery data on housing voucher recipients in Miami-Dade. This data provides info on the rent of the unit (before receipt of the voucher) for about 200,000 unique address from 2001-2008. We then supplement this data with more than 2 million historical rent listings on more than 300,000 unique

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<sup>13</sup>Districts must employ a liaison to assist in identifying students that are homeless. Districts must use existing data to help identify homeless students and schools are encouraged to reach out to community shelters, soup kitchens, transitional living facilities, and a host of other agencies, as well as develop partnerships with programs such as Head Start and Early Intervention, to assist in identifying children who are eligible for services under the McKinney-Vento Act.

dwellings from Altos Research, a provider of housing market data. We geocode records from both datasets to the parcel level, and link them to housing unit characteristics from the Miami-Dade County Property Appraiser. These records contain detailed housing unit characteristic such as square-footage, age of home (and a cubic in age), the assessed value, bedrooms, bathrooms, number of floors, building type (i.e. townhouse, single-family attached etc) indicators, and quality codes assigned by the assessor. Our hedonic model described in Appendix B includes the attributes above, as well as block-group median rents, which we calculate directly from our rent data, and neighborhood subdivision fixed effects to account for other unobserved characteristics of the neighborhood (such as proximity to the ocean, downtown, or south beach). The attributes in our hedonic model can explain roughly 70 percent of the cross-sectional variation in rents in our data. See Appendix B for further details on data construction and hedonic model estimation.

### 3.6 Neighborhood Data

We link address locations to neighborhood characteristics measured at the Census tract level. These include: poverty rates and rates of single-parent household from the ACS 2006-2010 five-year estimates. Violent crimes per 1,000 from the National Neighborhood Crime Study (NNCS), Wave 2 (Krivo et al., 2023).<sup>14</sup> Finally, we link to the predicted income rank in adulthood from the Opportunity Atlas (Chetty et al., 2018).

## 4 Empirical Strategy

The random initial order of the waiting lists for vouchers and public housing provides exogenous variation in offers of housing assistance. Since some children are in households randomly placed at the top of the voucher waiting list, while others are randomly sorted to the bottom, a natural strategy is to compare those placed at the top to those at the bottom. This strategy will recover a reduced-form “Intent to Treat” (ITT)-style parameter. However, because many households on the waiting list do not receive assistance, we are also interested in estimating the effect of leasing up with housing assistance. To do so, we use the random waiting list positions as instruments for assistance and estimate a Local Average Treatment

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<sup>14</sup>While the NNCS has multiple years of crime data for neighborhoods in several large jurisdictions in Miami-Dade County, including Miami and Hialeah, it lacks coverage in some suburban communities. To account for this, we regression impute neighborhood violent crimes for these uncovered tracts using several neighborhood characteristics from the ACS: the poverty rate, the fraction of single-parent households and the home ownership rate.

Effect (LATE): the effects of leasing with assistance for those induced into assistance by their waitlist position.<sup>15</sup>

To estimate the reduced-form effect of voucher and public housing waitlist position on child outcomes, we estimate the following equation:

$$Y_i = \beta_1 \text{Top Voucher}_i + \beta_2 \text{Top Public Housing}_i + \phi_x + \mu_z + X_i' \gamma + \varepsilon_i \quad (1)$$

Where  $Y_i$  is the outcome for child  $i$ ,  $\text{Top Voucher}_i$  is an indicator for being randomized to the top of the voucher waiting list, and  $\text{Top Public Housing}_i$  is an indicator for being randomized to the top of the public housing waiting list.  $\beta_1$  is the reduced-form effect of being at the top of the voucher list and  $\beta_2$  is the reduced-form effect of being at the top of the public housing waiting list. We include fixed effects for two pre-lottery factors which affect the mapping of public housing position to unit offers: the predicted number of bedrooms needed ( $\phi_x$ ) and the location in one of three geographic zones delineated by PHCD ( $\mu_z$ ). Additionally, since a small number of some individuals appear multiple times on the waiting list under different household applications, we include fixed effects for the number of applications. Finally, we control for year-quarter of birth to account for cohort differences. Throughout the analysis, we cluster the standard errors at the household level.

To estimate the effects of being admitted to public housing or leasing up with housing vouchers, we instrument for lease-up with the initial randomized position on each waiting list. Specifically, the empirical approach taken has the following form:

$$\text{Housing Voucher}_i = \pi_1 \text{Top Voucher}_i + \pi_2 \text{Top Public Housing}_i + \phi_x + \mu_z + X_i' \gamma + \varepsilon_i \quad (2)$$

$$\text{Public Housing}_i = \gamma_1 \text{Top Voucher}_i + \gamma_2 \text{Top Public Housing}_i + \phi_x + \mu_z + X_i' \delta + \nu_i \quad (3)$$

Where  $i$  indexes individuals, the outcome of the second stage is  $Y$  in equation 4. The first stage is estimated simultaneously, jointly predicting lease-up for public housing and housing vouchers. Equations 2 and 3 represent the first stage broken out by type of assistance. For both types of assistance, we instrument for leasing-up with indicators for being randomly assigned to the top of the public housing list ( $\text{Top Public Housing}$ ) and top of the voucher list ( $\text{Top Voucher}$ ). Again,  $\mu_z$  and  $\phi_x$  capture the fixed effects mentioned above. We also include a vector of pre-lottery controls including: child and household demographics, neighborhood characteristics ( $X$ ).<sup>16</sup>

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<sup>15</sup>Below we discuss how this LATE does not have the usual interpretation of the effect of a single treatment relative to a “no-treatment” control group.

<sup>16</sup>Our base controls include: child gender, race/ethnicity, age and gender of household head, number of

$$Y_i = \beta_1 \text{Housing Voucher}_i + \beta_2 \text{Public Housing}_i + \phi_x + \mu_z + X_i' \gamma + \varepsilon_i \quad (4)$$

In the simple case of a single treatment, just-identified IV will recover a well-defined Local Average Treatment Effect of the effect of receiving treatment relative to counterfactual of “no-treatment” under the assumptions of [Imbens and Angrist \(1994\)](#). However, with more than one endogenous variable, standard IV does not identify the effect of receiving either treatment relative to a well-defined counterfactual without further assumptions. This is because an instrument can potentially draw in new households from multiple margins: those that would choose the alternative treatment if they did not receive the instrumented treatment and those who would choose no-treatment. That is, some households may switch from public housing to vouchers if they get a voucher offer. This “switching” behavior, or substitution bias ([Heckman et al., 2000](#)), constitutes a violation of the exclusion restriction. This issue has been considered in a number of recent papers ([Mogstad and Torgovitsky, 2024](#); [Kirkeboen et al., 2016](#); [Kline and Walters, 2016](#); [Hull, 2018](#); [Mountjoy, 2022](#)). Instead, the parameters estimated in equation 4 correspond to the effect of vouchers (public housing) relative to a mix of “no treatment and public housing (vouchers).”

These studies ([Kline and Walters, 2016](#); [Hull, 2018](#); [Kirkeboen et al., 2016](#); [Mountjoy, 2022](#)) show that in the presence of multiple treatments (or multiple counterfactual states), estimating just-identified IV will recover a weighted average of “subLATEs” that are counterfactual-specific treatment effects. In Appendix D, we illustrate this result in our context and the assumptions required. For vouchers, when we estimate these weights ( $\omega$ ) in our sample, which appear in Appendix Table A.4, we find that roughly 83 percent of the weight is on the LATE for moving from no-assistance to a voucher ( $LATE_{0 \rightarrow V}$ ), and 17 percent is LATE capturing the effect of going from public housing to a voucher ( $LATE_{P \rightarrow V}$ ). To more cleanly isolate the effects of assistance relative to a counterfactual of “no assistance,” we also report the effects of receiving a voucher (public housing) for those unlikely to have received a public housing offer in the outcome window, which we refer to as the “No Assistance Alternative”. Using this “No Assistance Alternative” specification, the weight on  $LATE_{0 \rightarrow V}$  (i.e. the effect of voucher relative to a “no assistance” counterfactual) increases to 99 percent, and the weight on  $LATE_{P \rightarrow V}$  falls to 1 percent. That is, by restricting the sample to households whose voucher (public housing) position has little impact on their public housing (voucher) take-up, we can more clearly isolate the effects of vouchers (public housing) relative to a counterfactual of no assistance<sup>17</sup>

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kids in the household, and neighborhood poverty.

<sup>17</sup>Note that we focus on the weights here for the effects of vouchers, because we find in Table 2 that having a better public housing position generates almost no substitution away from vouchers

## 4.1 Evaluating Balance and Sample Characteristics

Before turning to results, we first examine whether the initial housing voucher and public housing positions are indeed random. Table 1 reports the mean baseline sample characteristics of children and households at the top of the voucher list (“Top Vouchers”) in Column (1), the top of the public housing list (“Top Public Housing”) in Column (2), and those outside the top of both lists (“Control”) in Column (3). We report the p-values from the joint significance tests between the groups in Column (4). Consistent with the random assignment described by the housing authority, the characteristics of children and families look quite similar across waiting list position groups, and we fail to reject that the groups are different for most covariates. Due to the age of children in our sample, we cannot observe pre-lottery test scores; however, we can evaluate balance on prior test scores for older siblings, which we report in Appendix Table A.1, and find that they are strongly balanced. Together, this evidence gives us confidence that the assignment process or initial waiting list positions was indeed random.

The sample in Table 1 is the analytical sample of children who linked to Florida Kindergarten Readiness Screener (FLKRS) data. A concern is that housing assistance could impact whether children appear in the screener data. Appendix Table A.2 explores this possibility. Since the screener data are from *after* random assignment, there is no *ex ante* guarantee that appearance in these data would be balanced. Fortunately, we find precise zeros when we estimate regressions of screener data availability on voucher and public housing waiting list position, suggesting that differential appearance in our outcome data should not bias our estimates. Finally, Appendix Table A.3, reports balance and summary statistics for the sample of children in the Medicaid analysis. In addition to illustrating balance along waiting list application measures, this table also confirms that children at the top of the voucher or public housing waiting lists have similar rates of prior Medicaid coverage, low-birth weight, asthma, respiratory problems, and similar cognitive development index values as children at the bottom of both waiting lists, prior to the lottery.

Our sample consists primarily of children ages 0-4, with most children being either 1 or 2 years old at the time of the lottery. Over 90 percent of children are in female-headed households, and most households include at least two minor children and only a single adult. The sample is overwhelmingly minority, with roughly 70 percent being Black or African-American, and 30 percent identifying as Hispanic. Most families reside in moderately high poverty neighborhoods as baseline, and families have high rates of homeless histories. In general, our sample is not dissimilar from prior studies of housing assistance ([Sanbonmatsu et al., 2011](#); [Wood et al., 2008](#)).

## 4.2 Waiting List Positions and Housing Assistance

Our IV specification relies on a strong relationship between a household’s randomized initial position on each waiting list and their receipt of housing assistance. Figure 2a provides graphical results that illustrate the relationship between randomized waiting list position for housing vouchers and the probability of voucher lease-up. Children in households near the “top” of the voucher waiting list (those with positions less than 9,200) are about 30 percentage points more likely to receive a voucher than children with initial positions greater than 9,200.

Figure 2b provides an analogous plot for public housing waiting list position and lease-up. Take-up rates for public housing are significantly lower, reflecting the fact that families in Miami-Dade exhibit preference for housing vouchers. Additionally, the relationship is noisier, reflecting the housing authority’s consideration of need for number of bedrooms and three geographic zones in making public housing offers. Still, children in households near the “top” of the public housing waiting list (those with positions less than 22,000) are about 16 percentage points more likely to lease-up in public housing than children with initial positions greater than 22,000.

The first-stage regression analog to these results appears in Table 2. In both cases, the randomized initial waiting list position strongly predicts take-up. The 30-percentage point increase in lease-up for vouchers is somewhat lower than estimates in previous studies of vouchers (Jacob et al., 2015; Wood et al., 2008) which found take-up rates ranging from 40 to 55 percent. Part of this difference is due to the fact that voucher offers were spread out over a couple years, and the housing authority was working from “stale” records in later years, making it difficult to find and contact families to notify them. Public housing take-up is meaningfully lower, and this appears in part due to a preference for vouchers: having a “top” voucher position reduces public housing take-up by nearly 5 percentage points while public housing position has no effect on voucher take-up. Nevertheless, both positions have a strong direct effect on take-up, generating first-stage F-statistics of 453 for vouchers and 405 for public housing.

## 5 Results

### 5.1 Effects on Housing and Neighborhood Conditions

Before exploring the effects of housing assistance on child outcomes, we start by examining the effects of housing assistance on housing and neighborhood conditions experienced by applicants. Previous research has generally found small effects of voucher receipt on measures such as neighborhood poverty rates for families living in private housing at baseline (Wood

et al., 2008; Jacob and Ludwig, 2012). This contrasts with the effects of a voucher for families residing in public housing at baseline, as in Moving to Opportunity (MTO), where voucher families experienced large reductions in neighborhood poverty and crime exposure (Sanbonmatsu et al., 2011).

Table 3 presents the reduced-form and IV estimates of changes in housing and neighborhood characteristics from pre-lottery and post-lottery locations. Columns (2)-(6) report results on several neighborhood characteristics, each measured at the Census tract level. We estimate that receipt of a housing voucher causes a family to move to a neighborhood with roughly 3 percentage points lower poverty rates (relative to baseline poverty rate of 21 percent). This estimated effect is slightly larger than prior experimental estimates of vouchers for families living in private housing (Wood et al., 2008; Jacob and Ludwig, 2012), but only about one-sixth of the effect of relocating from distressed public housing with a voucher from MTO (Sanbonmatsu et al., 2011; Chetty et al., 2016). Vouchers also cause families to move to safer neighborhoods. We estimate that receipt of a voucher leads a family to move to a neighborhood with 5.4 fewer violent crimes per one thousand (a 43-percent reduction relative to the baseline mean). We find no detectable effects on the quality of the neighborhood school, the fraction of single-parent households in the neighborhood, or the predicted income rank in adulthood from the Opportunity Atlas (Chetty et al., 2018).

In contrast to vouchers, participation in public housing, on average, entails moves to neighborhoods that are *more* impoverished. We estimate that moving into public housing increases neighborhood poverty rate exposure by 8 percentage points (36 percent of the baseline control mean). Although it is unsurprising that the tracts containing public housing have elevated poverty rates, it is striking that families in our sample are moving *from* pre-lottery neighborhoods with substantially lower poverty. Moving into public housing also increases exposure to violent crime. Families' lease-up in public housing causes them to move to neighborhoods with 8 more violent crimes per 1,000 (a 65-percent increase relative to the baseline neighborhood).

In Table 3, we also estimate the effects of housing assistance on changes in hedonic housing quality, as described in Section 3.5. Receipt of a voucher increases hedonic quality by roughly 10 percent. We estimate that admission to public housing causes families to move into units of lower hedonic quality, on average, which is marginally significant. Taken together, these results suggest that vouchers and public housing have divergent effects on the housing and neighborhood conditions of applicants. Public housing induces moves to somewhat worse quality housing in neighborhoods with higher poverty and crime, and vouchers lead to moves to better quality housing in less impoverished and safer neighborhoods.

## 5.2 Effects on Housing Instability and Homelessness

Another important dimension of housing consumption relates to housing *instability* or homelessness. While housing instability is often especially difficult to observe for families, we can use the data collected by McKinney-Vento coordinators on students experiencing housing instability to capture it. Prior analyses demonstrate that these data capture important changes in housing stability (Collinson et al., 2025) in related contexts. Section 3.4 describes the data in more detail.

Table 4 reports the effects on three measures of housing instability. Column (1) captures the impact on whether a child experienced any type of homelessness, including doubling-up. Columns (2) and (3) disaggregate housing instability into its two most common categories in the data: living in a shelter, or living doubled-up due to economic hardship.

Both housing vouchers and public housing appear to substantially reduce housing instability. Children in families receiving a housing voucher are 7.4 percentage points less likely to be recorded as living in shelter, hotel/motels, places not meant for habitation, or doubling-up. Similarly, children in families admitted to public housing are 8 percentage points less likely to be housing unstable. Both estimates represent large relative reductions with about 8 percent of the control group experiencing homelessness or doubling-up during that same time. The effects appear similar across the types of housing instability. Receipt of a housing voucher leads to a 3-percentage point reduction in living in a shelter. The estimate for public housing is nearly identical; however, it is more imprecise and not statistically significant. Similarly, both programs appear to decrease doubling-up by an estimated 5 percentage points in each case.

These findings that voucher receipt reduces homelessness are similar to previous studies using more selected samples, such as homeless families Gubits et al. (2015) or families that exit welfare (Wood et al., 2008). To our knowledge, we are the first to provide randomized evidence of the effects of public housing on children’s experiences of housing instability and homelessness.

## 5.3 Effects of Assistance on Child Development Outcomes

Having established how public housing and housing vouchers impact the housing and neighborhood conditions experienced by families, we now turn to documenting the effects of housing assistance on the developmental outcomes of children, using data from assessments at school entry, as well as Medicaid claims data on development-related diagnoses.

### 5.3.1 School Readiness

As described in Section 3.3, we draw on an assessment of early literacy skills as part of the Florida Assessment for Instruction in Reading which produces a reading score and a determination of kindergarten readiness, and from classroom observations derived from the Early Childhood Observation System (ECHOS).

We start by providing non-parametric evidence of housing assistance impacts on children’s outcomes. Figure 3a plots binned means of kindergarten readiness rates for children by their randomized initial housing voucher waiting list position. The treatment group (i.e. those with a “Top HCV” position) are represented in navy circles, while the control group (those outside the top) are in gold. The average values for both groups appear separately as solid lines. Children with a voucher waiting list position in the “top” group appear to be, on average, about 4 percentage points more likely to be “Kindergarten Ready.” For public housing, which appears in Appendix Table A.1a, there is no visually apparent difference in kindergarten readiness across waiting list positions.

Table 5 features the regression analog to these reduced-form plots. The top panel (panel A) reports our reduced form estimates of the causal effects of waiting list position on early literacy skills and kindergarten readiness. The bottom panel reports our IV estimates of the effects of housing assistance. Consistent with the reduced-form plots for vouchers, we estimate that being randomly-assigned to the top of the voucher waiting list (“Top HCV”) increases kindergarten readiness by 4 percentage points (column (2)). The IV estimate implies that families’ receipt of a housing voucher leads to a 13 percentage point increase in their children’s kindergarten readiness. This effect is substantial, representing a 22 percent increase relative to mean control mean of 61 percent. The effects appear slightly larger when we focus on households that were unlikely to have public housing as an alternative using our “No Assistance Alternative” specification (column (3)).

Table 5 also reports the standardized performance on the FAIR literacy assessment (“Std. Reading Score”). Being randomly assigned to the top of the voucher list increases literacy scores by 0.06 standard deviations. Re-scaling by the first stage, our IV estimates imply that receipt of a voucher boosts early literacy skills by 0.22 standard deviations (column (2), panel B). As we note in Section 4, the randomized voucher position may divert some people away from public housing, in which case this estimate combines the effects of vouchers relative to no-assistance and the effects of vouchers relative to public housing. In column (3) we report the no-assistance alternative, which isolates the effects relative to a no-assistance counterfactual. For children who otherwise would not receive assistance, the receipt of a voucher improves their literary scores by 0.29 standard deviations ( $p = 0.015$ ).

As an additional validation of our main results, we revisit the declining pattern of voucher

take-up found in Figure 2a. Those at the very top of the “Top Voucher” group have noticeably higher take-up of vouchers, than those further down the “Top Voucher” group. If our estimated effects reflect the *causal* effect of a voucher, then our reduced-form results should be larger among the group with a higher take-up rate than the group with a lower take-up rate. This is exactly what we find in Appendix Table A.5. In column (1), we show the differences in take-up, consistent with our plot, the earlier voucher waitlist positions have a higher take-up rate of 46 percent.<sup>18</sup> Column (2) of Panel A in the same table reports the reduced-form effects. The reduced-form point estimates appear considerably larger for those with very “Early” positions than those with “Later” positions, giving us further confidence the positive effects we find for vouchers are indeed causal.

While vouchers appear to improve early literacy and kindergarten readiness, we find no similar gains for children in families who lease-up in public housing. The point estimates for public housing are consistently close to zero for kindergarten readiness and standardized early literacy skills, and none are statistically significant. For example, the IV estimate for leasing-up in public housing on early literacy scores is -0.006 standard deviations and a 1.7 percentage point increase in kindergarten readiness. The estimates for public housing are slightly less precisely estimated than for voucher. Still, if public housing produced similar-in-magnitude effects as vouchers, we would be powered to detect them.

Next, we investigate how housing vouchers and public housing impact the likelihood that children meet developmental benchmarks. As described in 3.3 and detailed in Appendix C, these measures come from embedded observational assessments of children’s behaviors and activities in the classroom. Again, we start with non-parametric evidence on this question. In the bottom panel of Figure 3b we plot the probability that a child falls into the lowest category of performance, “Not Emerging or Demonstrating,” by the initial randomized voucher position. There is a clear downward shift among the children at the top of the voucher list, indicating that they are *less* likely to be flagged as not demonstrating the necessary skill development in classroom observations. However, we see no similar pattern for public housing, which appears in Appendix Figure A.1b.

Table 6 reports the regression estimates of the impacts of waitlist position and assistance on children’s likelihood of demonstrating developmental benchmarks during these assessments. Again, the top panel presents the reduced-form and the bottom panel reports the IV estimates. The reduced-form estimates indicate that children randomly-assigned to the top of the voucher list are 2.7 percentage points less likely to be flagged as not demonstrating the appropriate skills for kindergarten. This indicates that children in families that were more likely to

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<sup>18</sup>The housing authority appeared to work through the voucher list in batches, so there is a relatively small “Early” group that receives vouchers more quickly while the “Later” group is much larger in size.

receive a voucher also performed better on these assessments. We find that receipt of a voucher leads to a 10-percentage point reduction in the likelihood of receiving the lowest category of performance, “Not Emerging or Demonstrating.” At the same time, we find a similarly-sized *increase* in the likelihood of receiving the the middle category, “Emerging.” Children in voucher-receiving families are also 10 percentage points more likely to be classified as “Emerging” in developmental skills. The pattern of effects indicates that the impacts of vouchers on children are concentrated in the lower part of the skill distribution. With receipt of a voucher, children are more likely to move out of the bottom and into the middle categorization of skills; however, we find no effects on moving into the top categorization of skill development (“Demonstrating”). Similar to our results on kindergarten readiness and early literacy, the positive effects appear slightly larger when we estimate impacts only among children in families with no alternative housing assistance.

In contrast to the voucher results, public housing does not appear to improve children’s likelihood of reaching developmental benchmarks, as recorded in the observational assessment. None of our estimates for public housing are statistically significant. Relative to the results for early literacy skills, the point estimates for impacts on developmental benchmark measures more consistently point to improved outcomes. Again, if the point estimate for public housing on “Not Emerging or Demonstrating” were as large as the estimate for vouchers, we would have sufficient power to detect such an effect. Our point estimates for public housing do allow us to rule out even moderately sized *negative* effects on children’s skill building. This result is not trivial given the strong negative reputation public housing historically held in public consciousness (Olsen, 2003; Collinson et al., 2016), and the generally negative effects we find on housing and neighborhood quality in Section 5.1. It is also broadly consistent with earlier quasi-experimental studies of public housing, such as Currie and Yelowitz (2000), which found non-negative impacts on child outcomes.

### 5.3.2 School Readiness Effects by Gender

Given a large literature documenting gender differences in response to changes in family resources or environmental factors (Dahl and Lochner, 2012; Milligan and Stabile, 2011; Barr et al., 2022), we next examine the effects of housing assistance separately by gender. Columns (5) and (7) of Table 5 report estimates on early literacy skills for boys and girls, respectively. The positive effects of housing vouchers are driven by large, beneficial effects for boys. Boys in families randomized into receiving a voucher perform 0.55 standard deviations higher on the pre-reading skills assessment, and they are 29 percentage points (49 percent) more likely to reach the kindergarten readiness threshold. The results in Table 6 point to similar gender differences in the classroom-based observational assessments in columns (5)

and (7). “Treated” boys are also much less likely than their control group counterparts to be classified as “not emerging or demonstrating” necessary skills for kindergarten, a nearly 16-percentage point reduction (nearly 100-percent decrease relative to the control group). They are also 17 percentage points more likely to be in the “emerging” category, a 34-percent increase. Both sets of results for boys appear as non-parametric reduced-form plots in Figure 4a and 4b. Girls, on the other hand, show no consistent effects (positive or negative) from receipt of housing vouchers.

### 5.3.3 Heterogeneity by Family Size, Race, and Ethnicity

To better understand the role of housing assistance in children’s skill development, we explore additional heterogeneity in the effects of vouchers and public housing by family and child characteristics. In Figure 5, we report heterogeneity by the number of children in the household and race/ethnicity. The top panel of Figure 5 reports results for vouchers and the bottom panel reports public housing results. The positive developmental effects of housing vouchers are significantly larger for children living in households with multiple children. The point estimate is also larger for Black, non-Hispanic children than for Hispanic children though the difference is not statistically distinguishable. For public housing, we do not find any significant patterns of heterogeneity in the effects by household or family characteristics. In Appendix E, we examine several other dimensions of heterogeneity, including baseline neighborhood characteristics and baseline early childcare options; we do not find detectably different effects of vouchers or public housing in either domain.

### 5.3.4 Duration and Exposure

Given that the literature on neighborhood effects ([Chetty et al., 2016](#); [Chetty and Hendren, 2018](#)) has found a strong relationship between childhood exposure to new neighborhood environments and long-term outcomes such as earnings, it is worth exploring whether similar exposure or duration effects are at work in our setting. The typical child in our kindergarten readiness sample who received a voucher through the lottery had been receiving assistance for roughly two years before taking the readiness screener. In Appendix Table A.6, we explore how years with a housing voucher matter for the realization of improved reading scores and the likelihood of being ready for kindergarten. In the top panel of Appendix Table A.6, we report results of regressing screener outcomes on the years with a voucher and a separate indicator for receipt of a voucher, instrumenting for each with an indicator of being at the “top” of the voucher list and bins of voucher positions within the top group. The coefficient on voucher receipt, when years with a voucher are included, offers a test of anticipation or perhaps differences in grading environment. The fact that the coefficient is zero provides further support for the effects that we document above. We estimate that each additional year

of voucher receipt improves early literacy scores by 0.08 standard deviations and increases kindergarten readiness by roughly 5 percentage points. These results appear broadly in line with evidence of longer-run duration effects from assistance (Pollakowski et al., 2022) and neighborhood environment (Chetty et al., 2016; Chetty and Hendren, 2018).

### 5.3.5 Cognitive and Behavioral Health

We now turn to examining the impact of housing assistance on diagnoses of cognitive and behavioral health issues using the linked Medicaid records, described in Section 3.3.2. We focus on conditions plausibly affected by environmental factors, such as mild delays in reading, math, or speech, or impulse control challenges, rather than more severe intellectual disabilities or conditions. Table 7 reports reduced-form and IV estimates for the effects of waitlist position and housing assistance on diagnoses for developmental conditions, and our summary index.

Being randomly-assigned to the top of the voucher list reduces diagnoses for impulse control problems by 0.9 percentage points, and is statistically significant at conventional levels. Receipt of a voucher reduces diagnosed impulse control problems by 2.5 percentage points. This effect is a large relative reduction, as only 3 percent of the control group is diagnosed with the condition during the same period. For public housing, the sign is reversed, but the estimates are not statistically significant. Column (2) reports estimates of the effects on diagnoses for oppositional defiance disorder and column (3) reports estimates for impacts on attention deficit/hyper-activity disorder. None of the estimates are statistically significant for either condition. Column (4) displays the impacts on the diagnoses of reading, math, or speech delays. Here, we find evidence that children who were more likely to get a voucher were subsequently less likely to be diagnosed with such delays. The IV estimate implies that receiving a voucher reduces the likelihood of developmental delay by 3 percentage points compared to a control mean of 11 percentage points. The estimates for the effects of public housing have the opposite sign, but are not statistically significant.

Finally, we combine the diagnoses into a standardized index where higher values indicate fewer cognitive and behavioral health challenges. Column (5) reports the impacts on the index. We find that having a top voucher waiting list position causes a 0.043 standard deviation improvement in the index, with the IV estimate implying that voucher receipt leads to a 0.14 standard deviation improvement in cognitive and behavioral health. At the same time, we find little evidence that public housing improves or worsens children’s cognitive and behavioral health. The estimate for public housing is near zero and is not statistically significant.

### 5.3.6 Cognitive and Behavioral Health Effects by Gender

Panels B and C of table 7, report the impacts separately for boys and girls, respectively. For boys, we find some evidence that vouchers appear to reduce impulse control problems and oppositional defiance challenges, though most of the estimates are somewhat imprecise. For diagnoses of development delays, we find that the effects of public housing are *positive* for boys, and marginally significant, suggesting that admissions to public housing may lead to increased developmental delays. For vouchers, the coefficient is negative and suggests a four-percentage-point reduction, but is not statistically significant. Combining the effects across outcomes, we find that being on the top of the voucher list causes a 0.06-standard deviation improvement in behavioral health, implying a 0.2-standard deviation improvement from voucher receipt. At the same time, the impacts of public housing on the index appear much more muted.

We find less evidence for any effects of housing assistance on girl’s cognitive and behavioral health. In fact, the point estimates for almost every outcome are smaller in absolute magnitude for girls than for boys. The one exception is for ADHD diagnoses, where we find slightly larger estimates, although not statistically significant. Estimates for the effects of vouchers on the index measure are less than half the size of the effect for boys. Together, these patterns are similar to those that we document in Section 5.3.2: vouchers appear to improve cognitive and behavioral health with benefits concentrated among boys, while public housing has little effect, positive or negative, on average.

## 5.4 Robustness

This section presents results from several robustness exercises that address potential concerns with the empirical strategy or interpretation of results. First, we evaluate the sensitivity of our results to the precise cut-offs used in the definition of being at the “top” of each waiting list. Appendix Table A.7 reports our main results varying the cutoff in either direction relative to the cutoff used in our main specifications (bolded). The point estimates are quite stable across alternative definitions of being at the “top” of the voucher list and are consistently statistically significant. This exercise gives us confidence that these results are not the product of particular choices of “top” definition.

If the behavior of applicant households changes in response to waitlist position — through channels other than actual assistance receipt, including anticipatory effects — that would constitute a threat to our identification strategy, potentially violating the exclusion restriction. We evaluate whether anticipatory child investments could be affecting our results in two ways. First, we drop households with waitlist positions outside of the “top” but *near* the top, such

that they could plausibly receive assistance in future years outside of our outcome window.<sup>19</sup> These results appear alongside our main estimates in Appendix Tables A.10 and A.11. The reduced-form and IV coefficients are nearly identical to our main specification, suggesting that the presence of these “possibly treated” households is not impacting our estimates. Second, as a placebo test, we evaluate whether waitlist position impacts outcomes within a sample of unlikely-to-be-treated households. To do this, we restrict the sample to applicants outside of the “top” of their respective waiting list, and re-estimate reduced-form regressions of how position affects outcomes within this sample. These results appear in Appendix Table A.12. We find zero effect of waitlist position in this sample of unlikely-to-be-treated households. This gives us confidence that anticipatory effects are not impacting our results.<sup>20</sup>

Another concern is that the improvements in child development for voucher recipients do not reflect improvements in cognitive and noncognitive skills, but instead reflect differences across test administrators or graders in the schools children attend for kindergarten. There are several pieces of evidence that suggest that this potential concern is unlikely to be driving our results. First, from the top panel of Appendix Table A.6, as discussed in Section 5.3.4, we see no evidence of voucher effects when a child’s family receives a voucher just before they are scheduled to take the screener. Second, while we cannot directly observe the graders, we can compare the treatment and control children within the *same* school, which should remove any systematic differences in the graders. In principle, post-treatment school choice is endogenous and undesirable to condition on; however, it provides a useful check for interpreting our results. In Tables A.8 and A.9, we show that our main results are nearly unchanged after including school fixed effects, supporting the notion that differences in grading practices or grader severity are unlikely to drive our results. This exercise also reinforces our argument that our results are unlikely to be driven by changes in school quality. Finally, the cognitive and behavioral health improvements found in the linked Medicaid data show similar patterns of effects from an entirely different data source, providing further support for the observed effects reflecting *actual* changes in child development.

## 5.5 Persistence of Effects

Because early childhood development measures are typically unavailable in large experimental and quasi-experimental studies of program impact, we know little about the link between improvements on these measures and subsequent standardized test scores. In our context, we are able to observe reading and math test scores during grades three through

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<sup>19</sup>Note this approach is similar to [Jacob et al. \(2015\)](#) who exclude households with lottery positions in the middle, “possibly” treated range.

<sup>20</sup>The lack of anticipatory effects is unsurprising given recent evidence of parents’ lack of anticipatory investments in response to predictable loss of benefits ([Deshpande and Dizon-Ross, 2023](#))

eight for most of the early childhood sample (our focus in Section 5.3.1), as well as for all children in our sample taking the relevant statewide exams in the available years.

Table 8 shows the results for the children in our early childhood sample who are also observed in the later test score data in the top panel. Interestingly, benefits for the voucher group persist in the form of improved reading scores. The IV point estimate suggests that children whose families received vouchers are scoring 0.34 standard deviations higher on English Language Arts exams 7-8 years after the lottery. Though effects on math scores are statistically insignificant, the point estimates are positive and of non-trivial magnitude with the IV estimate implying a 0.13-standard deviation improvement. Combining math and reading scores, we estimate that children in families receiving a voucher score about 0.23-standard deviations higher than control group children seven to eight years after the lottery. This estimate is similar in magnitude to the effects we found on the developmental assessment measured at the beginning of kindergarten, suggesting that the developmental gains made before entering formal schooling persist several years later.

In the bottom panel of 8, we explore effects for all school-age children, regardless of age at the time of randomization, for whom we have test scores during our study period. Again, children in households at the top of the voucher waiting list see improved reading scores. The IV estimate for the receipt of a voucher implies a 0.21-standard deviation improvement in reading scores. We also find some improvements in math performance for children at the top of the voucher waiting list. Overall, we estimate that receipt of a voucher increases the combined math and reading score in the full sample by 0.16 standard deviations. At the same time, students whose families were at the top of the public housing waiting list do not experience detectable changes in their test scores.

## 6 Mechanisms and Discussion

### 6.1 Cash v. In-Kind: Non-Housing Consumption v. Housing Consumption

To better understand whether the documented effects on child development are driven by increased non-housing spending (i.e., “family resources”) or improvements in housing and neighborhood conditions, we explore treatment-effect heterogeneity by the family’s baseline housing situation. Public housing and housing vouchers operate with similar rent rules: families contribute a minimum of 30% of their adjusted income towards rent.<sup>21</sup> Consequently, families spending significantly more than 30% of their income towards rent *before* housing

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<sup>21</sup>Families with a voucher can “top up” and spend more than 30% of their adjusted income on rent if the market rent is above the locally-defined payment standard (price ceiling). However, they must not spend more than 40% of their income towards rent at initial lease-up. The majority of voucher holders spend do not “top up.” [Collinson and Ganong \(2018\)](#).

assistance receive a large increase in disposable income, while families spending 30% or less of their income on housing at baseline could experience a reduction in disposable income.

Figure 6 displays this intuition with simple representations of budget constraints before and after housing assistance (see [Collinson et al. \(2016\)](#) and [Olsen \(2003\)](#) for similar representations). In the top panel (a), we consider a household spending relatively little on housing at baseline, consuming low-levels of housing:  $h'_0$ . If we assume that this household will maximize the value of the voucher, then they will consume  $h_v$  when they receive a housing voucher (this level of housing quality corresponds to the local “payment standard,” or the maximum value of the voucher before the household pays more out-of-pocket). This household experiences a relatively large increase in housing consumption  $\Delta h'$ , but a comparatively small change in non-housing consumption  $\Delta c'$ . In contrast, panel b considers a household spending a high proportion of their budget on housing at baseline:  $h''_0$ . Assuming again that this household selects  $h_v$  with a voucher, they experience a large increase in non-housing consumption  $\Delta c''$ , but a small change in housing consumption,  $\Delta h''$ . Given this, we can assess heterogeneity in effects by baseline housing consumption to learn about the role of increased non-housing spending versus improved housing and neighborhood conditions.

Although we cannot directly observe baseline housing consumption,  $h_0$ , we can use our hedonic rent estimates for the baseline addresses described in Section 3.5 (and used in Section 5.1) as a proxy. We categorize households in our sample as “low” housing consumption at baseline if the hedonic estimate:  $\hat{h}_0$  is less than the payment standard (i.e.,  $h_v$ ), and “high” baseline housing consumption if the hedonic estimate is greater (or equal to) the payment standard. In the top panel of Figure 7, we report impacts on hedonic quality (i.e. housing consumption) separately for “low” baseline housing consumption households and “high” baseline housing consumption households. Consistent with the theoretical predictions in Figure 6, we find that receipt of a voucher produces much larger effects on housing consumption for households with “low” baseline consumption than those categorized as “high” housing consumption at baseline. The IV estimate for the “low” baseline consumption households is a 26-percent increase and is highly significant, while the effect for “high” baseline housing consumption is slightly negative and not significant.

Having isolated a subgroup that likely experiences a sizable increase in housing consumption and a subgroup that should experience a marked increase in non-housing consumption, we now turn to heterogeneity in child development impacts for these groups. The bottom panel of Figure 7 reports the effects of a housing voucher for the pooled sample, then separately by predicted baseline housing spending. We find that the effects of the voucher appear to be driven primarily by children in households with low baseline housing consumption. The IV point estimate for receipt of a voucher is 0.58 standard deviations, and highly significant

( $p < 0.001$ ). In contrast, the effects of a voucher for households with high baseline housing consumption is close to zero and not significant. This suggests that improvements in housing and neighborhood conditions may be particularly important in mediating the effects on child development.

One concern is that the large effect we find for the low housing spending group is simply an artifact of exploring effects across many subgroups. In Appendix Table A.13 we report a variety of adjusted p-values which correct for the many sub-groups that we explore. Even after accounting for the large number of groups tested, the effect for the low housing spending group remains highly significant. Another concern is that the “low” housing consumption at baseline group may just be lower income, in which case, we would be conflating heterogeneity in effects by housing consumption with heterogeneity by income. To evaluate this possibility, we construct predicted income in our data, and explore heterogeneity along predicted income.<sup>22</sup> We find that income-based heterogeneity does not account for the sharp pattern of heterogeneity that we document here. Together, these results point to the important role that changes in housing and neighborhood conditions play in explaining the observed effects rather than changes in non-housing consumption alone.

## 6.2 Comparing the Effects of Higher- and Lower-Quality Public Housing

In Section 5, we document how public housing and housing vouchers appear to differentially impact child development, on average, and have divergent effects on housing and neighborhood quality. This raises the question of whether moving into better quality public housing has different effects than moving into worse quality public housing. In this subsection we take an exploratory approach to investigate this possibility.

A natural empirical strategy would partition public housing developments into good or poor quality and estimate a regression of the form:

$$Y_i = \beta_1 \text{High Quality PH}_i + \beta_2 \text{Low Quality PH}_i + \nu_i \quad (5)$$

Even though public housing offers are random, the choice of development—and therefore selection into good or poor quality public housing developments—is non-random and could bias estimates of  $\beta_1$  or  $\beta_2$ . Thus, estimating the effects of moving into higher- and lower-quality public housing requires separate instruments for each type of treatment.<sup>23</sup>

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<sup>22</sup>In Appendix Figure A.7, we plot the estimates for households by estimated baseline income, where we use income observed for assisted households to predict income for all lottery households. Taken at face value, the results in Appendix Figure A.7 suggest that heterogeneity in effects by income is not driving the heterogeneity we find for baseline housing consumption.

<sup>23</sup>These instruments must also be margin-specific.

The process by which the housing authority made offers of public housing apartments provides a source of plausibly exogenous variation in the public housing choices available to waiting list families. In particular, as described in Section 2.2, families coming off the waiting list for public housing were to be offered one (or more) unit of the requisite size in each of three geographic leasing zones, depending on availability. While we do not observe the choice set offered to families, we can nevertheless use random variation in the waiting time for public housing in different zones interacted with pre-lottery zone of residence – as a proxy for unobserved zone preferences – to simulate the variation in public housing quality available to waiting list families as these receive housing offers.

To reduce the dimensionality of the development choice problem, and to align with the features of the public housing offer process, we categorize public housing developments as “High Quality” or “Low Quality” based on the leasing zone in which they are located. As we show in Appendix H, this binary classification comports with observable characteristics of the developments and neighborhoods. Public housing developments categorized as high quality are located in neighborhoods with lower poverty and violent crime, they have higher hedonic quality, and higher predicted income rank in adulthood, according to the Opportunity Atlas (Chetty et al., 2018).

To construct our instruments, we first flexibly estimate waiting times for each public housing development leasing zone using the randomized initial waiting list position and bedroom size. This yields a predicted waiting time:  $\hat{w}_{i,k}$  for each household  $i$  for each public housing leasing zone, indexed by  $k \in \{1, 2, 3\}$ .<sup>24</sup> Details on the construction of the estimated waiting times are provided in Appendix G. To generate additional variation, we also interact these randomized zone-specific waiting times with the baseline zone of residence ( $Z_i$ ) to instrument for the type of public housing development that families lease. Specifically, we start with the following system of equations:

$$Y_i = \sum_{j \in (\text{Hi-Qual PH}, \text{Low-Qual PH}, \text{HCV})} \beta_j D_{i,j} + \phi_x + \mu_z + X_i' \gamma + \varepsilon_i \quad (6)$$

and

$$D_{i,j} = \sum_{k=1}^3 \pi_{k,j} \hat{w}_{i,k} + \sum_{k=1}^3 \sum_{z=1}^3 \pi_{k,j,z} \mathbf{1}(Z_i = z) \times \hat{w}_{i,zk} + \gamma_j \text{Top HCV}_i + \phi_{xj} + \mu_{zj} + X_i' \delta_j + \nu_i \quad (7)$$

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<sup>24</sup>Technically, these predicted waiting times are  $\hat{w}(k)_{i,b,r}$  where bedroom needs are indexed by  $b$  and for bins of randomized waiting list position  $r$ , but for compactness and readability, we suppress these additional subscripts. See Appendix G for details.

Where for individual  $i$ , their treatment status  $D_{i,j}$  can take one of four mutually exclusive values indexed by  $j$  : no treatment (0), high-quality public housing (High-Qual PH), low-quality public housing (Low-Qual PH), or a housing voucher (HCV). The system described by equations 6 and 7 contains 12 possible instruments for high-quality and low-quality public housing. Given the well-known concerns about the bias of IV with possibly many weak instruments (Bound et al., 1995), and recent guidance emphasizing fewer instruments in over-identified settings (Mikusheva and Sun, 2024), we use the IV-LASSO of Belloni et al. (2012) to select among the waiting time instruments. Throughout, we also control for baseline zone fixed effects:  $\mu_z$ , fixed effects for number of bedrooms needed  $\phi_x$ , and, as in earlier specifications, a vector of pre-lottery child and household demographics and neighborhood characteristics ( $X_i$ ).<sup>25</sup>

This strategy is similar in spirit to the approach of using interactions of randomized treatment offers with site dummies or other baseline covariates to generate additional sources of variation to study counterfactual-specific treatment effects or the role of mediators in randomized experiments (Kling et al., 2007; Duncan et al., 2011; Kline and Walters, 2016). The intuition for our empirical strategy is that longer waiting times for high-quality public housing affect families’ propensity to lease up differently depending on their baseline zone of residence. So we compare child development outcomes for families experiencing shorter versus longer waits, across families initially living in high-quality zones (more likely to lease in high-quality zones) and those living in low-quality zones (less likely to lease in high-quality zones). The difference in how outcomes respond to waiting time variation for high/low-quality public housing between these groups identifies the causal effect of accessing high-quality versus low-quality public housing.

The IV-LASSO (Belloni et al., 2012) selects potentially different sets of “optimal instruments” for different outcomes, so the precise first stage specification varies with the outcome. For illustrative purposes, we report an example of a first stage in Appendix Table we A.20. Unsurprisingly, interactions of waiting times for the highest (lowest) quality zone in the south (north) with baseline residence in highest (lowest) quality zone are the strongest instruments for higher (lower) quality public housing. Appendix Table A.20 also illustrates that our

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<sup>25</sup>An alternate approach would fully saturate interactions of baseline zone with waiting times for each zone with the following first stage:

$$D_{i,j} = \sum_{k=1}^3 \sum_{z=1}^3 \pi_{zkj} \mathbf{1}(Z_i = z) \times \hat{w}_{i,zk} + \gamma_{\text{Top}} \text{HCV}_i + \phi_{xj} + \mu_{zj} + X_i' \delta_j + \nu_i \quad (8)$$

where  $k$  indexes the public housing zone, and  $z$  indexes the pre-lottery zone of residence. This creates nine instruments for higher- and lower-quality public housing. The results of this alternative specification appear in Appendix Table A.21. The results are similar to those in Table 9, however, we prefer to use the LASSO data-driven approach to selecting the instruments, as it generally yields more precise estimates.

instruments generate significant independent variation in moves to high- and low- quality public housing with an Angrist-Pischke partial F statistics of 154 for moving into lower-quality public housing and 47 for moves into higher-quality public housing.

However, in order for this approach to yield causal effects with a LATE interpretation, it requires that exclusion and monotonicity continue to hold *for each instrument*. This is non-trivial, and could be violated in our setting if, for example, changes in waiting time for high-quality zone public housing caused households to switch between low-quality public housing, vouchers, or no-treatment. Alternatively, if we were willing to assume constant treatment effects within public housing types, then our IV estimates could be interpreted as LATEs (Mogstad and Torgovitsky, 2024). Although we do not find significant heterogeneity in treatment effects *for public housing* in Section 5.3, researchers typically view constant effects as unrealistic. Given that this analysis requires either: stronger restrictions on how the instruments impact choice of assistance *or* constant effects, we emphasize that this analysis is exploratory, and therefore should be interpreted with caution. We also report the results of over-identification tests, which under a constant effects model, test instrument validity.

Table 9 summarizes the results across the main outcomes domains studied in Section 5. In columns (1) and (2), we focus on impacts on housing and neighborhood quality, as proxied by hedonic quality estimates and tract poverty rates, respectively. The point estimates for the effect of moving into lower-quality public housing both indicate a substantial worsening of neighborhood and housing quality. In particular, moving into the lower-quality public housing zone causes a 22-percentage point *increase* in poverty exposure, and reduces estimated housing consumption by 24 percent. In contrast, the coefficients for the effects of moving into a higher-quality public housing development are consistently opposite-signed. Taken at face value, these estimates imply that moving to better quality public housing developments leads to a 5-percentage point reduction in tract poverty exposure, and a slight uptick in housing quality, though not statistically different than zero. We can reject that the effects of moving to higher-quality public housing and lower quality public housing are the same for neighborhood poverty ( $p < 0.001$ ) and housing quality ( $p < 0.05$ ).<sup>26</sup>

We find broadly similar patterns when we evaluate the effects on early literacy. Columns (3) and (4) report the effects on early literacy scores and kindergarten readiness, respectively. Moving to lower quality public housing appears to worsen early literacy scores by -0.21 standard deviations, although the effect is not statistically significant. The effect of moving into better quality public housing appears to be positive; however, the estimates are imprecisely estimated, and we cannot technically reject that the effects are the same ( $p = 0.21$ ). For

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<sup>26</sup>It is worth noting that the overidentification test for neighborhood poverty rejects at the 10-percent levels ( $p = 0.094$ ), suggesting that constant effects may be unlikely to hold for the neighborhood poverty results.

kindergarten readiness, we find that moving into worse public housing causes an 16-percentage point reduction in being kindergarten ready, which is marginally significant. Whereas, the point estimate for moving into higher quality public housing implies a 19-percentage point *increase* in kindergarten readiness, but it is not statistically significant. Nevertheless, we can reject that the effects of high- and low-quality public housing on kindergarten readiness are the same at the 0.10 significance level. Moreover, the overidentification tests fails to reject instrument validity under constant effects.

Finally, we investigate the effects on the cognitive and behavioral health index used in Section 5.3.5. The estimates are reported in column (5). Moving to lower-quality public housing lowers the index of cognitive and behavioral health by 0.45 standard deviations, which is statistically significant. In contrast, the point estimate for moving to higher-quality public housing is positive, and the estimate is marginally significant. Moreover, we can reject that the effects of moving to low-quality public housing are the same as the effects of moving to high-quality public housing.

This analysis, while requiring strong assumptions, suggests that the effects of public housing appear to differ substantially based on the quality of housing and the neighborhood environment it provides. We view these results as largely consistent with the voucher analysis in Section 6.1, which found that the positive impact of vouchers largely accrued to households who would experience the greatest improvement in housing and neighborhood conditions. Both analyses attempt to isolate the contribution of changes in the consumption of the in-kind good, while holding fixed family resources, and both analyses highlight how improvements in child development seem to depend on the degree of improvement in housing and neighborhood environment.

### 6.3 Relationship to Existing Literature

How do these results compare to other studies of the effects of housing assistance? For vouchers, the closest study to ours is [Jacob et al. \(2015\)](#) (hereafter “JKL”), which uses a lottery of housing vouchers in Chicago linked to data on 3-8th grade test scores, high school completion, crime and hospitalizations. In contrast to our results, they conclude that there are few, if any, meaningful effects of vouchers on children. In Appendix I, we investigate potential explanations for why our voucher results differ from this work, which we briefly summarize here.

The major differences between our work and JKL can be broadly classified as differences in treatment context, data construction, and outcome measures. While both papers study the same “treatment” — federal housing vouchers — the housing authority we study is a county-level housing authority, which means it can more easily facilitate voucher lease-ups

in suburban jurisdictions (which may be of higher quality), whereas the Chicago Housing Authority studied by JKL is often more constrained to city neighborhoods. Consistent with these differences, families in our context use their vouchers to lease-up in observably “better” neighborhoods: Census tracts with lower crime and less poverty, while JKL finds no effect of voucher receipt on similar standard measures of neighborhood quality. Another difference compared to prior work is that our waitlist data allow us to *directly observe* the children listed on each application and their identifying information necessary for linking to outcomes. In contrast, JKL do not observe the children attached to assistance applications, and must use public assistance records to identify the children likely to be in the waiting list household. As a result, our data should be less likely to suffer from measurement error than prior studies. Additionally, we rely on statewide education data, which allows us to measure children’s outcomes even if they move across public school districts, whereas prior work relies on data from a single school district. Finally, while we do still find effects in elementary and early middle school test scores, consistent with some non-experimental estimates (Schwartz et al., 2020), much of our focus is on measures drawn from assessments taken upon entry into formal schooling, which is more proximal to children’s experiences of early-life conditions and resources, and therefore may be more likely to detect changes induced by housing programs.

Most studies of public housing focus on the effects of escaping troubled public housing with housing vouchers (Sanbonmatsu et al., 2011; Chetty et al., 2016; Jacob, 2004; Chyn, 2018; Haltiwanger et al., 2024), but a couple of studies explore the effects of being admitted to, or time spent in, public housing (Currie and Yelowitz, 2000; Pollakowski et al., 2022). Currie and Yelowitz (2000) use a split-sample IV, based on the gender composition of children and public housing bedroom rules, to estimate impacts on over-crowding and grade retention, as measured in Census data. Consistent with our results on housing instability, they find that moving into public housing relieves over-crowding. Pollakowski et al. (2022) use data on housing assistance and a research design comparing siblings who differ in their exposure to public housing to examine the effects of duration in public housing on children’s future earnings. Their results suggest that longer stays in public housing during adolescence are associated with increased earnings in adulthood. We provide new evidence using randomized variation in public housing on a range of housing and child development outcomes.

A growing literature explores the effects of changing neighborhoods — without accompanying changes in family resources — on children’s short- and long-run outcomes (Kling et al., 2007; Chetty et al., 2016; Chetty and Hendren, 2018; Chyn, 2018; Chyn et al., 2025).<sup>27</sup> Studies focused on longer-run outcomes such as earnings and college-going have consistently

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<sup>27</sup>These studies mostly focus on the effects of moving out of public housing with a voucher, and thus entail little change in the size of a transfer going to low-income families.

found positive effects of being exposed to less impoverished neighborhoods (Chetty et al., 2016; Chyn, 2018; Chyn et al., 2025), however, the literature finds more mixed evidence of neighborhood effects on cognitive development in childhood and adolescence (Jacob, 2004; Kling et al., 2007; Sanbonmatsu et al., 2011). If one of the main channels through which vouchers improve child development is better housing and neighborhood conditions, as our results suggest, it is hard to square this finding with interim evidence from MTO, which did not find consistent improvements in academic achievement for children who moved from poor quality public housing to better neighborhood environments with a voucher (Sanbonmatsu et al., 2006; Kling et al., 2007; Sanbonmatsu et al., 2011). One possibility is that there are complementarities between increases in family resources (not present in MTO) and improved housing and neighborhood environment. For example, increases in child-focused consumption (such as children’s books) or time with parents could be more valuable when a child has their own bedroom, or is moved to less-crowded housing, safer or less impoverished environments. Another explanation is that the development gains from improved housing and neighborhood environment accrue primarily to the youngest children (our focus), while much of the MTO evidence on academic achievement was drawn from samples with somewhat older children (Sanbonmatsu et al., 2006, 2011). Of course, other contextual differences in the outcome measures used or the precise populations studied could explain these divergent findings.

Still another related literature examines the effects of in-kind or cash subsidies to low-income families on children, without direct changes to housing or neighborhood conditions. Much of this literature on in-kind transfers features exposure designs and focuses on long-run outcomes such as adult health and labor market performance (Bailey et al., 2023; Hoynes et al., 2016; East et al., 2023). These long-run quasi-experimental studies mainly find positive effects on future health or labor market outcomes for children with greater access to the safety net. Whereas more recent experimental studies of cash transfers to low-income families have found generally muted effects on children’s outcomes (Noble et al., 2025; Krause et al., 2025). Our experimental evidence suggests that inkind transfers can lead to improvements in child development outcomes, but that these impacts are sensitive to the particular changes in consumption induced by the subsidy. Taken at face value, these differences in results across studies may indicate that consumption of particular goods such as housing, nutrition, or health care — those targeted by in-kind transfers — could be especially important for child development (Currie and Gahavri, 2008), either due to spillovers, internalities, or externalities.

## 7 Conclusion

This paper provides novel experimental evidence on how two major in-kind transfers — housing vouchers and public housing — affect early childhood development through their differential impacts on housing and neighborhood quality. We link individual child records from a unique housing assistance lottery in Miami-Dade County, Florida to housing and neighborhood quality, housing instability, and early childhood skills assessments, test scores, and Medicaid claims data to study impacts on a range of policy-relevant outcomes. Vouchers reduce housing instability and cause families to move to better-quality housing and neighborhoods. In turn, we find significant improvements in school readiness and early literacy skills among children whose families receive vouchers, likely capturing both cognitive and non-cognitive skill development. These gains persist into elementary/middle school. We also find evidence that children in families receiving vouchers experience reductions in cognitive and behavioral health challenges. Across each child development outcome, the effects are only realized among families receiving vouchers, and are concentrated among boys.

Notably, public housing shows no detectable effects on child development outcomes despite reducing housing instability. However, our exploratory analysis reveals substantial heterogeneity: children moving to higher-quality public housing developments appear to experience improved outcomes similar to voucher recipients, while those in lower-quality developments appear to suffer worse outcomes. Consistent with these results, our heterogeneity analysis of voucher holders suggests that gains in child development were concentrated among children whose families experienced the greatest improvements in housing and neighborhood quality from receiving a voucher. These patterns reinforce the notion that improvements in housing and neighborhood environment, rather than simply increasing family resources, seem to drive positive developmental effects.

Our work provides experimental evidence that social safety net programs can bolster children’s healthy development as detected both in their school readiness and reduced likelihood of exhibiting behavioral problems or delays, and illustrates how these effects depend on the particular changes in consumption induced by program participation. This finding connects to a wide literature studying the second-generation effects of in-kind ([Jacob et al., 2015](#); [Hoynes et al., 2016](#); [Bailey et al., 2023](#); [East et al., 2023](#)) and cash transfers ([Aizer et al., 2016](#); [Duncan et al., 2011](#); [Hawkins et al., 2025](#); [Barr et al., 2022](#)). The gains in school-entry cognitive and non-cognitive skill measures may also help bridge two literature streams, one finding that in-kind transfers can improve children’s long-run outcomes ([Hoynes et al., 2016](#); [Schwartz et al., 2020](#); [Barr et al., 2022](#)) and a second on the importance of resources and investment in the critical early childhood years for generating downstream effects ([Almond](#)

and Currie, 2011; Almond et al., 2018; Heckman, 2006; Heckman and Masterov, 2007).

The results offer several lessons for policy. First, vouchers appear similarly effective at reducing housing instability, and more effective at increasing housing and neighborhood quality and boosting early child development than public housing. Second, the effects of public housing depend on the quality and location of the public housing development. This finding suggests that improving the siting and quality of public housing may yield benefits comparable to those of providing vouchers. Similarly, removing the worst quality public housing in favor of vouchers — a focus of low-income housing policy the last two decades — seems likely to improve child outcomes on average.

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**Table 1: Randomization and Summary Stats**

|                      | Top<br>Vouchers<br>(HCV)<br>(1) | Top<br>Public Housing<br>(PH)<br>(2) | Control<br>(Bottom)<br>(3) | P-value from<br>joint test<br>(4) |
|----------------------|---------------------------------|--------------------------------------|----------------------------|-----------------------------------|
| Child Age            | 2.135                           | 2.084                                | 2.120                      | 0.266                             |
| Boy                  | 0.486                           | 0.508                                | 0.490                      | 0.203                             |
| Girl                 | 0.514                           | 0.492                                | 0.510                      | 0.203                             |
| Age, Head            | 27.534                          | 27.630                               | 27.667                     | 0.844                             |
| Disabled, Head       | 0.054                           | 0.060                                | 0.060                      | 0.755                             |
| Elderly, Head        | 0.005                           | 0.004                                | 0.005                      | 0.939                             |
| White (Not Hispanic) | 0.016                           | 0.017                                | 0.014                      | 0.469                             |
| Hispanic             | 0.270                           | 0.294                                | 0.290                      | 0.318                             |
| Black                | 0.713                           | 0.685                                | 0.691                      | 0.243                             |
| Asian                | 0.001                           | 0.002                                | 0.003                      | 0.537                             |
| Female, Head         | 0.916                           | 0.911                                | 0.922                      | 0.295                             |
| Ever Homeless        | 0.076                           | 0.078                                | 0.075                      | 0.920                             |
| Tract Poverty        | 0.228                           | 0.221                                | 0.218                      | 0.030                             |
| Has Email            | 0.700                           | 0.710                                | 0.719                      | 0.423                             |
| Total Adults         | 1.230                           | 1.208                                | 1.231                      | 0.148                             |
| Total Children       | 2.273                           | 2.270                                | 2.201                      | 0.063                             |
| Predicted Beds       | 2.720                           | 2.728                                | 2.708                      | 0.740                             |
| N                    | 1,438                           | 3,037                                | 6,660                      |                                   |

*Notes:* This table reports the mean characteristics for the for children who link to some Florida Kindergarten Readiness Screener data by voucher and public housing waitlist positions. Column (1) reports the mean baseline characteristics for children who's household was randomly assigned to the top of the voucher list. Column (2) reports the means for children who's household was randomly assigned to the top of the public housing waiting list. Column (3) for children from households at the bottom of both waiting lists. Column (4) reports p-values from a joint test of the significance of difference between columns (1) -(2) and column (3). For ease of interpretation, the groups are constructed to be mutually exclusive. P-values are calculated using robust standard errors clustered at the household level.

**Table 2: First Stage**

|              | HCV<br>(1)          | PH<br>(2)            |
|--------------|---------------------|----------------------|
| Top HCV      | 0.304***<br>(0.014) | −0.048***<br>(0.006) |
| Top PH       | −0.002<br>(0.005)   | 0.160***<br>(0.008)  |
| Control Mean | 0.007               | 0.021                |
| N            | 10,864              | 10,864               |

*Notes:* This table reports the first stage results. Column 1 reports the impact of being at the top of the housing voucher waiting list (“Top HCV”) and being at the top of the public housing waiting list (“Top PH”) on lease-up with a housing voucher. Column 2 reports the impact of being at the top of the housing voucher waiting list (“Top HCV”) and being at the top of the public housing waiting list (“Top PH”) on admission to public housing. The sample consists of children less than 5 on the waiting list for housing assistance, who link to a developmental outcome described in section 3. Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Table 3: Impact on Housing and Neighborhood Quality**

|                              | Housing<br>Quality       | Neighborhood Quality     |                            |                      |                           | Opportunity<br>Atlas    |
|------------------------------|--------------------------|--------------------------|----------------------------|----------------------|---------------------------|-------------------------|
|                              | $\Delta \ln(\hat{R}ent)$ | $\Delta$ Poverty<br>Rate | $\Delta$ School<br>Quality | $\Delta$ Crime       | $\Delta$ Single<br>Parent | $\Delta$ Income<br>Rank |
|                              | (1)                      | (2)                      | (3)                        | (4)                  | (5)                       | (6)                     |
| <i>Panel A: Reduced Form</i> |                          |                          |                            |                      |                           |                         |
| Top HCV                      | 0.042***<br>(0.013)      | −0.016***<br>(0.004)     | 0.020<br>(0.026)           | −2.475***<br>(0.494) | −0.004<br>(0.004)         | 0.002<br>(0.002)        |
| Top PH                       | −0.017*<br>(0.010)       | 0.016***<br>(0.003)      | 0.014<br>(0.020)           | 1.615***<br>(0.369)  | 0.007**<br>(0.003)        | −0.002<br>(0.001)       |
| <i>Panel B: IV (LATE)</i>    |                          |                          |                            |                      |                           |                         |
| HCV                          | 0.098***<br>(0.035)      | −0.031***<br>(0.011)     | 0.065<br>(0.073)           | −5.405***<br>(1.372) | −0.005<br>(0.010)         | 0.004<br>(0.005)        |
| PH                           | −0.072*<br>(0.042)       | 0.078***<br>(0.014)      | 0.072<br>(0.098)           | 7.921***<br>(1.776)  | 0.036**<br>(0.014)        | −0.010<br>(0.006)       |
| Control Mean                 | 7.148                    | 0.218                    | −0.653                     | 12.371               | 0.266                     | 0.360                   |

*Notes:* This table reports the effects of waitlist position (reduced-form) and housing assistance (IV) on housing consumption and neighborhood quality. All outcomes are measured as changes between post-lottery and pre-lottery addresses. All neighborhood quality outcomes are measured at the census tract-level. Column 1 reports impacts on housing quality, defined as hedonic rent, described in section 3.5. Column 6 impacts on neighborhood predicted income rank in adulthood from the Opportunity Atlas from [Chetty et al. \(2018\)](#). Panel A reports the ITT estimates from estimating equation 1 and Panel B reports the LATE estimates. “HCV” reports the coefficient on voucher waitlist position (reduced-form) or voucher lease-up (IV). “PH” reports the coefficient on public housing waitlist position (reduced-form) or public housing lease-up (IV). Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Table 4: Impact on Housing Instability and Homelessness**

|                              | Homeless or<br>Doubled-up<br>(1) | Shelter<br>(2)      | Doubled-up<br>(3)   |
|------------------------------|----------------------------------|---------------------|---------------------|
| <i>Panel A: Reduced Form</i> |                                  |                     |                     |
| Top HCV                      | −0.018**<br>(0.008)              | −0.008**<br>(0.004) | −0.013*<br>(0.007)  |
| Top PH                       | −0.014**<br>(0.006)              | −0.005<br>(0.004)   | −0.008*<br>(0.005)  |
| <i>Panel B: IV (LATE)</i>    |                                  |                     |                     |
| HCV                          | −0.074***<br>(0.028)             | −0.033**<br>(0.015) | −0.052**<br>(0.023) |
| PH                           | −0.085**<br>(0.039)              | −0.032<br>(0.023)   | −0.053*<br>(0.031)  |
| Control Mean                 | 0.084                            | 0.027               | 0.057               |
| Controls                     | Y                                | Y                   | Y                   |
| N                            | 11,131                           | 11,131              | 11,131              |

*Notes:* This table reports the effects of waitlist position (reduced-form) and housing assistance (IV) on homelessness and housing instability. Column 1 reports impacts on any form of homelessness and doubling-up as described in 3.4. Columns 2 and 3 report impacts on the two common housing instability statuses in the McKinney-Vento data, which are living in shelter or transitional housing (i.e “Shelter”) and living doubled-up (“Doubled-up”). Panel A reports the ITT estimates from estimating equation 1 and Panel B reports the LATE estimates. “HCV” reports the coefficient on voucher waitlist position (reduced -orm) or voucher lease-up (IV). “PH” reports the coefficient on public housing waitlist position (reduced-form) or public housing lease-up (IV). Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Table 5: Impacts on Early Literacy and Kindergarten Readiness**

|                               | All                    |                       |                                 | Boys                   |                       | Girls                  |                       |
|-------------------------------|------------------------|-----------------------|---------------------------------|------------------------|-----------------------|------------------------|-----------------------|
|                               | Control<br>Mean<br>(1) | Full<br>Sample<br>(2) | No<br>Assistance<br>Alt.<br>(3) | Control<br>Mean<br>(4) | Full<br>Sample<br>(5) | Control<br>Mean<br>(6) | Full<br>Sample<br>(7) |
| <i>Panel A: Reduced Form</i>  |                        |                       |                                 |                        |                       |                        |                       |
| <b>Kindergarten Readiness</b> |                        |                       |                                 |                        |                       |                        |                       |
| Top HCV                       | 0.608<br>(0.488)       | 0.040***<br>(0.014)   | 0.046**<br>(0.019)              | 0.571<br>(0.495)       | 0.083***<br>(0.020)   | 0.645<br>(0.479)       | −0.005<br>(0.019)     |
| Top PH                        | 0.612<br>(0.487)       | 0.002<br>(0.010)      | 0.004<br>(0.011)                | 0.581<br>(0.493)       | −0.002<br>(0.015)     | 0.641<br>(0.480)       | 0.008<br>(0.014)      |
| <b>Std. Reading Score</b>     |                        |                       |                                 |                        |                       |                        |                       |
| Top HCV                       | −0.005<br>(1.004)      | 0.066**<br>(0.029)    | 0.095**<br>(0.038)              | −0.093<br>(1.038)      | 0.156***<br>(0.040)   | 0.080<br>(0.962)       | −0.026<br>(0.041)     |
| Top PH                        | 0.004<br>(0.999)       | −0.001<br>(0.021)     | 0.004<br>(0.023)                | −0.078<br>(1.033)      | 0.015<br>(0.030)      | 0.082<br>(0.959)       | −0.017<br>(0.029)     |
| <i>Panel B: IV (LATE)</i>     |                        |                       |                                 |                        |                       |                        |                       |
| <b>Kindergarten Readiness</b> |                        |                       |                                 |                        |                       |                        |                       |
| HCV                           | 0.608<br>(0.488)       | 0.135***<br>(0.048)   | 0.141**<br>(0.059)              | 0.571<br>(0.495)       | 0.284***<br>(0.074)   | 0.645<br>(0.479)       | −0.008<br>(0.063)     |
| PH                            | 0.612<br>(0.487)       | 0.017<br>(0.064)      | 0.026<br>(0.064)                | 0.581<br>(0.493)       | −0.016<br>(0.089)     | 0.641<br>(0.480)       | 0.048<br>(0.090)      |
| <b>Std. Reading Score</b>     |                        |                       |                                 |                        |                       |                        |                       |
| HCV                           | −0.005<br>(1.004)      | 0.218**<br>(0.100)    | 0.291**<br>(0.120)              | −0.093<br>(1.038)      | 0.553***<br>(0.152)   | 0.080<br>(0.962)       | −0.098<br>(0.132)     |
| PH                            | 0.004<br>(0.999)       | −0.006<br>(0.134)     | 0.025<br>(0.134)                | −0.078<br>(1.033)      | 0.082<br>(0.186)      | 0.082<br>(0.959)       | −0.111<br>(0.185)     |
| Controls                      | N                      | Y                     | Y                               | N                      | Y                     | N                      | Y                     |
| N                             | 9,456                  | 10,863                | 9,454                           | 4,671                  | 5,351                 | 4,785                  | 5,510                 |

*Notes:* This table reports the effects of waitlist position (reduced-form) and housing assistance (IV) on early child outcomes. “Kindergarten Readiness” is an indicator for having scored at or above the threshold to be determined ready for kindergarten. “Std. Reading Score” is a standardized score from evaluations of early literacy skills. Column 1 reports the mean among those randomized to the bottom of the waiting list (“Control Mean”). Column 2 reports the ITT estimates from equation 1 in Panel A and LATE estimates in Panel B. “HCV” reports the coefficient on voucher waitlist position (reduced form) or voucher lease-up (IV). “PH” reports the coefficient on public housing waitlist position (reduced form) or public housing lease-up (IV). Column 3 reports the effects using the “No Alternative Assistance” specification described in 4.2. Columns 4-5 and 6-7 repeats the specifications in columns (1) and (2) separately for boys and girls, respectively. Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

update table notes

**Table 6: Impacts on Children’s Observed Developmental Benchmarks**

|                                      | All                    |                       |                                 | Boys                   |                       | Girls                  |                       |
|--------------------------------------|------------------------|-----------------------|---------------------------------|------------------------|-----------------------|------------------------|-----------------------|
|                                      | Control<br>Mean<br>(1) | Full<br>Sample<br>(2) | No<br>Assistance<br>Alt.<br>(3) | Control<br>Mean<br>(4) | Full<br>Sample<br>(5) | Control<br>Mean<br>(6) | Full<br>Sample<br>(7) |
| <i>Panel A: Reduced Form</i>         |                        |                       |                                 |                        |                       |                        |                       |
| <b>Not Emerging or Demonstrating</b> |                        |                       |                                 |                        |                       |                        |                       |
| Top HCV                              | 0.148<br>(0.355)       | −0.027***<br>(0.009)  | −0.033***<br>(0.012)            | 0.169<br>(0.375)       | −0.047***<br>(0.014)  | 0.128<br>(0.334)       | −0.009<br>(0.013)     |
| Top PH                               | 0.147<br>(0.354)       | −0.008<br>(0.007)     | −0.009<br>(0.008)               | 0.164<br>(0.370)       | −0.002<br>(0.011)     | 0.131<br>(0.338)       | −0.016*<br>(0.010)    |
| <b>Emerging</b>                      |                        |                       |                                 |                        |                       |                        |                       |
| Top HCV                              | 0.505<br>(0.500)       | 0.033**<br>(0.014)    | 0.055***<br>(0.019)             | 0.501<br>(0.500)       | 0.056***<br>(0.020)   | 0.508<br>(0.500)       | 0.010<br>(0.020)      |
| Top PH                               | 0.510<br>(0.500)       | −0.006<br>(0.010)     | 0.006<br>(0.011)                | 0.512<br>(0.500)       | −0.017<br>(0.015)     | 0.507<br>(0.500)       | 0.006<br>(0.014)      |
| <b>Demonstrating</b>                 |                        |                       |                                 |                        |                       |                        |                       |
| Top HCV                              | 0.357<br>(0.479)       | 0.003<br>(0.013)      | −0.002<br>(0.018)               | 0.328<br>(0.469)       | 0.009<br>(0.019)      | 0.386<br>(0.487)       | −0.002<br>(0.019)     |
| Top PH                               | 0.354<br>(0.478)       | 0.013<br>(0.010)      | 0.007<br>(0.011)                | 0.326<br>(0.469)       | 0.011<br>(0.014)      | 0.381<br>(0.486)       | 0.014<br>(0.014)      |
| <i>Panel B: IV (LATE)</i>            |                        |                       |                                 |                        |                       |                        |                       |
| <b>Not Emerging or Demonstrating</b> |                        |                       |                                 |                        |                       |                        |                       |
| HCV                                  | 0.148<br>(0.355)       | −0.099***<br>(0.032)  | −0.100***<br>(0.039)            | 0.169<br>(0.375)       | −0.164***<br>(0.051)  | 0.128<br>(0.334)       | −0.044<br>(0.042)     |
| PH                                   | 0.147<br>(0.354)       | −0.052<br>(0.044)     | −0.050<br>(0.045)               | 0.164<br>(0.370)       | −0.013<br>(0.065)     | 0.131<br>(0.338)       | −0.105*<br>(0.061)    |
| <b>Emerging</b>                      |                        |                       |                                 |                        |                       |                        |                       |
| HCV                                  | 0.505<br>(0.500)       | 0.103**<br>(0.048)    | 0.167***<br>(0.058)             | 0.501<br>(0.500)       | 0.172**<br>(0.072)    | 0.508<br>(0.500)       | 0.038<br>(0.064)      |
| PH                                   | 0.510<br>(0.500)       | −0.034<br>(0.064)     | 0.036<br>(0.064)                | 0.512<br>(0.500)       | −0.103<br>(0.088)     | 0.507<br>(0.500)       | 0.037<br>(0.092)      |
| <b>Demonstrating</b>                 |                        |                       |                                 |                        |                       |                        |                       |
| HCV                                  | 0.357<br>(0.479)       | 0.024<br>(0.046)      | −0.005<br>(0.055)               | 0.328<br>(0.469)       | 0.045<br>(0.068)      | 0.386<br>(0.487)       | 0.007<br>(0.062)      |
| PH                                   | 0.354<br>(0.478)       | 0.080<br>(0.061)      | 0.039<br>(0.062)                | 0.326<br>(0.469)       | 0.068<br>(0.083)      | 0.381<br>(0.486)       | 0.092<br>(0.089)      |
| Controls                             | N                      | Y                     | Y                               | N                      | Y                     | N                      | Y                     |
| N                                    | 9,697                  | 11,131                | 9,694                           | 4,810                  | 5,506                 | 4,887                  | 5,623                 |

*Notes:* This table reports the effects of waitlist position (reduced form) and housing assistance (IV) on developmental assessments from embedded classroom observations. “Not Emerging or Demonstrating” is an indicator for the lowest categorization of development during classroom observations, taking the value zero otherwise. “Emerging” is an indicator if the child is coded as showing some indication of the necessary skill development. “Demonstrating” is an indicator which takes the value 1 if the child shows the highest level of skill-development, and zero otherwise. Column 1 reports the mean among those randomized to the bottom of the waiting list. Column 2 reports the ITT estimates from estimating equation 1 in Panel A and LATE estimates in Panel B. “HCV” reports the coefficient on voucher waitlist position (reduced form) or voucher lease-up (IV). “PH” reports the coefficient on public housing waitlist position (reduced form) or public housing lease-up (IV). Column 3 reports the effects using the “No Alternative Assistance” specification described in 4.2. Columns 4-5 and 6-7 repeats the specifications in columns (1) and (2) separately for boys and girls, respectively. Columns 4-5 and 6-7 report control group means and treatment effects separately for boys and girls, respectively. Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ . update table notes

**Table 7: Impacts on Children’s Cognitive and Behavioral Health**

|                       | Impulse<br>Control/<br>Disturbance<br>of Conduct | Oppositional<br>Defiance/<br>Emotional<br>Disturbance | Attention<br>Deficit<br>Hyper-Active | Developmental<br>Delays in<br>Reading,<br>Math, or<br>Speech | Index              |
|-----------------------|--------------------------------------------------|-------------------------------------------------------|--------------------------------------|--------------------------------------------------------------|--------------------|
|                       | (1)                                              | (2)                                                   | (3)                                  | (4)                                                          | (5)                |
| <b>Panel A: All</b>   |                                                  |                                                       |                                      |                                                              |                    |
| <i>Reduced Form</i>   |                                                  |                                                       |                                      |                                                              |                    |
| Top HCV               | −0.009**<br>(0.004)                              | −0.002<br>(0.003)                                     | −0.003<br>(0.006)                    | −0.012*<br>(0.007)                                           | 0.043**<br>(0.022) |
| Top PH                | 0.005<br>(0.003)                                 | −0.003<br>(0.002)                                     | −0.002<br>(0.004)                    | 0.008<br>(0.005)                                             | −0.007<br>(0.017)  |
| <i>IV (LATE)</i>      |                                                  |                                                       |                                      |                                                              |                    |
| HCV                   | −0.025*<br>(0.013)                               | −0.008<br>(0.009)                                     | −0.014<br>(0.020)                    | −0.032<br>(0.024)                                            | 0.140*<br>(0.076)  |
| PH                    | 0.030<br>(0.021)                                 | −0.018<br>(0.013)                                     | −0.014<br>(0.029)                    | 0.049<br>(0.035)                                             | −0.045<br>(0.114)  |
| Control Means         | 0.031                                            | 0.014                                                 | 0.071                                | 0.110                                                        | −0.005             |
| N                     | 15,638                                           | 15,638                                                | 15,638                               | 15,638                                                       | 15,638             |
| <b>Panel B: Boys</b>  |                                                  |                                                       |                                      |                                                              |                    |
| <i>Reduced Form</i>   |                                                  |                                                       |                                      |                                                              |                    |
| Top HCV               | −0.011*<br>(0.006)                               | −0.006<br>(0.004)                                     | 0.003<br>(0.010)                     | −0.015<br>(0.011)                                            | 0.060*<br>(0.034)  |
| Top PH                | 0.006<br>(0.005)                                 | −0.004<br>(0.003)                                     | −0.007<br>(0.007)                    | 0.014*<br>(0.008)                                            | −0.008<br>(0.027)  |
| <i>IV (LATE)</i>      |                                                  |                                                       |                                      |                                                              |                    |
| HCV                   | −0.035<br>(0.023)                                | −0.028*<br>(0.015)                                    | 0.002<br>(0.037)                     | −0.040<br>(0.040)                                            | 0.214<br>(0.131)   |
| PH                    | 0.040<br>(0.033)                                 | −0.026<br>(0.021)                                     | −0.048<br>(0.047)                    | 0.094*<br>(0.055)                                            | −0.050<br>(0.178)  |
| Control Means         | 0.041                                            | 0.020                                                 | 0.095                                | 0.141                                                        | −0.117             |
| N                     | 7,853                                            | 7,853                                                 | 7,853                                | 7,853                                                        | 7,853              |
| <b>Panel C: Girls</b> |                                                  |                                                       |                                      |                                                              |                    |
| <i>Reduced Form</i>   |                                                  |                                                       |                                      |                                                              |                    |
| Top HCV               | −0.006<br>(0.004)                                | 0.003<br>(0.004)                                      | −0.010<br>(0.006)                    | −0.009<br>(0.008)                                            | 0.028<br>(0.027)   |
| Top PH                | 0.004<br>(0.004)                                 | −0.001<br>(0.002)                                     | 0.003<br>(0.005)                     | 0.001<br>(0.006)                                             | −0.009<br>(0.020)  |
| <i>IV (LATE)</i>      |                                                  |                                                       |                                      |                                                              |                    |
| HCV                   | −0.015<br>(0.013)                                | 0.009<br>(0.012)                                      | −0.028<br>(0.021)                    | −0.028<br>(0.026)                                            | 0.082<br>(0.087)   |
| PH                    | 0.023<br>(0.025)                                 | −0.009<br>(0.016)                                     | 0.020<br>(0.035)                     | 0.005<br>(0.042)                                             | −0.057<br>(0.136)  |
| Control Means         | 0.021                                            | 0.009                                                 | 0.047                                | 0.079                                                        | 0.108              |
| N                     | 7,785                                            | 7,785                                                 | 7,785                                | 7,785                                                        | 7,785              |
| Controls              | Y                                                | Y                                                     | Y                                    | Y                                                            | Y                  |

*Notes:* This table reports the effects of waitlist position (reduced form) and housing assistance (IV) on cognitive and behavioral health diagnoses described in section 3.3.2. Each column reports effects on a different outcome, based on three-digit ICD-9 codes. Column 5 reports effects on a standardized index of cognitive and behavioral health, with higher values indicating better health. The top of each panel reports the ITT estimates from estimating equation 1 and the bottom reports the LATE estimates. Panel A reports for all children less than 5 at the time of random assignment. Panel B and C reports the results separately for boys and girls, respectively. Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Table 8: Impacts on Elementary and Middle School Test Scores**

|                                        | Reading<br>Scores<br>(1) | Math<br>Scores<br>(2) | Combined<br>Scores<br>(3) |
|----------------------------------------|--------------------------|-----------------------|---------------------------|
| <b>Panel A: Early Childhood Sample</b> |                          |                       |                           |
| <i>Reduced Form</i>                    |                          |                       |                           |
| Top HCV                                | 0.085**<br>(0.038)       | 0.035<br>(0.040)      | 0.060*<br>(0.035)         |
| Top PH                                 | 0.026<br>(0.028)         | 0.001<br>(0.030)      | 0.014<br>(0.026)          |
| <i>IV (LATE)</i>                       |                          |                       |                           |
| HCV                                    | 0.340**<br>(0.143)       | 0.127<br>(0.150)      | 0.234*<br>(0.132)         |
| PH                                     | 0.167<br>(0.179)         | 0.008<br>(0.192)      | 0.087<br>(0.166)          |
| N                                      | 7,097                    | 7,097                 | 7,097                     |
| <b>Panel B: Full Child Sample</b>      |                          |                       |                           |
| <i>Reduced Form</i>                    |                          |                       |                           |
| Top HCV                                | 0.057**<br>(0.023)       | 0.037*<br>(0.022)     | 0.047**<br>(0.021)        |
| Top PH                                 | -0.001<br>(0.017)        | -0.016<br>(0.016)     | -0.009<br>(0.015)         |
| <i>IV (LATE)</i>                       |                          |                       |                           |
| HCV                                    | 0.212**<br>(0.091)       | 0.116<br>(0.087)      | 0.164**<br>(0.082)        |
| PH                                     | -0.010<br>(0.127)        | -0.121<br>(0.123)     | -0.066<br>(0.115)         |
| N                                      | 20,882                   | 20,882                | 20,882                    |

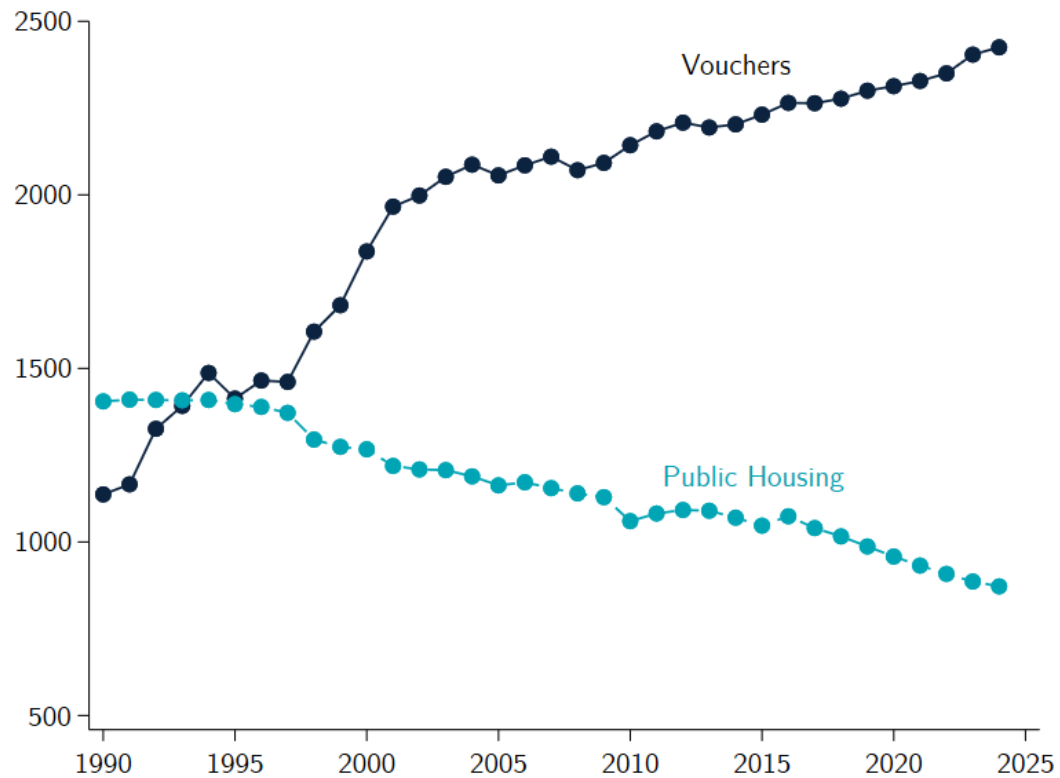
*Notes:* This table reports the effects of waitlist position (reduced form) and housing assistance (IV) on math and reading test scores in grades 3-8. Column 1 reports impacts on standardized reading scores, Column 2 reports impacts on standardized math scores, and Column 3 reports the impacts on combined reading and math scores. Panel A reports the ITT estimates and LATE estimates for the sample of children in tables 5 that have valid follow-up test score data. Panel B repeats the specifications in Panel A, but for the entire sample of children that we observe in the relevant test score years. Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Table 9: The Effects of Public Housing by Public Housing Development Quality**

|                             | Housing<br>Quality       | Neighborhood<br>Quality  | Early<br>Literacy        |                           | Behavioral<br>Health |
|-----------------------------|--------------------------|--------------------------|--------------------------|---------------------------|----------------------|
|                             | $\Delta \ln(\hat{R}ent)$ | $\Delta$ Poverty<br>Rate | Std.<br>Reading<br>Score | Kindergarten<br>Readiness | Development<br>Index |
|                             | (1)                      | (2)                      | (3)                      | (4)                       | (5)                  |
| Lower Quality PH            | -0.244***<br>(0.062)     | 0.226***<br>(0.022)      | -0.209<br>(0.215)        | -0.163*<br>(0.105)        | -0.447**<br>(0.181)  |
| Higher Quality PH           | 0.004<br>(0.072)         | -0.048**<br>(0.023)      | 0.368<br>(0.295)         | 0.190<br>(0.137)          | 0.490*<br>(0.284)    |
| P-value: Lower=Higher       | 0.032                    | 0.000                    | 0.211                    | 0.099                     | 0.026                |
| HCV                         | 0.080***<br>(0.030)      | -0.024**<br>(0.009)      | 0.206**<br>(0.097)       | 0.122***<br>(0.047)       | 0.132*<br>(0.074)    |
| P-value: Lower=HCV          | 0.000                    | 0.000                    | 0.076                    | 0.011                     | 0.002                |
| P-value: Higher=HCV         | 0.315                    | 0.329                    | 0.603                    | 0.640                     | 0.221                |
| P-value: Overidentification | 0.208                    | 0.094                    | 0.724                    | 0.850                     | 0.143                |
| N                           | 7,371                    | 10,921                   | 10,862                   | 10,862                    | 15,638               |

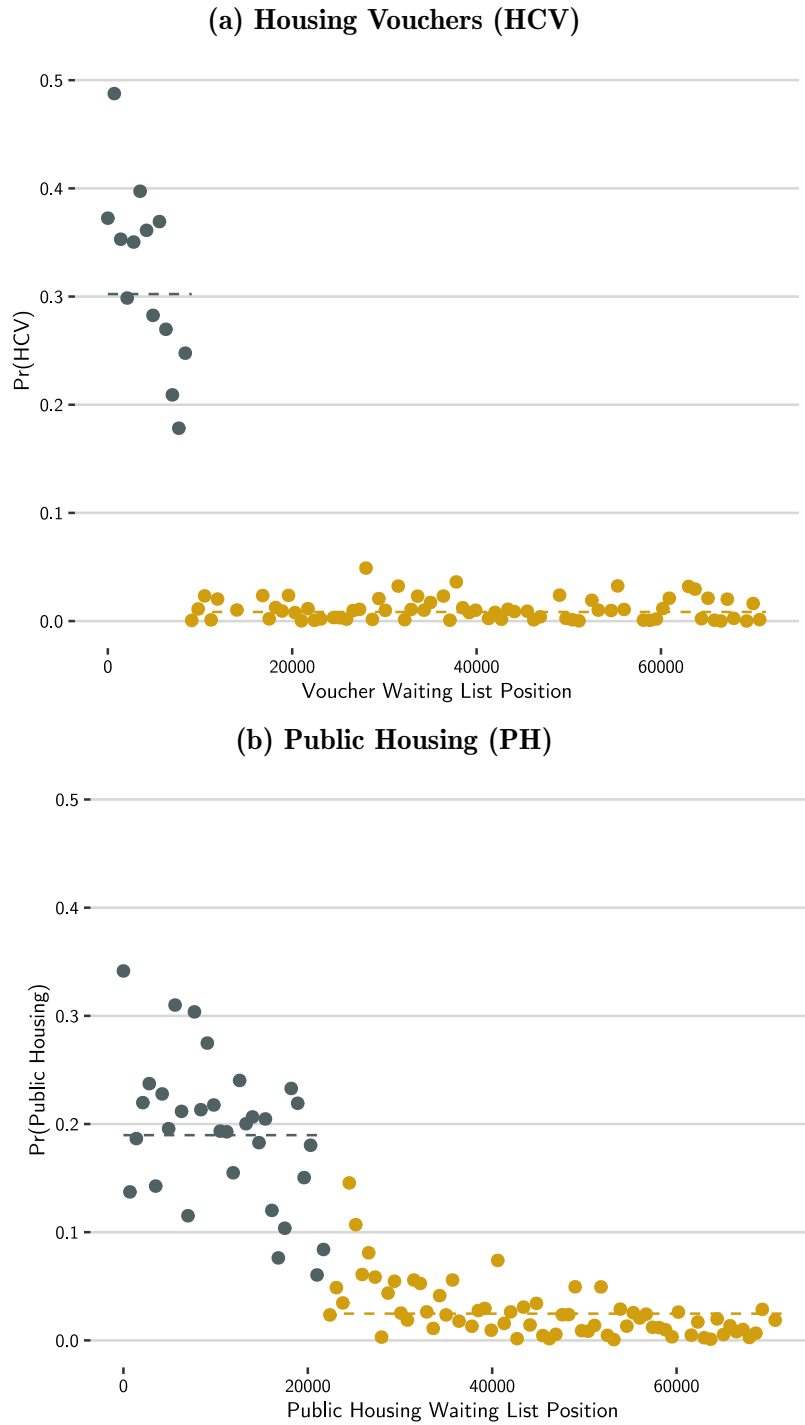
*Notes:* This table reports IV estimates of the effects of moving into higher quality public housing (“Higher Quality PH”) and the effects of moving into lower quality public housing (“Lower Quality PH”), along with the effects of leasing-up with a housing voucher (“HCV”) as described in section 6.2. P-value: Higher=Lower reports a test of equality of coefficients in the effects of Higher and Lower Quality Public Housing. We also report the resulting p-value from an overidentification test (Hansen’s J statistic). Column 1 reports impacts on changes in hedonic quality, described in 3.5. Column 2 reports effects on changes in Census tract poverty. Column 3 shows effects on standardized reading success scores, and column 4 shows impacts on kindergarten readiness. Column 5 reports impacts on the cognitive and behavioral health index described in 3.3.2. standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Figure 1: Federal Housing Assistance Over Time by Program Type**



*Notes:* This figure plots the number of assisted households/units, in thousands, over time by the housing choice voucher program (navy), and public housing program (teal). Source: [Collinson et al. \(2019\)](#) and HUD's A Picture of Subsidized Households.

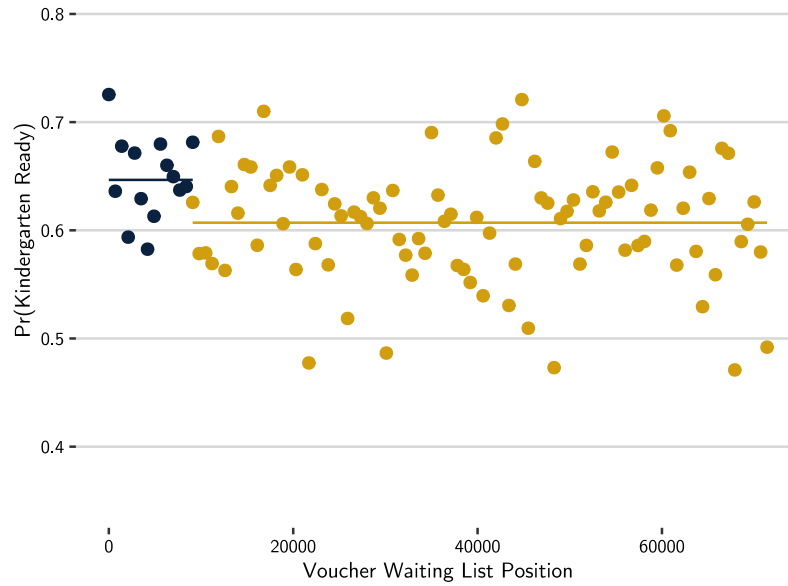
**Figure 2: Housing Assistance Receipt by Waiting List Positions**



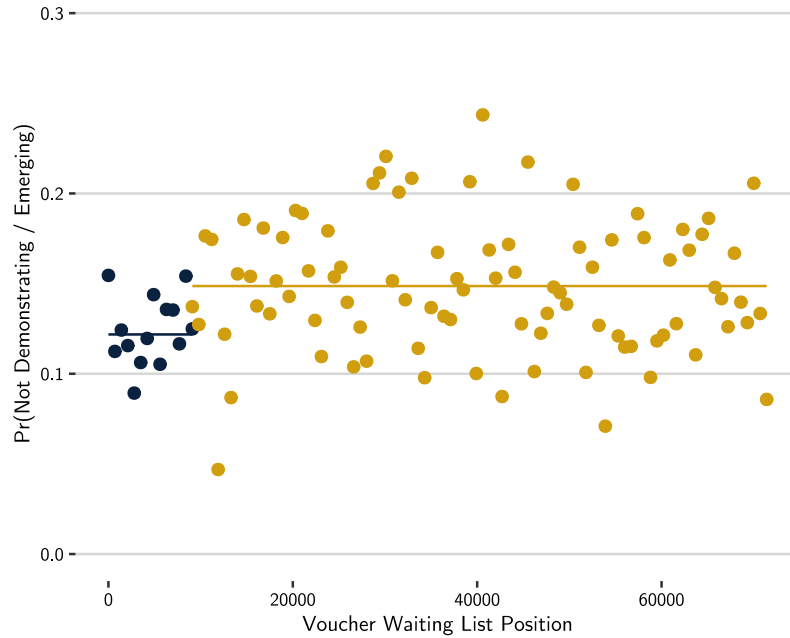
*Notes:* This figure plots the probability of leasing-up with a voucher by voucher waiting list position (panel a), and the probability of leasing-up in public housing by public housing waiting list position (panel b). Each dot is a 700 positions bin of the analytical sample. The navy circles represent binned means for those at the “top” of their respective waiting list, and gold circles represent control group children outside the top.

**Figure 3: Child Development Outcomes by Voucher Lottery Position**

**(a) Kindergarten Readiness**



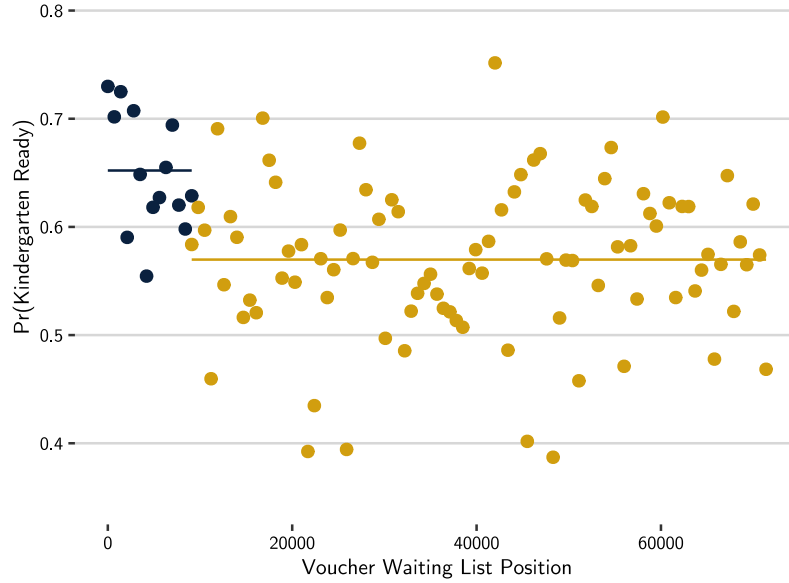
**(b) Not Emerging or Demonstrating Skills**



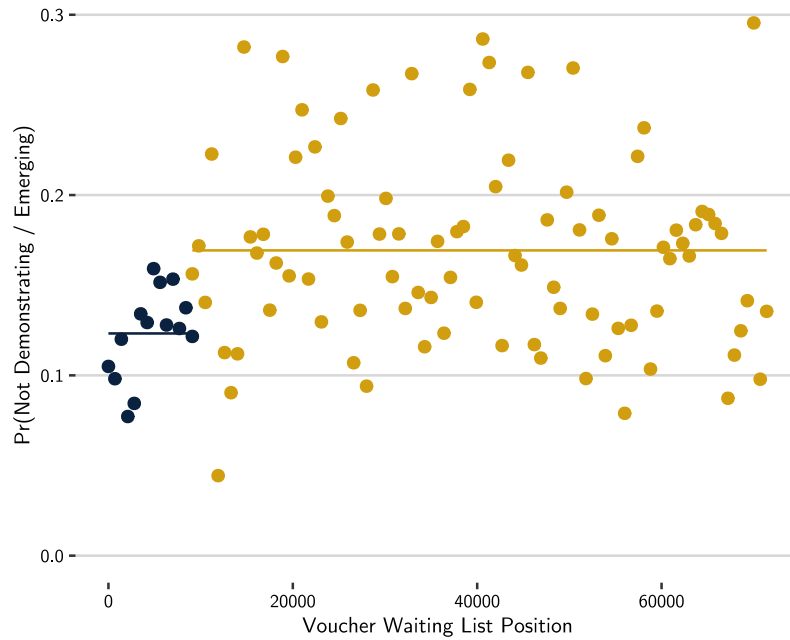
*Notes:* This figure plots mean child development outcomes by bins of the randomized initial positions on the voucher list for the sample. In panel A the outcome is kindergarten readiness. In panel B the outcome is whether the child is determined to be “Not Emerging or Demonstrating” in the ECHOS assessment. Each dot represents 700 slots of voucher positions. The solid blue line is the average of the y-variable for children at the “top” of the voucher list (i.e. those with positions less than 9,200 on the voucher list), the solid gold line is the average y-variable for children in positions greater than 9,200.

**Figure 4: Child Development Outcomes by Voucher Lottery Position (Boys)**

**(a) Kindergarten Readiness**

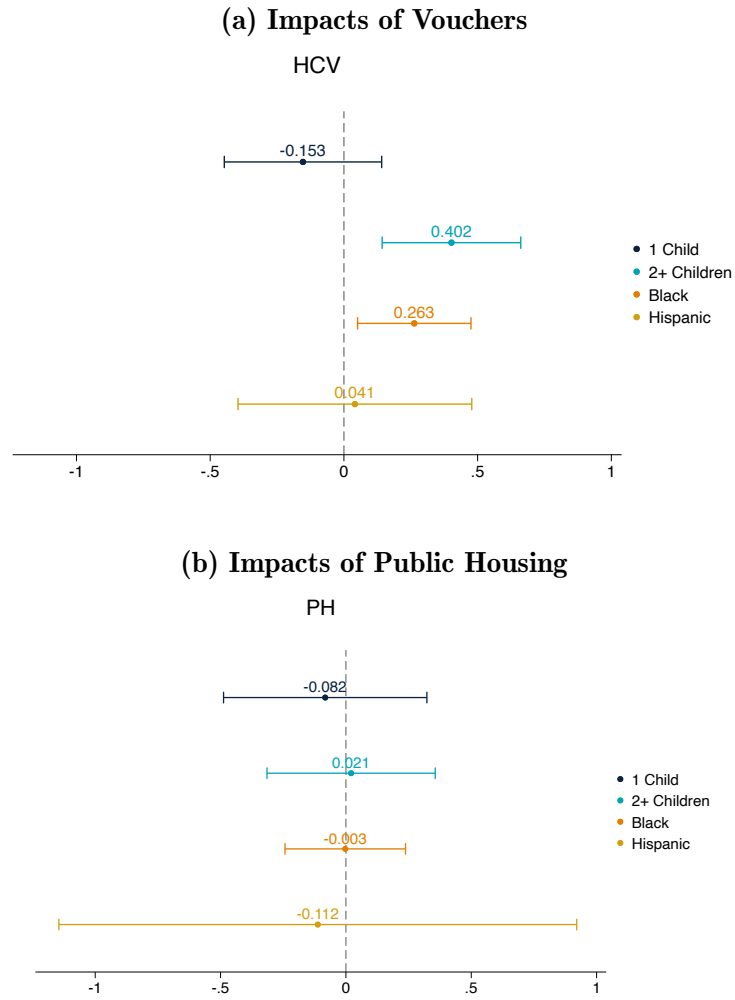


**(b) Not Emerging or Demonstrating Skills**



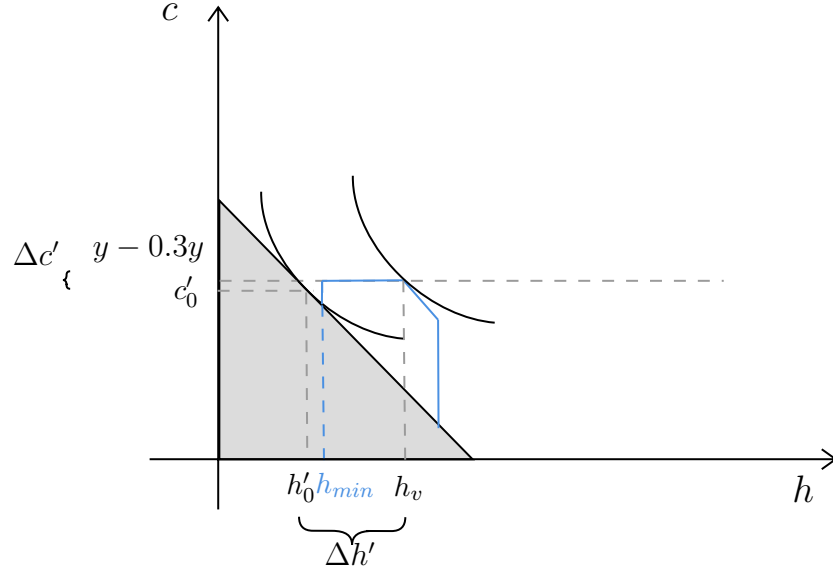
*Notes:* This figure plots mean child development outcomes by bins of the randomized initial positions on the voucher list for the sample of boys. Each dot represents 700 slots of voucher positions. The solid blue line is the average of the y-variable for children at the “top” of the voucher list (i.e. those with positions less than 9,200 on the voucher list), the solid gold line is the average y-variable for children in positions greater than 9,200.

**Figure 5: Heterogeneity by Demographics**

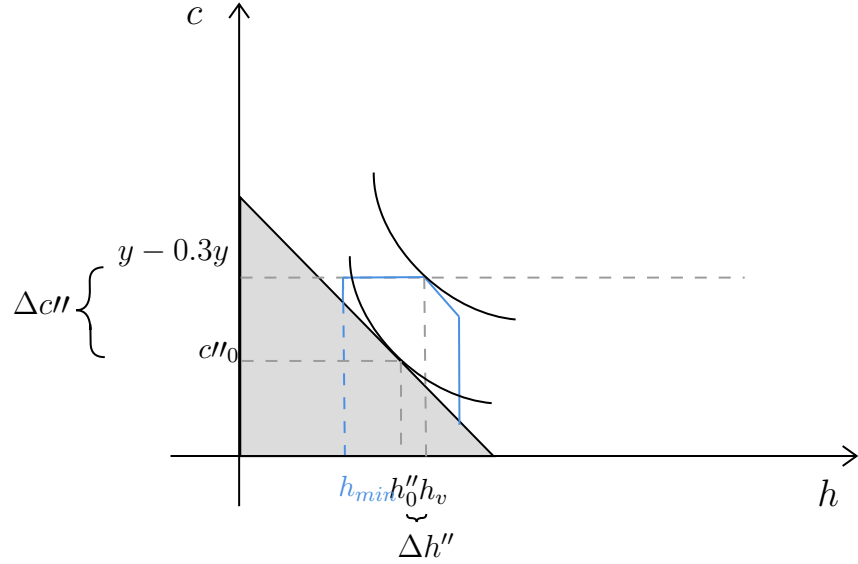


*Notes:* This figure plots the IV estimates for the effects of housing vouchers (“HCV”) and public housing (“PH”) on children’s standardized early literacy skills for several subgroups based on baseline characteristics. Whiskers are 95% confidence intervals.

**Figure 6: Theoretical Impacts of Voucher on Consumption**



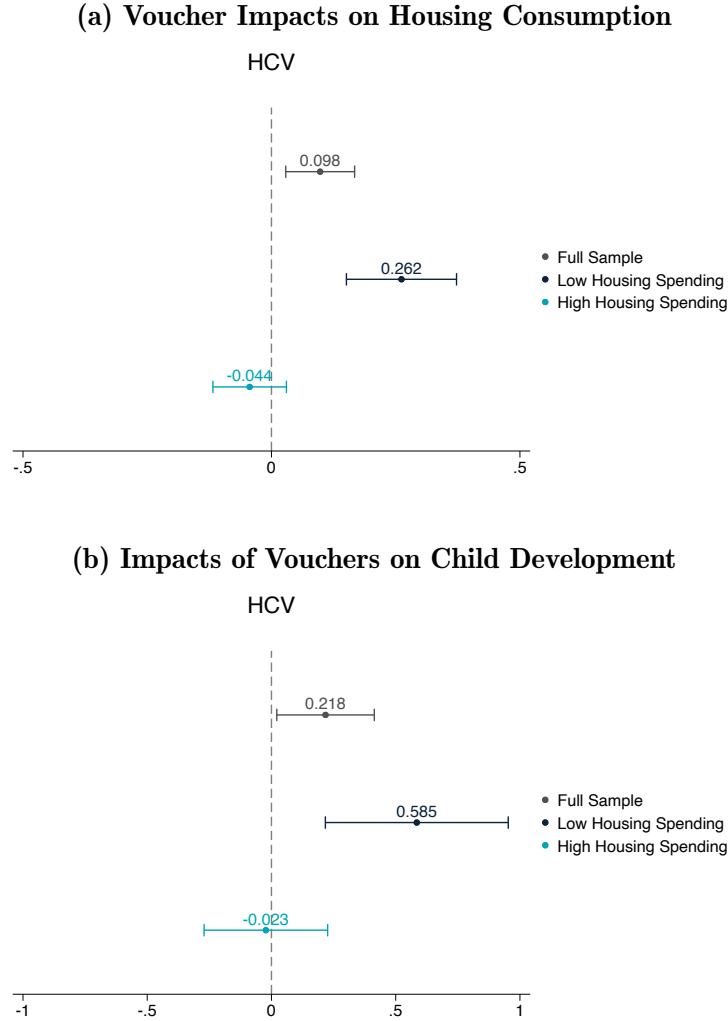
**(a) Low Baseline Housing Consumption**



**(b) High Baseline Housing Consumption**

*Notes:* This figure characterizes the theoretical consumption trade-off facing a household with and without a housing voucher. The grey shaded area is the budget region before receiving a voucher. The blue line represents the augmented portion of the budget constraint with a housing voucher. The top panel illustrates the changes in housing and non-housing consumption for a household with low spending on housing before voucher receipt. The bottom illustrates the changes in non-housing spending for a household with high initial spending on housing.

**Figure 7: Heterogeneity in Voucher Impacts on Housing Consumption and Child Development**



*Notes:* This figure plots the IV estimates for the effects of housing vouchers (“HCV”). In a panel a) the outcome is the change in hedonic housing consumption (described in 3.5) separately for households who spend relatively less on housing at baseline (hedonic rent less than FMR), and those that spend more relatively more on housing at baseline (hedonic rent greater than FMR). Panel b) reports the effects of vouchers on children’s standardized early literacy skills for households who spend relatively less on housing at baseline (hedonic rent less than FMR), and those that spend more relatively more on housing at baseline (hedonic rent greater than FMR). Results for the full sample are in gray, the results for the “Low Housing Spending” group are in black, and the “High Housing Spending” group is in teal. Whiskers are 95% confidence intervals

## Appendix

**Appendix Table A.1: Balance: Waiting List Position and Pre-Lottery Test Scores for Older Siblings**

|                        | Control Mean<br>(1) | Treatment Effect<br>(2) |
|------------------------|---------------------|-------------------------|
| <b>Std. Test Score</b> |                     |                         |
| Top HCV                | 0.001<br>(1.000)    | −0.006<br>(0.024)       |
| Top PH                 | −0.003<br>(1.002)   | 0.011<br>(0.017)        |
| Controls               | N                   | N                       |
| N                      | 19838               | 19838                   |

*Notes:* This table reports the results from regressing averaged pre-lottery test scores on waiting list position in a sample of older children. Column 1 reports the mean and standard deviation of standardized test scores for children whose families are outside the top of the voucher list and outside of the top of the public housing list. Column 2 reports the results from regressing test scores (pre-lottery) on indicators for being at the top of the voucher list (“Top HCV”) and being at the top of the public housing list (“Top PH”). Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.2: Waiting List Positions and Early Childhood Assessment Data Availability**

|                        | Control Mean<br>(1) | Treatment Effect<br>(2) |
|------------------------|---------------------|-------------------------|
| <b>Has FAIR-K Data</b> |                     |                         |
| Top HCV                | 0.724<br>(0.447)    | 0.004<br>(0.010)        |
| Top PH                 | 0.725<br>(0.446)    | −0.001<br>(0.007)       |
| <b>Has ECHOS Data</b>  |                     |                         |
| Top HCV                | 0.741<br>(0.438)    | 0.001<br>(0.009)        |
| Top PH                 | 0.743<br>(0.437)    | −0.003<br>(0.007)       |
| Controls               | N                   | N                       |
| N                      | 14911               | 14911                   |

*Notes* This table reports the relationship between randomized initial waiting list positions and whether a child has non-missing FAIR-K data and non-missing ECHOS Data, as described in Section 5.3.1. The top panel assesses balance in linking to a valid FAIR assessment. Column (1) reports the control mean and Column (2) reports the results from regressing an indicator for non-missing assessment data on an indicator for being at the top of the voucher list (“Top HCV”) and an indicator for being at the top of the public housing list (“Top PH”). Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.3: Randomization and Summary Stats (Medicaid Sample)**

|                      | Top<br>Vouchers<br>(HCV)<br>(1) | Top<br>Public Housing<br>(PH)<br>(2) | Control<br>(Bottom)<br>(3) | P-value from<br>joint test<br>(4) |
|----------------------|---------------------------------|--------------------------------------|----------------------------|-----------------------------------|
| Child Age            | 2.560                           | 2.548                                | 2.571                      | 0.636                             |
| Boy                  | 0.496                           | 0.508                                | 0.501                      | 0.635                             |
| Girl                 | 0.504                           | 0.492                                | 0.499                      | 0.635                             |
| Age, Head            | 27.851                          | 28.140                               | 28.054                     | 0.423                             |
| Disabled, Head       | 0.056                           | 0.065                                | 0.064                      | 0.421                             |
| Elderly, Head        | 0.004                           | 0.005                                | 0.005                      | 0.702                             |
| White (Not Hispanic) | 0.014                           | 0.016                                | 0.015                      | 0.830                             |
| Hispanic             | 0.284                           | 0.302                                | 0.294                      | 0.469                             |
| Black                | 0.701                           | 0.679                                | 0.687                      | 0.357                             |
| Asian                | 0.001                           | 0.001                                | 0.002                      | 0.631                             |
| Female, Head         | 0.917                           | 0.913                                | 0.919                      | 0.623                             |
| Ever Homeless        | 0.082                           | 0.080                                | 0.077                      | 0.781                             |
| Tract Poverty        | 0.226                           | 0.220                                | 0.217                      | 0.040                             |
| Has Email            | 0.706                           | 0.708                                | 0.717                      | 0.563                             |
| Total Adults         | 1.223                           | 1.213                                | 1.232                      | 0.229                             |
| Total Children       | 2.291                           | 2.307                                | 2.239                      | 0.058                             |
| Predicted Beds       | 2.721                           | 2.756                                | 2.731                      | 0.450                             |
| Any Prior Coverage   | 0.934                           | 0.932                                | 0.926                      | 0.313                             |
| Low Birth Weight     | 0.045                           | 0.053                                | 0.047                      | 0.283                             |
| Asthma               | 0.122                           | 0.121                                | 0.117                      | 0.812                             |
| Respiratory Problems | 0.190                           | 0.187                                | 0.182                      | 0.688                             |
| Depression           | 0.118                           | 0.116                                | 0.112                      | 0.710                             |
| Development Index    | -0.001                          | 0.004                                | -0.002                     | 0.890                             |
| N                    | 2,005                           | 4,294                                | 9,340                      |                                   |

*Notes:* This table reports the mean characteristics for the children in the medicaid sample by voucher and public housing waitlist positions. Column (1) reports the mean baseline characteristics for children who's household was randomly assigned to the top of the voucher list. Column (2) reports the means for children who's household was randomly assigned to the top of the public housing waiting list. Column (3) for children from households at the bottom of both waiting lists. Column (4) reports p-values from a joint test of the significance of difference between columns (1) -(2) and column (3), clustering standard errors at the household level.

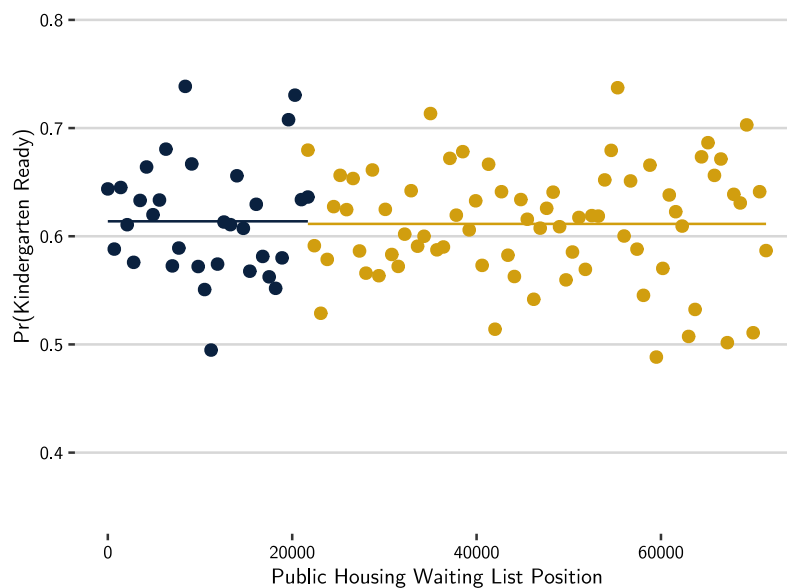
**Appendix Table A.4: Complier Counterfactual Shares**

|                             | Main<br>Specification<br>(1) | No Assistance<br>Alternative<br>(2) |
|-----------------------------|------------------------------|-------------------------------------|
| Panel A: Vouchers           |                              |                                     |
| Share $_{0 \rightarrow V}$  | 0.831                        | 0.986                               |
| Share $_{PH \rightarrow V}$ | 0.169                        | 0.014                               |
| Panel B: Public Housing     |                              |                                     |
| Share $_{0 \rightarrow PH}$ | 0.994                        | 1.012                               |
| Share $_{V \rightarrow PH}$ | 0.006                        | −0.012                              |

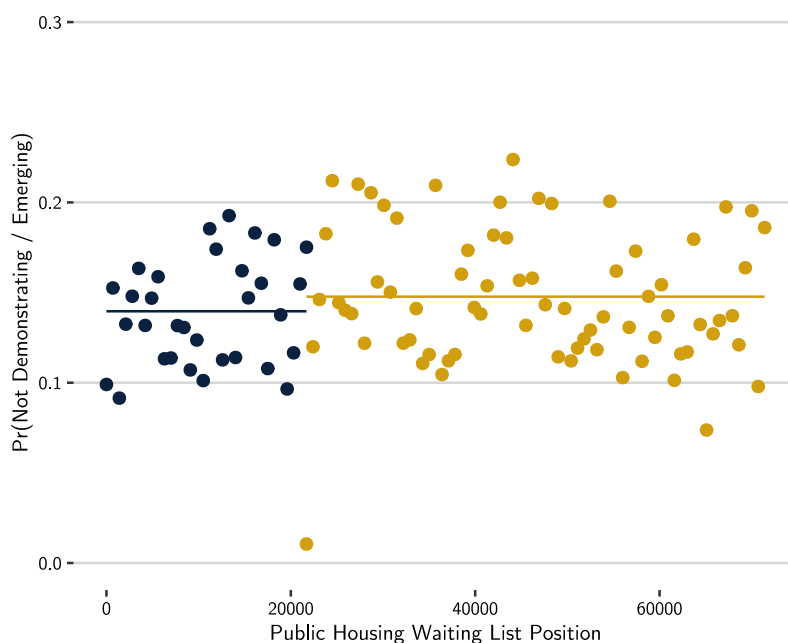
*Notes:* This table reports the estimated share of the complier populations drawn from different counterfactual states. These are the weights on the respective subLATEs derived in Appendix D. The subscripts denote the relevant transition:  $0 \rightarrow V$  indicates the share of the overall LATE that is coming from those who would otherwise get no-assistance, while  $P \rightarrow V$  indicates the share of the overall LATE that is coming from those diverted from public housing and towards vouchers. .

## Appendix Figure A.1: Child Development Outcomes by Public Housing Lottery Position

(a) Kindergarten Readiness



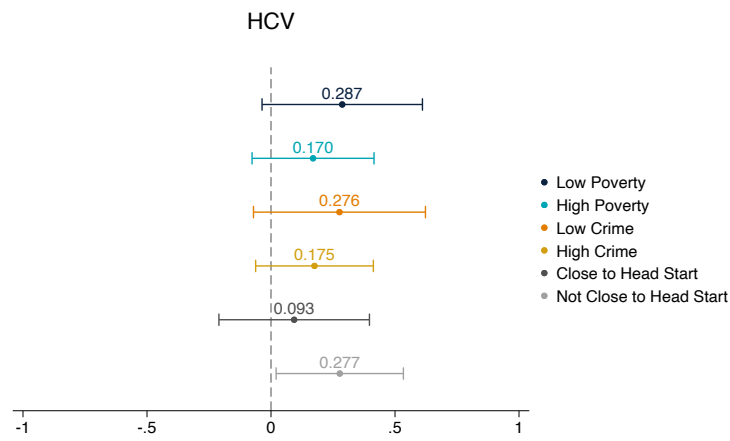
(b) Not Emerging or Demonstrating Skills



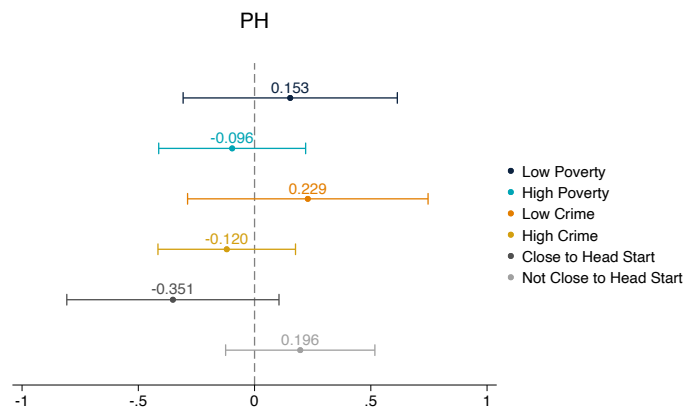
*Notes:* This figure plots mean child development outcomes by bins of the randomized initial positions on the public housing list for the sample. In panel A the outcome is kindergarten readiness. In panel B the outcome is whether the child is determined to be “Not Emerging or Demonstrating” in the ECHOS assessment. Each dot represents 700 slots of public housing positions. The dashed blue line is the average of the y-variable for children below position 22,000 on the public housing list, the dashed gold line is the average y-variable for children in positions above 22,000.

## Appendix Figure A.2: Heterogeneity by Neighborhood Environment

### (a) Impacts of Vouchers



### (b) Impacts of Public Housing



*Notes:* This figure plots the IV estimates for the effects of housing vouchers (“HCV”) and public housing (“PH”) on children’s standardized early literacy skills for several subgroups of baseline neighborhood conditions. Whiskers are 95% confidence intervals.

**Appendix Table A.5: Impacts of Vouchers on Early Literacy and Kindergarten Readiness for Early Positions v. Later Positions**

|                              | HCV Take-up<br>(1)  | Kindergarten<br>Readiness<br>(2) | Std. Reading<br>Score<br>(3) |
|------------------------------|---------------------|----------------------------------|------------------------------|
| <i>Panel A: Reduced Form</i> |                     |                                  |                              |
| Early List Position          | 0.463***<br>(0.062) | 0.137***<br>(0.045)              | 0.190*<br>(0.100)            |
| Later List Position          | 0.293***<br>(0.015) | 0.034**<br>(0.014)               | 0.058*<br>(0.030)            |
| <i>Panel B: IV (LATE)</i>    |                     |                                  |                              |
| Early HCV                    |                     | 0.301***<br>(0.113)              | 0.421*<br>(0.231)            |
| Late HCV                     |                     | 0.119**<br>(0.051)               | 0.199*<br>(0.106)            |
| Controls                     | Y                   | Y                                | Y                            |
| N                            | 10,863              | 10,863                           | 10,863                       |

*Notes:* This table examines the impact of voucher waiting list position on voucher take-up (Column 1), as well as the reduced-form impact of voucher waiting list position on kindergarten readiness (Column 2) and early literacy skills (Column 3). In particular, this table compares the effect of having a position near the very top (“Early List Position”) to the effect of having a position within the top, but outside the very top (“Later List Position”). Panel A reports the reduced-form. Panel B reports IV effects, where we separately estimate the effects of earlier and later *voucher receipt*. Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.6: Duration of Assistance Impacts**

|                                               | Std.<br>Reading<br>(1) | Kindergarten<br>Readiness<br>(2) |
|-----------------------------------------------|------------------------|----------------------------------|
| <b>Panel A. Receipt and Duration Estimate</b> |                        |                                  |
| Years with HCV                                | 0.083<br>(0.092)       | 0.067<br>(0.044)                 |
| HCV                                           | −0.004<br>(0.243)      | −0.033<br>(0.117)                |
| <b>Panel B. Duration Estimate</b>             |                        |                                  |
| Years with HCV                                | 0.082**<br>(0.035)     | 0.055***<br>(0.017)              |
| Controls                                      | Y                      | Y                                |
| N                                             | 10,863                 | 10,863                           |

*Notes:* This table reports estimates of the effects of assistance duration on standardized reading scores (Column 1) and an indicator of kindergarten readiness (Column 2). “Years with HCV” is the number of years between voucher receipt and the assessment, and “HCV” indicates any voucher receipt. Panel A reports IV estimates from a specification with two endogenous variables: an indicator for voucher receipt (“HCV”) and a continuous measure of years with the voucher (“Years with HCV”), where we instrument for each with an indicator “Top HCV” and separate dummies for earlier and later positions *within* the Top HCV. Panel B drops the indicator for voucher receipt. Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.7: Robustness to Top Voucher Cutoff**

|                              | < 8600               | <8800                | <9000                | <b>&lt;9200</b>             | <9400                | <9600                | <9800                |
|------------------------------|----------------------|----------------------|----------------------|-----------------------------|----------------------|----------------------|----------------------|
|                              | (1)                  | (2)                  | (3)                  | (4)                         | (5)                  | (6)                  | (7)                  |
| <i>Panel A: Reduced Form</i> |                      |                      |                      |                             |                      |                      |                      |
| Kindergarten Readiness       | 0.041***<br>(0.014)  | 0.037**<br>(0.014)   | 0.039***<br>(0.014)  | <b>0.040***</b><br>(0.014)  | 0.040***<br>(0.014)  | 0.038***<br>(0.014)  | 0.041***<br>(0.014)  |
| Std Reading                  | 0.063**<br>(0.030)   | 0.055*<br>(0.030)    | 0.062**<br>(0.029)   | <b>0.066**</b><br>(0.029)   | 0.063**<br>(0.029)   | 0.063**<br>(0.029)   | 0.069**<br>(0.028)   |
| Not Emerging/ Demonstrating  | -0.027***<br>(0.010) | -0.027***<br>(0.010) | -0.028***<br>(0.009) | <b>-0.027***</b><br>(0.009) | -0.027***<br>(0.009) | -0.026***<br>(0.009) | -0.026***<br>(0.009) |
| Emerging                     | 0.036**<br>(0.014)   | 0.037***<br>(0.014)  | 0.036***<br>(0.014)  | <b>0.033**</b><br>(0.014)   | 0.034**<br>(0.014)   | 0.034**<br>(0.014)   | 0.033**<br>(0.014)   |
| Demonstrating                | -0.000<br>(0.014)    | -0.003<br>(0.014)    | 0.001<br>(0.013)     | <b>0.003</b><br>(0.013)     | 0.002<br>(0.013)     | -0.000<br>(0.013)    | -0.001<br>(0.013)    |
| <i>Panel B: IV (LATE)</i>    |                      |                      |                      |                             |                      |                      |                      |
| Kindergarten Readiness       | 0.132***<br>(0.047)  | 0.118**<br>(0.047)   | 0.130***<br>(0.047)  | <b>0.135***</b><br>(0.048)  | 0.138***<br>(0.049)  | 0.134***<br>(0.049)  | 0.147***<br>(0.050)  |
| Std Reading                  | 0.197**<br>(0.098)   | 0.173*<br>(0.098)    | 0.199**<br>(0.099)   | <b>0.218**</b><br>(0.100)   | 0.210**<br>(0.102)   | 0.215**<br>(0.103)   | 0.242**<br>(0.105)   |
| Not Emerging/ Demonstrating  | -0.095***<br>(0.031) | -0.095***<br>(0.032) | -0.099***<br>(0.032) | <b>-0.099***</b><br>(0.032) | -0.102***<br>(0.033) | -0.101***<br>(0.033) | -0.102***<br>(0.034) |
| Emerging                     | 0.109**<br>(0.046)   | 0.113**<br>(0.046)   | 0.113**<br>(0.047)   | <b>0.103**</b><br>(0.048)   | 0.110**<br>(0.049)   | 0.112**<br>(0.049)   | 0.113**<br>(0.050)   |
| Demonstrating                | 0.012<br>(0.044)     | 0.004<br>(0.044)     | 0.017<br>(0.045)     | <b>0.024</b><br>(0.046)     | 0.021<br>(0.047)     | 0.013<br>(0.047)     | 0.012<br>(0.048)     |

*Notes:* This table reports our main reduced-form (Panel A) and IV (Panel B) results for all child development screener outcomes under a variety of alternative “top” definitions for voucher position. Our main estimates are in **bold** in column (4). We then vary the cutoff by 200 spots above and below our selected cutoff. Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.8: Robustness: Including School Fixed Effects (FAIR Outcomes)**

|                               | Control Mean<br>(1) | Main<br>Specification<br>(2) | With School<br>Fixed Effects<br>(3) |
|-------------------------------|---------------------|------------------------------|-------------------------------------|
| <i>Panel A: Reduced Form</i>  |                     |                              |                                     |
| <b>Kindergarten Readiness</b> |                     |                              |                                     |
| Top HCV                       | 0.608<br>(0.488)    | 0.040***<br>(0.014)          | 0.035**<br>(0.014)                  |
| Top PH                        | 0.612<br>(0.487)    | 0.002<br>(0.010)             | −0.000<br>(0.010)                   |
| <b>Std. Reading Score</b>     |                     |                              |                                     |
| Top HCV                       | −0.005<br>(1.004)   | 0.066**<br>(0.029)           | 0.059**<br>(0.030)                  |
| Top PH                        | 0.004<br>(0.999)    | −0.001<br>(0.021)            | −0.004<br>(0.022)                   |
| <i>Panel B: IV (LATE)</i>     |                     |                              |                                     |
| <b>Kindergarten Readiness</b> |                     |                              |                                     |
| HCV                           | 0.608<br>(0.488)    | 0.135***<br>(0.048)          | 0.112**<br>(0.047)                  |
| PH                            | 0.612<br>(0.487)    | 0.017<br>(0.064)             | −0.001<br>(0.064)                   |
| <b>Std. Reading Score</b>     |                     |                              |                                     |
| HCV                           | −0.005<br>(1.004)   | 0.218**<br>(0.100)           | 0.184*<br>(0.097)                   |
| PH                            | 0.004<br>(0.999)    | −0.006<br>(0.134)            | −0.021<br>(0.134)                   |
| Controls                      | -                   | Y                            | Y                                   |
| School FE                     | -                   | N                            | Y                                   |
| N                             | 9,456               | 10,863                       | 10,546                              |

*Notes:* This table repeats the analysis in Table 5, but includes school fixed effects, as discussed in Section 5.4. Columns (1) and (2) are repeated from Table 5 for ease of comparison. Column (3) reports the estimates with school fixed effects. This table reports the effects of waitlist position (reduced form) and housing assistance (IV) on Kindergarten Readiness and Standardized Early Literacy scores. Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.9: Robustness: Including School Fixed Effects (ECHOS Outcomes)**

|                                      | Control Mean<br>(1) | Main<br>Specification<br>(2) | With School<br>Fixed Effects<br>(3) |
|--------------------------------------|---------------------|------------------------------|-------------------------------------|
| <i>Panel A: Reduced Form</i>         |                     |                              |                                     |
| <b>Not Emerging or Demonstrating</b> |                     |                              |                                     |
| Top HCV                              | 0.148<br>(0.355)    | -0.027***<br>(0.009)         | -0.027***<br>(0.010)                |
| Top PH                               | 0.147<br>(0.354)    | -0.008<br>(0.007)            | -0.004<br>(0.007)                   |
| <b>Emerging</b>                      |                     |                              |                                     |
| Top HCV                              | 0.505<br>(0.500)    | 0.033**<br>(0.014)           | 0.030**<br>(0.014)                  |
| Top PH                               | 0.510<br>(0.500)    | -0.006<br>(0.010)            | -0.005<br>(0.010)                   |
| <b>Demonstrating</b>                 |                     |                              |                                     |
| Top HCV                              | 0.357<br>(0.479)    | 0.003<br>(0.013)             | 0.008<br>(0.013)                    |
| Top PH                               | 0.354<br>(0.478)    | 0.013<br>(0.010)             | 0.009<br>(0.010)                    |
| <i>Panel B: IV (LATE)</i>            |                     |                              |                                     |
| <b>Not Emerging or Demonstrating</b> |                     |                              |                                     |
| HCV                                  | 0.148<br>(0.355)    | -0.099***<br>(0.032)         | -0.089***<br>(0.032)                |
| PH                                   | 0.147<br>(0.354)    | -0.052<br>(0.044)            | -0.027<br>(0.046)                   |
| <b>Emerging</b>                      |                     |                              |                                     |
| HCV                                  | 0.505<br>(0.500)    | 0.103**<br>(0.048)           | 0.090*<br>(0.046)                   |
| PH                                   | 0.510<br>(0.500)    | -0.034<br>(0.064)            | -0.031<br>(0.063)                   |
| <b>Demonstrating</b>                 |                     |                              |                                     |
| HCV                                  | 0.357<br>(0.479)    | 0.024<br>(0.046)             | 0.034<br>(0.044)                    |
| PH                                   | 0.354<br>(0.478)    | 0.080<br>(0.061)             | 0.055<br>(0.059)                    |
| Controls                             | -                   | Y                            | Y                                   |
| School FE                            | -                   | N                            | Y                                   |
| N                                    | 9,697               | 11,131                       | 10,805                              |

*Notes:* This table repeats the analysis in Table 6, but includes school fixed effects. Columns (1) and (2) are repeated from Table 6 for ease of comparison. Column (3) reports the estimates with school fixed effects. This table reports the effects of waitlist position (reduced form) and housing assistance (IV) on Kindergarten Readiness and Standardized Early Literacy scores. Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.10: Robustness: Effects of Vouchers on Early Literacy & Kindergarten Readiness, Exclude “Possibly Treated” Voucher Households**

|                               | Control Mean<br>(1) | Full Sample<br>(2)  | Excluding<br>Near Top<br>(3) |
|-------------------------------|---------------------|---------------------|------------------------------|
| <i>Panel A: Reduced Form</i>  |                     |                     |                              |
| <b>Kindergarten Readiness</b> |                     |                     |                              |
| Top HCV                       | 0.608<br>(0.488)    | 0.040***<br>(0.014) | 0.043***<br>(0.014)          |
| <b>Std. Reading Score</b>     |                     |                     |                              |
| Top HCV                       | −0.005<br>(1.004)   | 0.066**<br>(0.029)  | 0.067**<br>(0.029)           |
| <i>Panel B: IV (LATE)</i>     |                     |                     |                              |
| <b>Kindergarten Readiness</b> |                     |                     |                              |
| HCV                           | 0.608<br>(0.488)    | 0.135***<br>(0.048) | 0.147***<br>(0.049)          |
| <b>Std. Reading Score</b>     |                     |                     |                              |
| HCV                           | −0.005<br>(1.004)   | 0.218**<br>(0.100)  | 0.235**<br>(0.102)           |
| Controls                      | N                   | Y                   | Y                            |
| N                             | 9,456               | 10,863              | 9,513                        |

*Notes:* This table repeats the analysis in Table 5, but drops children in families with voucher lottery positions that are “possibly treated” (i.e. those that could receive voucher offers in the years following our outcome window period). This table reports the effects of waitlist position (reduced form) and housing assistance (IV) on kindergarten readiness and early literacy skills. Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.11: Robustness: Effects of Vouchers on Developmental Benchmarks,  
Exclude “Possibly Treated” Voucher Households**

|                                      | Control Mean<br>(1) | Full Sample<br>(2)   | Excluding<br>Near Top<br>(3) |
|--------------------------------------|---------------------|----------------------|------------------------------|
| <i>Panel A: Reduced Form</i>         |                     |                      |                              |
| <b>Not Emerging or Demonstrating</b> |                     |                      |                              |
| Top HCV                              | 0.148<br>(0.355)    | −0.027***<br>(0.009) | −0.029***<br>(0.009)         |
| <b>Emerging</b>                      |                     |                      |                              |
| Top HCV                              | 0.505<br>(0.500)    | 0.033**<br>(0.014)   | 0.036**<br>(0.014)           |
| <b>Demonstrating</b>                 |                     |                      |                              |
| Top HCV                              | 0.357<br>(0.479)    | 0.003<br>(0.013)     | 0.001<br>(0.014)             |
| <i>Panel B: IV (LATE)</i>            |                     |                      |                              |
| <b>Not Emerging or Demonstrating</b> |                     |                      |                              |
| HCV                                  | 0.148<br>(0.355)    | −0.099***<br>(0.032) | −0.109***<br>(0.033)         |
| <b>Emerging</b>                      |                     |                      |                              |
| HCV                                  | 0.505<br>(0.500)    | 0.103**<br>(0.048)   | 0.107**<br>(0.049)           |
| <b>Demonstrating</b>                 |                     |                      |                              |
| HCV                                  | 0.357<br>(0.479)    | 0.024<br>(0.046)     | 0.024<br>(0.047)             |
| Controls                             | N                   | Y                    | Y                            |
| N                                    | 9,697               | 11,131               | 9,755                        |

*Notes:* This table repeats the analysis in Table 6, but drops children in families with voucher lottery positions that are “possibly treated” (i.e. those that could receive voucher offers in the years following our outcome window period). This table reports the effects of waitlist position (reduced form) and housing assistance (IV) on observation-based development benchmarks. Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.12: Robustness: Testing for Anticipation Effects on Early Literacy Skills and Kindergarten Readiness**

|                               | Control Mean<br>(1) | Placebo Sample<br>(2) |
|-------------------------------|---------------------|-----------------------|
| <i>Reduced Form</i>           |                     |                       |
| <b>Kindergarten Readiness</b> |                     |                       |
| Near Top HCV                  | 0.608<br>(0.488)    | 0.019<br>(0.014)      |
| Near Top PH                   | 0.612<br>(0.487)    | −0.004<br>(0.014)     |
| <b>Std. Reading Score</b>     |                     |                       |
| Near Top HCV                  | −0.005<br>(1.004)   | 0.008<br>(0.031)      |
| Near Top PH                   | 0.004<br>(0.999)    | 0.005<br>(0.029)      |
| Controls                      | N                   | Y                     |
| N                             | 9,456               | 7,499                 |

*Notes:* This table repeats the analysis in Table 5, but excludes households at the top of the voucher and public housing waiting lists, and instead looks at the effects of being “near the top” of the the voucher list and the public housing list respectively. Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.13: Multiple Hypothesis Testing Corrected P-Values for Subgroups**

|                         | Conventional<br>P-Value<br>(1) | Familywise<br>Error Rate<br>(2) | Bonferroni<br>(3) | Holm<br>(4) |
|-------------------------|--------------------------------|---------------------------------|-------------------|-------------|
| Boy                     | 0.0007                         | 0.0043                          | 0.0133            | 0.0127      |
| Girl                    | 0.4333                         | 0.8677                          | 1.0000            | 1.0000      |
| One Adult               | 0.0050                         | 0.0493                          | 0.1000            | 0.0850      |
| Multi-Adult             | 0.6167                         | 0.9350                          | 1.0000            | 1.0000      |
| One Child               | 0.2557                         | 0.8100                          | 1.0000            | 1.0000      |
| Multi-Child             | 0.0017                         | 0.0163                          | 0.0333            | 0.0300      |
| Hispanic Head           | 0.9590                         | 0.9590                          | 1.0000            | 0.9590      |
| Black Head              | 0.0110                         | 0.1073                          | 0.2200            | 0.1760      |
| Low Poverty             | 0.0793                         | 0.5093                          | 1.0000            | 1.0000      |
| High Poverty            | 0.1513                         | 0.6350                          | 1.0000            | 1.0000      |
| Any Childcare Close     | 0.0903                         | 0.5180                          | 1.0000            | 1.0000      |
| Not Any Childcare Close | 0.0950                         | 0.5330                          | 1.0000            | 1.0000      |
| Close to Gold-Seal      | 0.1493                         | 0.6517                          | 1.0000            | 1.0000      |
| Not Close to Gold-Seal  | 0.0870                         | 0.5187                          | 1.0000            | 1.0000      |
| Close to Head-Start     | 0.2880                         | 0.8160                          | 1.0000            | 1.0000      |
| Not Close to Head-Start | 0.0570                         | 0.4133                          | 1.0000            | 0.8550      |
| Low Housing Spending    | 0.0003                         | 0.0003                          | 0.0067            | 0.0067      |
| High Housing Spending   | 0.8250                         | 0.9630                          | 1.0000            | 1.0000      |
| Low Income              | 0.0973                         | 0.5097                          | 1.0000            | 0.9733      |
| High Income             | 0.3007                         | 0.7757                          | 1.0000            | 1.0000      |

*Notes:* This table reports conventional p-values, Family-Wise Error Rates, Bonferroni-corrected p-values, and Holm p-values for the effects of vouchers on standardized reading success across tested subgroups.

**Appendix Table A.14: Impacts on Children’s Cognitive and Behavioral Health, Balanced Panel and High Predicted Coverage Robustness**

|                                         | Impulse<br>Control/<br>Disturbance<br>of Conduct | Oppositional<br>Defiance/<br>Emotional<br>Disturbance | Attention<br>Deficit<br>Hyper-<br>Active | Developmental<br>Delays in<br>Reading,<br>Math, or<br>Speech | Index              |
|-----------------------------------------|--------------------------------------------------|-------------------------------------------------------|------------------------------------------|--------------------------------------------------------------|--------------------|
|                                         | (1)                                              | (2)                                                   | (3)                                      | (4)                                                          | (5)                |
| <b>Panel A: Balanced Panel Coverage</b> |                                                  |                                                       |                                          |                                                              |                    |
| <i>Reduced Form</i>                     |                                                  |                                                       |                                          |                                                              |                    |
| Top HCV                                 | −0.013***<br>(0.005)                             | −0.002<br>(0.003)                                     | −0.005<br>(0.007)                        | −0.017**<br>(0.008)                                          | 0.061**<br>(0.028) |
| Top PH                                  | 0.007*<br>(0.004)                                | −0.004<br>(0.003)                                     | −0.005<br>(0.006)                        | 0.008<br>(0.007)                                             | −0.008<br>(0.022)  |
| <i>IV (LATE)</i>                        |                                                  |                                                       |                                          |                                                              |                    |
| HCV                                     | −0.031**<br>(0.015)                              | −0.009<br>(0.010)                                     | −0.019<br>(0.023)                        | −0.042<br>(0.026)                                            | 0.175**<br>(0.085) |
| PH                                      | 0.040*<br>(0.023)                                | −0.020<br>(0.014)                                     | −0.028<br>(0.031)                        | 0.046<br>(0.037)                                             | −0.040<br>(0.121)  |
| Control Means                           | 0.039                                            | 0.017                                                 | 0.088                                    | 0.133                                                        | −0.084             |
| N                                       | 11,599                                           | 11,599                                                | 11,599                                   | 11,599                                                       | 11,599             |
| <b>Panel B: Predicted High Coverage</b> |                                                  |                                                       |                                          |                                                              |                    |
| <i>Reduced Form</i>                     |                                                  |                                                       |                                          |                                                              |                    |
| Top HCV                                 | −0.009**<br>(0.004)                              | −0.001<br>(0.003)                                     | −0.004<br>(0.006)                        | −0.012*<br>(0.007)                                           | 0.045*<br>(0.023)  |
| Top PH                                  | 0.006*<br>(0.003)                                | −0.003<br>(0.002)                                     | −0.003<br>(0.005)                        | 0.009*<br>(0.006)                                            | −0.009<br>(0.018)  |
| <i>IV (LATE)</i>                        |                                                  |                                                       |                                          |                                                              |                    |
| HCV                                     | −0.023*<br>(0.014)                               | −0.008<br>(0.010)                                     | −0.018<br>(0.021)                        | −0.031<br>(0.024)                                            | 0.142*<br>(0.079)  |
| PH                                      | 0.038*<br>(0.022)                                | −0.021<br>(0.014)                                     | −0.020<br>(0.031)                        | 0.062*<br>(0.037)                                            | −0.060<br>(0.120)  |
| Control Means                           | 0.033                                            | 0.015                                                 | 0.075                                    | 0.115                                                        | −0.024             |
| N                                       | 14,406                                           | 14,406                                                | 14,406                                   | 14,406                                                       | 14,406             |
| Controls                                | Y                                                | Y                                                     | Y                                        | Y                                                            | Y                  |

*Notes:* This table reports the effects of waitlist position (reduced-form) and housing assistance (IV) on cognitive and behavioral health diagnoses described in Section 3.3.2. Each column reports effects on a different outcome, based on three-digit ICD-9 codes. Column (5) reports effects on a standardized index of cognitive and behavioral health, with higher values indicating better health. The top of each panel reports the ITT estimates from estimating equation 1 and the bottom reports the LATE estimates. Panel A reports the results for a balanced panel of children with coverage in all years. Panel B reports the results for the sample of children with predicted high coverage. Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.15: Impacts on Children's Cognitive and Behavioral Health, No Prior Diagnosis Sample**

|                                    | Impulse Control/<br>Disturbance<br>of Conduct | Oppositional<br>Defiance/<br>Emotional<br>Disturbance | Attention<br>Deficit<br>Hyper-Active | Developmental<br>Delays in<br>Reading, Math<br>or Speech | Index              |
|------------------------------------|-----------------------------------------------|-------------------------------------------------------|--------------------------------------|----------------------------------------------------------|--------------------|
|                                    | (1)                                           | (2)                                                   | (3)                                  | (4)                                                      | (5)                |
| <b>Panel A: All</b>                |                                               |                                                       |                                      |                                                          |                    |
| <i>Reduced Form</i>                |                                               |                                                       |                                      |                                                          |                    |
| Top HCV                            | -0.009**<br>(0.004)                           | -0.002<br>(0.003)                                     | -0.003<br>(0.006)                    | -0.012*<br>(0.007)                                       | 0.043**<br>(0.022) |
| Top PH                             | 0.005<br>(0.003)                              | -0.003<br>(0.002)                                     | -0.002<br>(0.004)                    | 0.008<br>(0.005)                                         | -0.007<br>(0.017)  |
| <i>IV (LATE)</i>                   |                                               |                                                       |                                      |                                                          |                    |
| HCV                                | -0.025*<br>(0.013)                            | -0.008<br>(0.009)                                     | -0.014<br>(0.020)                    | -0.032<br>(0.024)                                        | 0.140*<br>(0.076)  |
| PH                                 | 0.030<br>(0.021)                              | -0.018<br>(0.013)                                     | -0.014<br>(0.029)                    | 0.049<br>(0.035)                                         | -0.045<br>(0.114)  |
| Control Means                      | 0.031                                         | 0.014                                                 | 0.071                                | 0.110                                                    | -0.005             |
| Controls                           | Y                                             | Y                                                     | Y                                    | Y                                                        | Y                  |
| N                                  | 15,638                                        | 15,638                                                | 15,638                               | 15,638                                                   | 15,638             |
| <b>Panel B: No Prior Diagnosis</b> |                                               |                                                       |                                      |                                                          |                    |
| <i>Reduced Form</i>                |                                               |                                                       |                                      |                                                          |                    |
| Top HCV                            | -0.009**<br>(0.004)                           | -0.001<br>(0.003)                                     | -0.005<br>(0.006)                    | -0.014**<br>(0.006)                                      | 0.043**<br>(0.021) |
| Top PH                             | 0.005<br>(0.003)                              | -0.003<br>(0.002)                                     | -0.002<br>(0.004)                    | 0.009*<br>(0.005)                                        | -0.014<br>(0.016)  |
| <i>IV (LATE)</i>                   |                                               |                                                       |                                      |                                                          |                    |
| HCV                                | -0.024*<br>(0.013)                            | -0.008<br>(0.009)                                     | -0.020<br>(0.020)                    | -0.039*<br>(0.022)                                       | 0.130*<br>(0.072)  |
| PH                                 | 0.032<br>(0.021)                              | -0.018<br>(0.013)                                     | -0.012<br>(0.029)                    | 0.058*<br>(0.034)                                        | -0.089<br>(0.109)  |
| Control Means                      | 0.031                                         | 0.014                                                 | 0.071                                | 0.110                                                    | -0.005             |
| Controls                           | Y                                             | Y                                                     | Y                                    | Y                                                        | Y                  |
| N                                  | 15,570                                        | 15,628                                                | 15,557                               | 14,731                                                   | 14,655             |

*Notes:* This table reports the effects of waitlist position (reduced form) and housing assistance (IV) on cognitive and behavioral health diagnoses described in Section 3.3.2. Each column reports effects on a different outcome, based on three-digit ICD-9 codes. Panel A repeats the results from Panel A of table 7. Panel B reports the results for the sample of children with no pre-lottery diagnosis for the listed condition. Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.16: Impacts on Children’s Medicaid Coverage**

|         | Covered in Any Post-Lottery Year |                     | Share of Post-Lottery Years Covered |                     |
|---------|----------------------------------|---------------------|-------------------------------------|---------------------|
|         | Control Mean<br>(1)              | Reduced Form<br>(2) | Control Mean<br>(3)                 | Reduced Form<br>(4) |
| Top HCV | 0.936<br>(0.245)                 | −0.005<br>(0.005)   | 0.860<br>(0.289)                    | −0.010<br>(0.007)   |
| Top PH  | 0.936<br>(0.245)                 | −0.004<br>(0.004)   | 0.859<br>(0.290)                    | −0.002<br>(0.005)   |

*Notes:* This table reports the effects of waitlist position (reduced form) on Medicaid coverage in the post lottery period. In Columns (1) and (2), the outcome is an indicator for being covered by Medicaid at any point in the post-lottery outcome window. In Columns (3) and (4), the outcome is the share of post-lottery years that the child is covered by Medicaid. Columns (1) and (3) report the mean among those randomized to the bottom of the waiting list (“Control Mean”). Columns (2) and (4) report the ITT estimates from estimating equation 1. Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

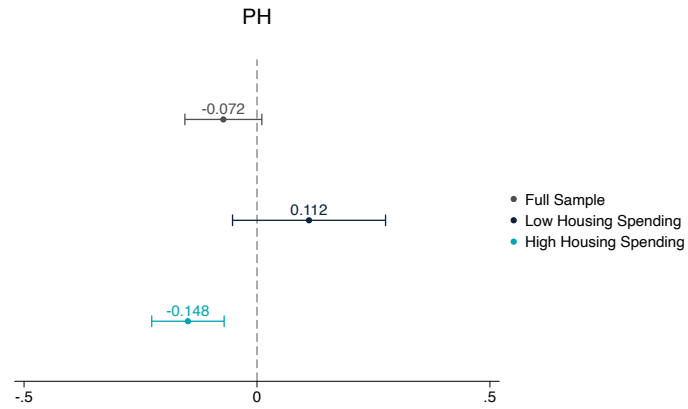
**Appendix Table A.17: Impacts on Doctor's Visits**

|         | Number of Doctor's Office Visits |                     | Any Doctor's Office Visits |                     |
|---------|----------------------------------|---------------------|----------------------------|---------------------|
|         | Control Mean<br>(1)              | Reduced Form<br>(2) | Control Mean<br>(3)        | Reduced Form<br>(4) |
| Top HCV | 16.177<br>(20.352)               | 0.121<br>(0.489)    | 0.595<br>(0.491)           | 0.004<br>(0.012)    |
| Top PH  | 16.010<br>(20.273)               | 0.408<br>(0.356)    | 0.594<br>(0.491)           | 0.003<br>(0.009)    |

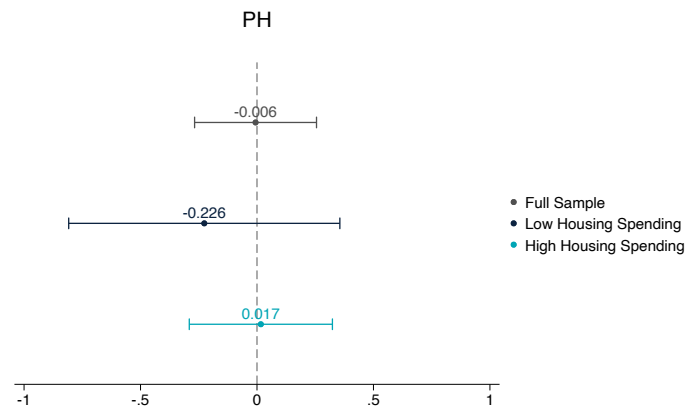
*Notes:* This table reports the effects of waitlist position (reduced form) on physician visits in the post lottery period. In Columns (1) and (2), the outcome is the number of visits to a physician's office in the post-lottery outcome window. In Columns (3) and (4), the outcome is any indicator for any primary care visit in the post-lottery years. Columns (1) and (3) report the mean among those randomized to the bottom of the waiting list ("Control Mean"). Columns (2) and (4) report the ITT estimates from estimating equation 1. Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

## Appendix Figure A.3: Heterogeneity in Public Housing Impacts on Housing Consumption Impacts and Child Development

### (a) Public Housing Impacts on Housing Consumption



### (b) Public Housing Impacts on Child Development



*Notes:* This figure plots the IV estimates for the effects of housing vouchers (“HCV”) and public housing (“PH”) on children’s standardized early literacy skills for households who spend relatively less on housing at baseline (hedonic rent less than FMR), and those that spend more relatively more on housing at baseline (hedonic rent greater than FMR). Whiskers are 95% confidence intervals

**Appendix Table A.18: Correlation of Sub-groups for Heterogeneity**

|                   | 1 Adult | 2+ Kids | Black | Hispanic | High Pov. | Any   | Gold-Seal | Head-Start | Low Housing Spend | Low Inc. |
|-------------------|---------|---------|-------|----------|-----------|-------|-----------|------------|-------------------|----------|
| 1 Adult           | 1       | -.042   | .228  | -.219    | .022      | -.012 | -.016     | .005       | -.101             | -.095    |
| 2+ Kids           | -.042   | 1       | -.047 | .048     | .04       | .008  | .029      | 0          | .251              | .065     |
| Black             | .228    | -.047   | 1     | -.864    | .093      | -.064 | -.097     | -.042      | .012              | -.053    |
| Hispanic          | -.219   | .048    | -.864 | 1        | -.095     | .072  | .088      | .039       | -.007             | .039     |
| High Pov.         | .022    | .04     | .093  | -.095    | 1         | .101  | .15       | .231       | .242              | .025     |
| Any               | -.012   | .008    | -.064 | .072     | .101      | 1     | .626      | .504       | .025              | .01      |
| Gold-Seal         | -.016   | .029    | -.097 | .088     | .15       | .626  | 1         | .588       | .026              | -.008    |
| Head-Start        | .005    | 0       | -.042 | .039     | .231      | .504  | .588      | 1          | .038              | .011     |
| Low Housing Spend | -.101   | .251    | .012  | -.007    | .242      | .025  | .026      | .038       | 1                 | .088     |
| Low Inc.          | -.095   | .065    | -.053 | .039     | .025      | .01   | -.008     | .011       | .088              | 1        |

*Notes:* This table is a correlation matrix of subgroups we consider in our heterogeneity analysis. Importantly, none of the major sub-groups are particularly correlated, suggesting that the heterogeneity in treatment effects that we find are more likely to reflect differences among the group in-question and not a mix of different groups.

**Appendix Table A.19: Comparing Characteristics of “High-Quality” and “Low-Quality” Public Housing**

|                   | Hedonic<br>Quality<br>(Rent \$)<br>(1) | Poverty<br>Rate<br>(2) | Crime<br>Rate<br>(3) | School<br>Test Scores<br>(4) | Inc Rank<br>(5) |
|-------------------|----------------------------------------|------------------------|----------------------|------------------------------|-----------------|
| Higher Quality PH | 1278                                   | 0.291                  | 17.2                 | -0.881                       | 0.334           |
| Lower Quality PH  | 1072                                   | 0.443                  | 31.5                 | -0.885                       | 0.295           |

*Notes:* This table reports the mean characteristics of Family public housing developments categorized as “High Quality” and “Low Quality”. Column (1) reports the mean hedonic rent estimate using the procedure outlined in section 3.5. Column (2) reports the mean Census tract poverty rate. Column (3) reports the violent crime rate per 10,000 as described in section 3.6. Column (4) characterizes the average standardized proficiency score of the neighborhood school. Column (5) reports the mean predicted income rank in adulthood for a child born to parent’s at the 25th percentile from the Opportunity Atlas [Chetty et al. \(2018\)](#).

**Appendix Table A.20: First stage for “High-Quality” and “Low-Quality” Public Housing**

|                                           | Lower Quality<br>PH<br>(1) | Higher Quality<br>PH<br>(2) | HCV<br>(3)          |
|-------------------------------------------|----------------------------|-----------------------------|---------------------|
| South Zone Wait Time x (Zone = South)     | −0.001<br>(0.001)          | −0.020***<br>(0.002)        | 0.002*<br>(0.001)   |
| Central Zone Wait Time x (Zone = Central) | −0.005***<br>(0.002)       | −0.009***<br>(0.002)        | −0.001<br>(0.002)   |
| North Zone Wait Time x (Zone = North)     | −0.019***<br>(0.001)       | −0.007***<br>(0.001)        | 0.000<br>(0.001)    |
| Top HCV                                   | −0.021***<br>(0.006)       | −0.010<br>(0.006)           | 0.294***<br>(0.012) |
| AP Partial F Statistic                    | 154                        | 47                          | 302                 |
| SW F Statistic                            | 70                         | 44                          | 282                 |
| N                                         | 15,639                     | 15,639                      | 15,639              |

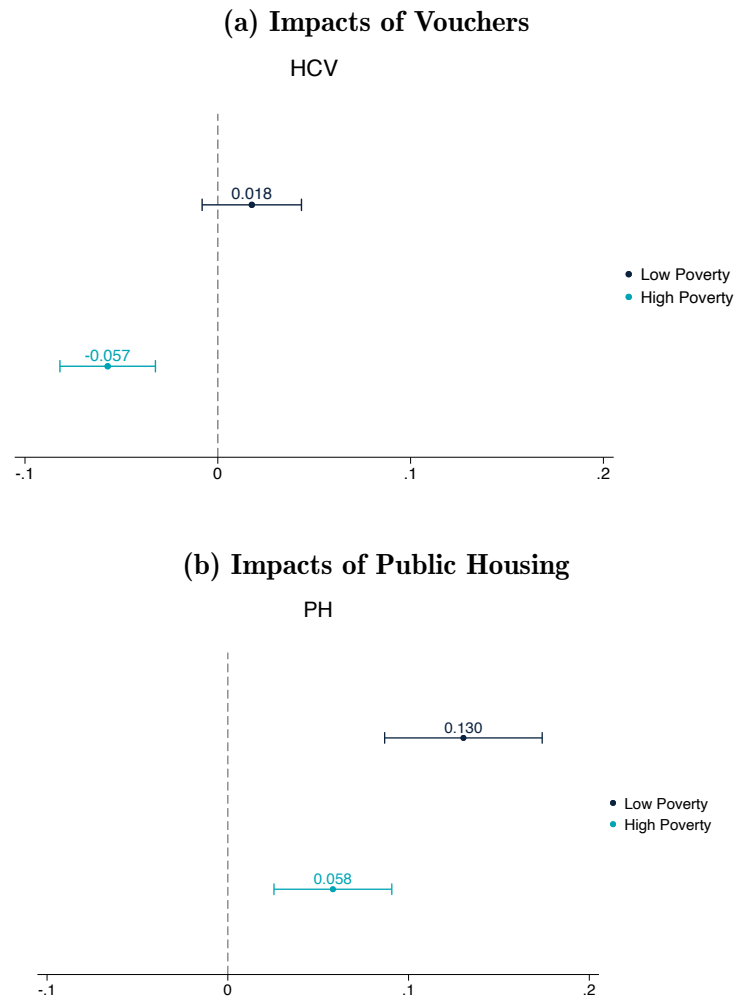
*Notes:* This table reports the results of an example first stage for the analysis described in Section 6.2. Estimated zone specific waiting times are measured in years. Column (1) reports the effects of the instruments on admission to “Lower-Quality” public housing. Column (2) reports the effects of the instruments on admission to “Higher-Quality” public housing. Column (3) reports the effects of the instruments on lease-up with a housing vouchers. The bottom rows of the table report the Angrist-Pischke Partial F-statistic (“AP Partial F Statistic”), and the Sanderson-Windmeijer F-statistic (“SW F Statistic”). Robust standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

**Appendix Table A.21: The Effects of Public Housing by Public Housing Development Quality (Alternative Specification with Many Instruments)**

|                             | Housing<br>Quality       | Neighborhood<br>Quality  | Early<br>Literacy        | Behavioral<br>Health      |                      |
|-----------------------------|--------------------------|--------------------------|--------------------------|---------------------------|----------------------|
|                             | $\Delta \ln(\hat{R}ent)$ | $\Delta$ Poverty<br>Rate | Std.<br>Reading<br>Score | Kindergarten<br>Readiness | Development<br>Index |
|                             | (1)                      | (2)                      | (3)                      | (4)                       | (5)                  |
| Lower Quality PH            | -0.234***<br>(0.074)     | 0.212***<br>(0.025)      | -0.156<br>(0.217)        | -0.144<br>(0.105)         | -0.582***<br>(0.193) |
| Higher Quality PH           | -0.008<br>(0.085)        | -0.037<br>(0.029)        | 0.312<br>(0.301)         | 0.167<br>(0.137)          | 0.637**<br>(0.292)   |
| P-value: Lower=Higher       | 0.096                    | 0.000                    | 0.297                    | 0.136                     | 0.005                |
| HCV                         | 0.081**<br>(0.036)       | -0.026**<br>(0.011)      | 0.210**<br>(0.099)       | 0.123***<br>(0.048)       | 0.127*<br>(0.077)    |
| P-value: Lower=HCV          | 0.000                    | 0.000                    | 0.109                    | 0.015                     | 0.000                |
| P-value: Higher=HCV         | 0.321                    | 0.703                    | 0.742                    | 0.761                     | 0.092                |
| P-value: Overidentification | 0.358                    | 0.103                    | 0.720                    | 0.804                     | 0.293                |
| N                           | 7,370                    | 10,921                   | 10,863                   | 10,863                    | 15,638               |

*Notes:* This table reports IV estimates of the effects of moving into higher quality public housing (“Higher Quality PH”) and the effects of moving into lower quality public housing (“Lower Quality PH”), along with the effects of leasing-up with a housing voucher (“HCV”) using an alternative specification found in equation 7. P-value: Higher=Lower reports a test of equality of coefficients in the effects of Higher and Lower Quality Public Housing. We also report the resulting p-value from an overidentification test (Hansen’s J statistic). Column 1 reports impacts on changes in hedonic quality, described in 3.5. Column 2 reports effects on changes in Census tract poverty. Column 3 shows effects on standardized reading success scores, and column 4 shows impacts on kindergarten readiness. Column 7 reports impacts on the cognitive and behavioral health index described in 3.3.2. standard errors are reported in parentheses, clustered at the household-level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

## Appendix Figure A.4: Impacts on Neighborhood Poverty by Baseline Neighborhood Poverty



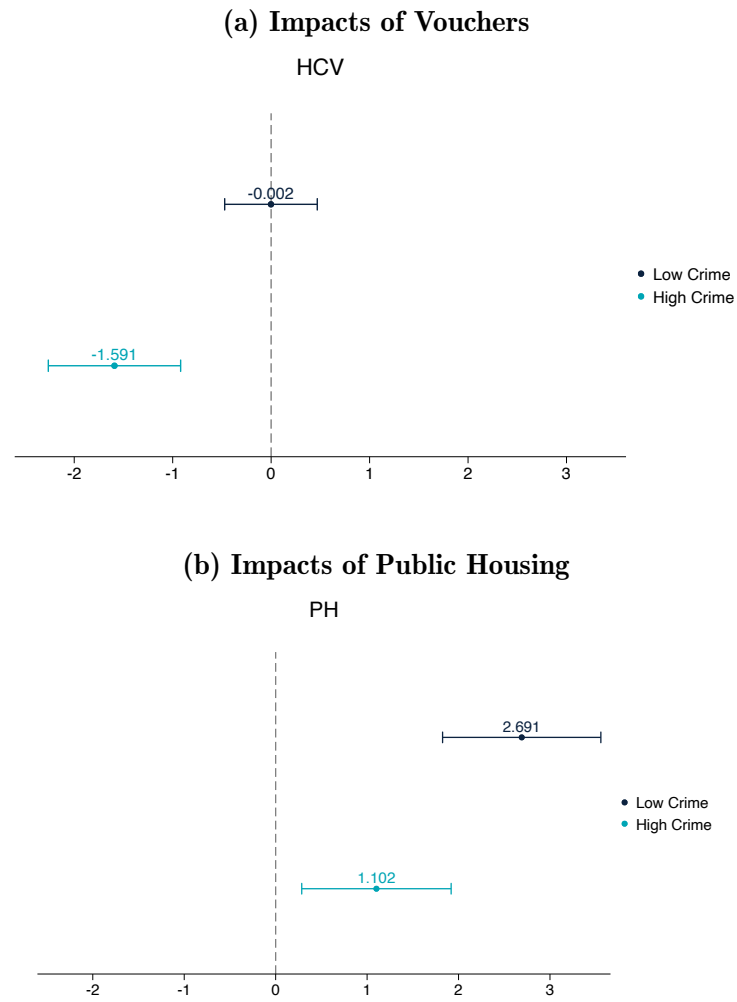
*Notes:* This figure plots the IV estimates for the effects of housing vouchers (“HCV”) and public housing (“PH”) on the change in neighborhood poverty rates by baseline poverty exposure. Whiskers are 95% confidence intervals.

**Appendix Table A.22: Effects of Vouchers on Early Literacy Skills and Kindergarten Readiness, Re-weighted to match the Race/Ethnic Composition of [Jacob et al. \(2015\)](#)**

|                               | Control Mean<br>(1) | Full Sample<br>(2)  | No Assistance<br>Alternative<br>(3) |
|-------------------------------|---------------------|---------------------|-------------------------------------|
| <i>Panel A: Reduced Form</i>  |                     |                     |                                     |
| <b>Kindergarten Readiness</b> |                     |                     |                                     |
| Top HCV                       | 0.608<br>(0.488)    | 0.042***<br>(0.016) | 0.044**<br>(0.021)                  |
| Top PH                        | 0.612<br>(0.487)    | 0.002<br>(0.012)    | 0.006<br>(0.013)                    |
| <b>Std. Reading Score</b>     |                     |                     |                                     |
| Top HCV                       | -0.005<br>(1.004)   | 0.082**<br>(0.033)  | 0.112***<br>(0.042)                 |
| Top PH                        | 0.004<br>(0.999)    | -0.001<br>(0.025)   | 0.012<br>(0.027)                    |
| <i>Panel B: IV (LATE)</i>     |                     |                     |                                     |
| <b>Kindergarten Readiness</b> |                     |                     |                                     |
| HCV                           | 0.608<br>(0.488)    | 0.134**<br>(0.052)  | 0.126**<br>(0.062)                  |
| PH                            | 0.612<br>(0.487)    | 0.015<br>(0.060)    | 0.029<br>(0.061)                    |
| <b>Std. Reading Score</b>     |                     |                     |                                     |
| HCV                           | -0.005<br>(1.004)   | 0.253**<br>(0.107)  | 0.321***<br>(0.124)                 |
| PH                            | 0.004<br>(0.999)    | 0.003<br>(0.124)    | 0.055<br>(0.124)                    |
| Controls                      | N                   | Y                   | Y                                   |
| JKL Weights                   | N                   | Y                   | Y                                   |
| N                             | 9,456               | 10,814              | 9,407                               |

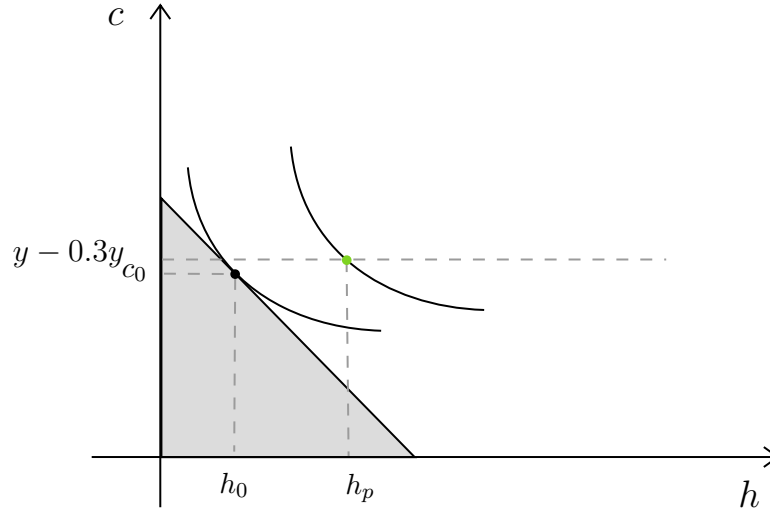
*Notes:* This table repeats the analysis in Table 5 for the full sample, but re-weights the sample to match the racial and ethnic composition of the lottery study in [Jacob et al. \(2015\)](#). Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

## Appendix Figure A.5: Impacts on Neighborhood Crime by Baseline Neighborhood Crime

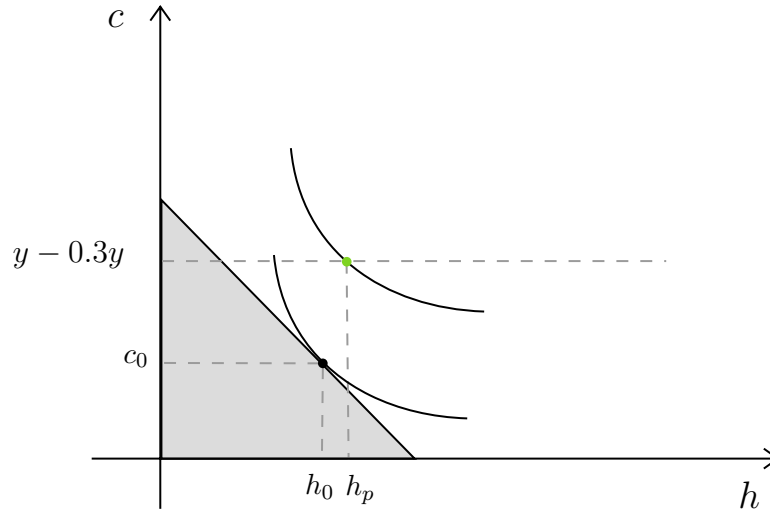


*Notes:* This figure plots the IV estimates for the effects of housing vouchers (“HCV”) and public housing (“PH”) on the change in neighborhood violent crime exposure by baseline neighborhood violent crime. Whiskers are 95% confidence intervals

**Appendix Figure A.6: Theoretical Impacts of Public Housing on Consumption**



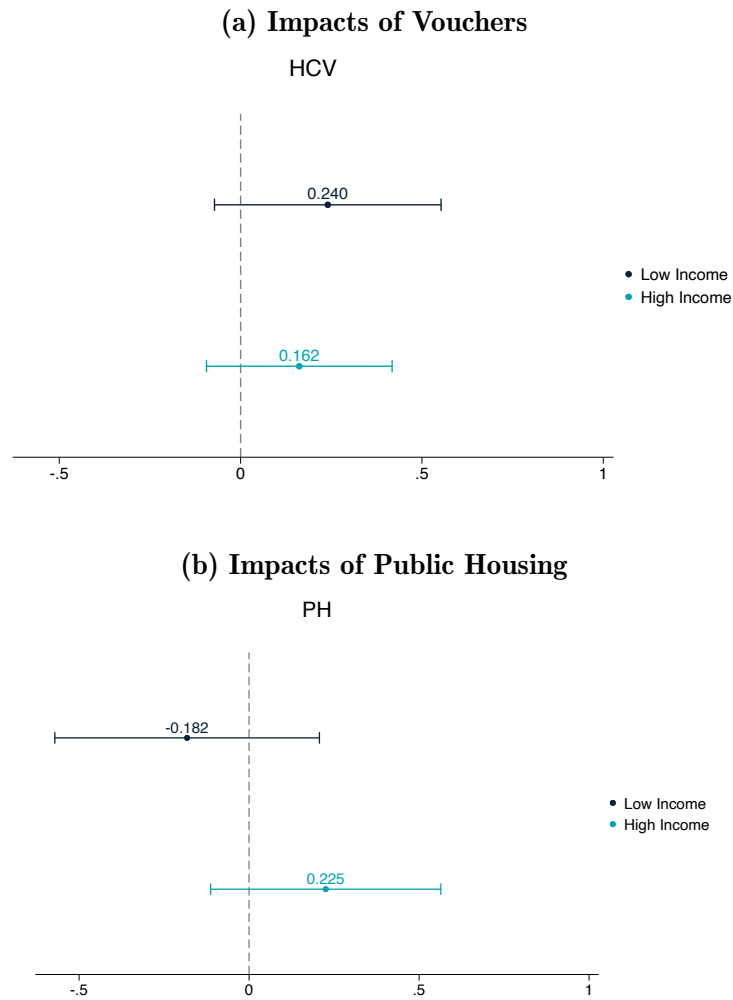
**(a) Low Baseline Housing Consumption**



**(b) High Baseline Housing Consumption**

*Notes:* This figure characterizes the theoretical consumption trade-off facing a household with and without public housing. The grey shaded area is the budget region. The quality of public housing ( $h_p$ ) is placed to be similar to the voucher maximum quality level in 6. Empirically, we find that  $h_p$  is probably quite a bit lower than what we indicate here. The top panel illustrates the changes in housing and non-housing consumption for a household with low spending on housing before public housing admission. The bottom illustrates the changes in non-housing spending for a household with high initial spending on housing.

**Appendix Figure A.7: Impacts on Early Literacy by Estimated Baseline Income**



*Notes:* This figure plots the IV estimates for the effects of housing vouchers (“HCV”) and public housing (“PH”) on the standardized measure of reading success separately by estimated baseline income. Whiskers are 95% confidence intervals.

## A Address History and Housing Data Details

This appendix describes how we construct address histories for our lottery sample. As we describe in Section 3.1, the waiting list application data provides address information for all households in the lottery at the time of waiting list opening in July 2008. This means we have complete baseline address information for the entire sample from a common source. To track follow-up addresses, we rely on two sources of information. The first is Florida voter registration records for 2012 and 2015. We link these to the household heads of our waiting list records using name and date of birth. In our sample of families with young children, we link 60 percent of household heads to a voter registration record. Our second source of address information is Infutor data, which we also link to the household heads in our sample using name and date of birth. Similarly, we link 68 percent of household heads to a post-lottery Infutor record. Across both sources, we can link nearly 80 percent of our household heads to a post-lottery address.

To determine which source of address information better captures the locational outcomes of lottery families, we evaluate whether the voter records or Infutor records do a better job of correctly assigning the address *for assisted households*, which we can validate using the housing assistance records. The assistance records offer a ground truth against which to compare sources. We find that the voter registration records are 1.5 times more likely to place the household at the correct address than the Infutor records. Therefore, we first assign a household the parcel and Census tract on the voter registration file, if available, and otherwise use the parcel data and Census tracts of the Infutor addresses.

Another complicating factor is uncertainty in move-in dates. While the voter records have known calendar vintages (in our case 2012 and 2015), they do not have precise move-in dates. Infutor has effective dates, which are supposed to reflect the date that of move-in, but are known to also be influenced by changes in their underlying data suppliers. Since moves typically are recorded with some lag, we assign the addresses from the latest observed record within the 2009-2015 calendar window.

A final concern is that appearance in any of these outcome data sources could, in principle, be affected by receiving housing assistance, which could bias our estimate for housing or neighborhood outcomes. To evaluate whether this is a significant threat, we regress indicators for appearance in the address data sources on indicators for being at the top of the voucher and top of the public housing waiting lists. The results appear in Appendix Table A.23. The top panel reports the results for appearance in the voter records, the middle panel reports the results for appearance in the Infutor data, and the bottom panel reports the results for appearance in either. Being at the top of the voucher or public housing lists appears to have

no meaningful effect on the likelihood of appearing in either move source, individually, or collectively across sources. In all cases, the reduced form coefficients are economically small (a percentage point or less) and none of the estimates are statistically significant. This gives us confidence that differential appearance in the move data by treatment status is not biasing our results.

**Appendix Table A.23: Waiting List Positions and Address Data Availability**

|                                         | Control Mean<br>(1) | Difference<br>(2) |
|-----------------------------------------|---------------------|-------------------|
| <b>Has Voter Data</b>                   |                     |                   |
| Top HCV                                 | 0.596<br>(0.491)    | -0.005<br>(0.014) |
| Top PH                                  | 0.596<br>(0.491)    | -0.002<br>(0.010) |
| <b>Has Infutor Data</b>                 |                     |                   |
| Top HCV                                 | 0.683<br>(0.465)    | -0.004<br>(0.014) |
| Top PH                                  | 0.685<br>(0.465)    | -0.002<br>(0.010) |
| <b>Has Either Voter or Infutor Data</b> |                     |                   |
| Top HCV                                 | 0.791<br>(0.407)    | -0.013<br>(0.012) |
| Top PH                                  | 0.789<br>(0.408)    | -0.002<br>(0.009) |
| Controls                                | N                   | N                 |
| N                                       | 14911               | 14911             |

*Notes* This table reports the relationship between randomized initial waiting list positions and whether a households has non-missing voter-record data, non-missing Infutor Data, or non-missing values for the union of the two sources. The top panel assesses balance in linking to a voter record, the middle panel to an Infutor record, and the bottom panel is for the union of the two measures. Column (1) reports the control mean and Column (2) reports the results from regressing an indicator for non-missing data on an indicator for being at the top of the voucher list (“Top HCV”) and an indicator for being at the top of the public housing list (“Top PH”). Robust standard errors are reported in parentheses, clustered at the household level. Statistical significance is denoted by: \*  $p \leq 0.1$ ; \*\*  $p \leq 0.05$ ; \*\*\*  $p \leq 0.01$ .

## B Hedonic Rent Estimation and Data Details

As described in section 3.5, we use hedonic rent estimates to measure housing quality changes for waiting list families. To construct these estimates, we begin with two sources of rent data. The first is contract rents from historical housing voucher recipients, covering the period 2001-2008. The second is rental listing data from Altos Research for the period 2012-2018. We start by deflating both series by the Miami metro CPI to \$2012, and then multiply the Altos rents by the ratio of contract rent to asking rent in the ACS to adjust for the differences in rent measurement. We geocode both data sets to the parcel level in Miami-Dade County. We then calculate mean rents at the parcel-by-bedroom size level. For single-family dwellings, this simply collapses to the parcel level. This provides parcel-by-bedroom rent cells that form the left-hand side of our hedonic rent estimates.

We assemble parcel-level data on housing attributes from the Miami-Dade County Property Appraiser. Using the characteristics below in the Appraisers's data, we construct the following variables for use in our hedonic model (Rosen, 1974; Black, 1999) :

| Characteristic               | Variable                                                                                                        | Source   |
|------------------------------|-----------------------------------------------------------------------------------------------------------------|----------|
| Square Feet                  | Square feet per unit                                                                                            | Assessor |
| Effective Year Built         | Building Age, Age cubed                                                                                         | Assessor |
| Number of Bedrooms           | Bedrooms per unit                                                                                               | Assessor |
| Number of Bathrooms          | Bathrooms per unit                                                                                              | Assessor |
| Assessed Value               | Ln(assessed value/unit)                                                                                         | Assessor |
| Building Type                | Indicators for each type: Single-family detached, single-family attached, garden apartments, midrise, high-rise | Assessor |
| Assessor Quality Scale (1-5) | Indicators for quality level                                                                                    | Assessor |
| Neighborhood rent            | Block-group median rent                                                                                         | Altos    |
| Neighborhood subdivision     | Subdivision fixed effects                                                                                       | Assessor |

We estimate the following regression, where  $j$  indexes parcels, and  $n$  indexes neighborhood subdivisions:

$$\ln(\text{Rent}_{j,n}) = X_j\beta + \mu_n + \varepsilon_j \quad (\text{A.1})$$

Where  $\text{Rent}_{j,n}$  is the mean monthly rent for the parcel, and  $X_j$  is a vector of the characteristics listed in B, with  $\mu_n$  capturing the neighborhood subdivision fixed effects. This regression yields an  $R^2$  of 0.70, suggesting that these we can explain about 70 percent of the cross-sectional variation in rents. We then construct the fitted values from the regression, which becomes our measure of hedonic housing quality ( $\hat{\text{Rent}}_j$ ). When comparing estimated hedonic quality to FMRs, we exponentiate the estimated rent level.

## C Kindergarten Readiness Data Details

The Florida Department of Education linked children listed on the the housing authority applications to all school readiness assessments for the years 2009-2014 using probabilistic matching on first name, last name, and date of birth. Among children listed on housing applications living in Miami-Dade County who were younger than four years old at the time of random assignment, we estimate that approximately 87% match to an assessment. At the time, the Florida Kindergarten Readiness Screener was comprised of two parts: the Florida Assessment for Instruction in Reading (FAIR), measuring early literacy development, and the Early Childhood Observation System (ECHOS), a classroom-embedded, observational assessment of school readiness across a broad range of cognitive and noncognitive skills.

### C.1 Early Literacy Assessment

The FAIR screening is administered by schools. The “Broad Screen” evaluates progress on two dimensions of early literacy, letter naming and phonemic awareness. Performance on the screener is then used to calculate a “Probability of Reading Success” score. This score ranges from 1 to 100, and a student is judged to be “Kindergarten Ready” if they score a 67 or above. The probability of reading success scores are calibrated to reflect the likelihood of the child scoring at or above proficiency on the reading section of the Stanford Early School Achievement Test if taken at the end of the year.

We use two outcome measures from the linked FAIR screening: first, an indicator for whether the child scored as being “Kindergarten Ready,” and second, we standardize the raw probability of reading success score to facilitate comparisons with other studies (“Standardized Score”).

### C.2 Observational Assessment

The ECHOS is also administered in schools within 30 days of kindergarten entry, but is an embedded assessment, intended to be administered during the course of natural classroom activities. The ECHOS evaluates progress on 19 benchmarks across seven domains: Language and Literacy, Mathematics, Social and Personal Skills, Science, Social Studies, Physical Development and Fitness, and Creative Arts. Teachers or assessment administrators use a matrix to code student progress along each item.

From the ECHOS data, we receive the overall readiness designation which falls into one of the three categories: “Demonstrating” – the child clearly demonstrated skills appropriate for beginning kindergarten; “Emerging/Progressing” – the child demonstrated some of the skills

**Appendix Figure A.8: FAIR Assessment: Probability of Reading Success Scorecard**

| AP1                           | Kindergarten - Assessment Period 1 - Probability of Reading Success |     |     |     |     |     |     |     |     |     |     |     |
|-------------------------------|---------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|                               | Phonemic Awareness - Total Correct                                  |     |     |     |     |     |     |     |     |     |     |     |
| Letter Naming - Total Correct | Total Correct                                                       | 0   | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  |
|                               | 0                                                                   | .09 | .11 | .14 | .17 | .20 | .24 | .29 | .34 | .39 | .45 | .51 |
|                               | 1                                                                   | .12 | .14 | .17 | .21 | .25 | .30 | .35 | .40 | .46 | .52 | .57 |
|                               | 2                                                                   | .15 | .18 | .22 | .26 | .31 | .36 | .41 | .47 | .53 | .59 | .64 |
|                               | 3                                                                   | .19 | .22 | .27 | .31 | .37 | .42 | .48 | .54 | .60 | .65 | .70 |
|                               | 4                                                                   | .23 | .28 | .32 | .38 | .43 | .49 | .55 | .61 | .66 | .71 | .76 |
|                               | 5                                                                   | .28 | .33 | .39 | .45 | .50 | .56 | .62 | .67 | .72 | .77 | .80 |
|                               | 6                                                                   | .34 | .40 | .46 | .51 | .57 | .63 | .68 | .73 | .77 | .81 | .84 |
|                               | 7                                                                   | .41 | .47 | .53 | .58 | .64 | .69 | .74 | .78 | .82 | .85 | .88 |
|                               | 8                                                                   | .48 | .54 | .59 | .65 | .70 | .75 | .79 | .83 | .86 | .88 | .90 |
|                               | 9                                                                   | .55 | .61 | .66 | .71 | .76 | .80 | .83 | .86 | .89 | .91 | .93 |
|                               | 10                                                                  | .62 | .67 | .72 | .76 | .80 | .84 | .87 | .89 | .91 | .93 | .94 |

appropriate; or “Not Yet Demonstrating” – the child did not demonstrate appropriate skill development during the screening. With these data, we examine effects of housing assistance on a child’s inclusion in each of the three distinct categories.

Appendix Figure A.9: Example of ECHOS Assessment Matrix

Class Record Form



TEACHER NAME: \_\_\_\_\_

| Domain/<br>Benchmark                                                                                  | Indicators                                                   | Student Name | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|
| 1. Concepts of Print<br>Knows how to use a book (1-A-1)                                               | A. Shows curiosity about all aspects of print                |              |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |
|                                                                                                       | B. Identifies front and back of book and where story begins  |              |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |
|                                                                                                       | C. Recognizes that the purpose of print is to tell the story |              |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |
| 2. Oral Language and Vocabulary<br>Shares information about events that happen outside school (1-E-1) | A. Shares events with class                                  |              |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |
|                                                                                                       | B. Describes an event in one or two simple sentences         |              |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |
|                                                                                                       | C. Elaborates on an event in great detail                    |              |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |

The following table presents the 19 ECHOS benchmarks, their indicators, and the assignment of children's ratings based on which indicators they demonstrate in classroom observation.

Appendix Table A.24: ECHOS Benchmarks, Indicators, and Ratings by Learning Domain

| Domain & Benchmark                                                                                          | Indicators                                                                                                                                                                  | Ratings                                                                               |
|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Concepts of Print</b><br><i>Knows how to use a book</i>                                                  | A. Shows curiosity about all aspects of print<br>B. Identifies front and back of book and where story begins<br>C. Recognizes that the purpose of print is to tell a story  | Emerging/Progressing: A or B<br>Demonstrating: B and C                                |
| <b>Oral Language and Vocabulary</b><br><i>Shares information about events that happen outside of school</i> | A. Shares events with class<br>B. Describes an event in one or two simple sentences<br>C. Elaborates on an event in great detail                                            | Emerging/Progressing: A or B<br>Emerging/Progressing: A or B<br>Demonstrating: B or C |
| <b>Comprehension</b><br><i>Retells a story or part of a story that has been read to the class</i>           | A. Listens attentively when the teacher reads books in class<br>B. Retells one part of the story accurately<br>C. Retells whole story or event experienced in class         | Emerging/Progressing: A or B<br>Demonstrating: C                                      |
| <b>Comprehension</b><br><i>Demonstrates understanding of story elements</i>                                 | A. Answers literal questions about story elements<br>B. Makes predictions based on illustrations or portions of stories<br>C. Answers inferential or cause-effect questions | Emerging/Progressing: A or B<br>Demonstrating: B or C                                 |
| <b>Writing</b><br><i>Distinguishes between “kid’s writing” and conventional writing</i>                     | A. Rereads own scribble writing<br>B. Recognizes difference between scribbles and conventional writing<br>C. Uses phonetic and conventional spelling                        | Emerging/Progressing: A or B<br>Demonstrating: B or C                                 |
| <b>Number Sense and Operations</b><br><i>Counts objects using one-to-one correspondence</i>                 | A. Counts five objects<br>B. Provides correct number of objects (up to 10) on request<br>C. Creates a collection of up to 20 items                                          | Emerging/Progressing: A or B<br>Demonstrating: B or C                                 |

| Domain & Benchmark                                                                                  | Indicators                                                                                                                      | Ratings                                                |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Geometry</b><br><i>Identifies two-dimensional shapes and their uses</i>                          | A. Recognizes and names shapes<br>B. Builds/draws triangles and rectangles<br><br>C. Describes shapes by counting sides         | Emerging/Progressing: A<br>Demonstrating: B or C       |
| <b>Algebraic Thinking</b><br><i>Recognizes, creates, and analyzes patterns</i>                      | A. Recognizes and creates repeating patterns<br>B. Copies patterns<br><br>C. Recognizes and iterates the unit/core of a pattern | Emerging/Progressing: A or B<br>Demonstrating: C       |
| <b>Data Analysis</b><br><i>Analyzes and represents data to answer questions</i>                     | A. Sorts objects and counts groups<br>B. Creates chart of play choices<br><br>C. Answers questions from charts/graphs           | Emerging/Progressing: A or B<br>Demonstrating: C       |
| <b>Responsible Decision Making</b><br><i>Uses classroom materials purposefully and respectfully</i> | A. Takes care of property<br>B. Asks permission<br><br>C. Uses/stores equipment appropriately                                   | Emerging/Progressing: A or B<br>Demonstrating: B and C |
| <b>Social Problem Solving</b><br><i>Talks to and plays cooperatively with others</i>                | A. Joins play activity<br>B. Asks others to join; converses about events<br><br>C. Shares, takes turns, and plans roles         | Emerging/Progressing: A or B<br>Demonstrating: C       |
| <b>Approaches to Learning</b><br><i>Shows curiosity about new topics and ideas</i>                  | A. Observes and comments<br>B. Asks questions<br><br>C. Discusses ideas with others                                             | Emerging/Progressing: A or B<br>Demonstrating: C       |
| <b>Scientific Inquiry</b>                                                                           | A. Asks questions                                                                                                               | Emerging/Progressing: A or B                           |

| Domain & Benchmark                                  | Indicators                                 | Ratings                      |
|-----------------------------------------------------|--------------------------------------------|------------------------------|
| <i>Uses inquiry, problem solving, and reasoning</i> | B. Observes and names properties           | Demonstrating: C             |
|                                                     | C. Conducts experiments during play        |                              |
| <b>Production, Distribution, Consumption</b>        | A. Identifies community helpers            | Emerging/Progressing: A or B |
| <i>Explores types of work and community roles</i>   | B. Role-plays jobs                         | Demonstrating: C             |
|                                                     | C. Draws or writes about service providers |                              |
| <b>Civic Ideals and Participation</b>               | A. Identifies rules                        | Emerging/Progressing: A or B |
| <i>Understands rules and authority figures</i>      | B. Shares chores                           | Demonstrating: B and C       |
|                                                     | C. Discusses rules and safety              |                              |
| <b>Fitness</b>                                      | A. Runs on playground                      | Emerging/Progressing: A or B |
| <i>Engages in large-muscle activity</i>             | B. Climbs with repetition                  | Demonstrating: C             |
|                                                     | C. Jumps, skips, throws                    |                              |
| <b>Fine Motor Skills</b>                            | A. Uses some computer keys                 | Emerging/Progressing: A or B |
| <i>Uses hands and fingers to perform tasks</i>      | B. Cuts with scissors                      | Demonstrating: B or C        |
|                                                     | C. Uses tools like stapler, tape, markers  |                              |
| <b>Dance</b>                                        | A. Claps/marches to beat                   | Emerging/Progressing: A or B |
| <i>Creates movement for music</i>                   | B. Moves to indicate tempo/dynamics        | Demonstrating: C             |
|                                                     | C. Moves purposefully to music             |                              |
| <b>Visual Arts</b>                                  | A. Mixes colors/media                      | Emerging/Progressing: A or B |
| <i>Applies techniques to create original art</i>    | B. Selects techniques for projects         | Demonstrating: B or C        |
|                                                     | C. Plans and creates original art          |                              |

## D LATEs under Alternative Treatment Options

We consider an environment where children can receive one of three possible treatment states  $D_k$  where  $k \in \{0, V, P\}$ , which are mutually exclusive and exhaustive

$$D = D_0 + D_V + D_P = 1 \quad (\text{A.2})$$

And outcomes depend on  $D_k$  as follows:

$$Y = D_0 Y_0 + D_V Y_V + D_P Y_P \quad (\text{A.3})$$

Where  $Y_V$  is the potential outcome  $Y$  for a child whose family receives a voucher,  $Y_P$  is potential outcome realization for a child whose family receives public housing, and  $Y_0$  is the child's outcome under no-treatment. Let  $Z_V$  denote a binary instrument for being at the top of the voucher waiting list. Consider the following system of questions:

$$Y = \beta_0 + \beta_V D_V + \varepsilon \quad (\text{A.4})$$

$$D_V = \gamma_0 + \gamma_V Z_V + \varepsilon \quad (\text{A.5})$$

In the presence of multiple treatments (or multiple counterfactual alternatives) the usual LATE assumptions of [Imbens and Angrist \(1994\)](#) are not sufficient to identify interpretable causal effects. With 3 treatment states, there are 9 possible combinations of potential treatment states:  $\{(0, 0), (0, V), (V, V), (V, 0), (P, V), (P, P), (V, P), (0, P), (P, 0)\}$ . As with the usual independence assumption and excludability of the instrument, the assistance never-takers  $(0, 0)$ , voucher always-takers  $(V, V)$ , and public housing always-takes  $(P, P)$  are permitted treatment choice patterns, but will cancel out in the numerator of the Wald estimator. The monotonicity assumption of IA eliminates voucher defiers  $\{(V, 0), (V, P)\}$ . This leaves four groups whose treatment status changes in response to the instrument:  $\{(0, V), (P, V), (P, 0), (0, P)\}$ , by extending the monotonicity assumption to  $D(1) \neq D(0) \Rightarrow D(1) = V$ , it leaves just two groups  $\{(0, V), (P, V)\}$ . [Kline and Walters \(2016\)](#) use this assumptions in the context of Headstart when comparing to other care alternatives. In our setting, this assumes that being randomly assigned to the top of the voucher waiting list only changes treatment status in the direction of receiving a voucher. This permits switching from no-treatment to vouchers or from public housing to vouchers, but rules out switches between alternatives  $\{(0, P), (P, 0)\}$ . That is, a higher voucher position should not cause someone to go from untreated to public housing, or from public housing to untreated.

Under the IA assumptions [Imbens and Angrist \(1994\)](#) and the extended monotonicity

assumption above, and ignoring other covariates, the IV estimate will yield the following (see Mountjoy (2022) for a full proof):

$$\begin{aligned} \beta_V = & \frac{[Y_V - Y_0|D(1) = V, D(0) = 0]E[D_0|Z_V = 1] - E[D_0|Z_V = 0]}{E[D_V|Z_V = 1] - E[D_V|Z_V = 0]} \dots \\ & + \frac{[Y_V - Y_P|D(1) = V, D(0) = P]E[D_P|Z_V = 1] - E[D_P|Z_V = 0]}{E[D_V|Z_V = 1] - E[D_V|Z_V = 0]} \end{aligned} \quad (\text{A.6})$$

This is a weighted average of two subLATEs: the effects of vouchers for those who would otherwise get no-treatment ( $LATE_{0 \rightarrow V}$ ) and the effect of vouchers who would otherwise get public housing ( $LATE_{P \rightarrow V}$ ), as follows:

$$= \omega E[Y_V - Y_0|D(1) = V, D(0) = 0] + (1 - \omega) E[Y_V - Y_P|D(1) = V, D(0) = P] \quad (\text{A.7})$$

where the weights are as follows:

$$\omega = -\frac{E[D_0|Z_V = 1] - E[D_0|Z_V = 0]}{E[D_V|Z_V = 1] - E[D_V|Z_V = 0]}, \quad (1 - \omega) = \frac{E[D_P|Z_V = 1] - E[D_P|Z_V = 0]}{E[D_V|Z_V = 1] - E[D_V|Z_V = 0]} \quad (\text{A.8})$$

The appendix table A.4 reports the estimated weights in our main specification in column 1, and the weights under the “No Assistance Alternative” specification in column 2.

## E Additional Heterogeneity

### E.0.1 Baseline Neighborhood Conditions

A large literature documents the effects of neighborhood conditions on the long-run outcomes of children (Chetty et al., 2016; Chetty and Hendren, 2018; Chyn, 2018). At the same time, the evidence of neighborhoods impacts on cognitive development measured earlier in life are more mixed (Kling et al., 2007; Jacob, 2004; Sanbonmatsu et al., 2011). To evaluate whether changes in neighborhood conditions are driving the effects we find, we examine heterogeneity in effects by baseline neighborhood characteristics. This exercise exploits the fact that the effects of receiving assistance will depend, in part, on the type of neighborhood the household is living in at baseline. As an example, Appendix Figure A.4 plots the effects of voucher on neighborhood poverty (as in section 5.1) separately for households living in Census tracts with poverty rates below 20% (“Low poverty”) and households living in Census tracts with poverty rates at or above 20% (“High poverty”). For families living in high poverty places at

baseline, receipt of a voucher leads to larger 6.6 percentage point reductions in neighborhood poverty than those already living in low poverty at baseline who experience no meaningful reduction in neighborhood poverty exposure. For public housing, families living in low poverty at baseline experience the larger *increases* in poverty than those already living in higher poverty neighborhoods before housing assistance.<sup>28</sup>

Figure A.2 summarizes heterogeneity in child development impacts by baseline neighborhood conditions: poverty and violent crime. Panel A reports the effects of housing vouchers. The point estimates for vouchers are 0.28 SD for children in families starting in low-poverty, and 0.17 SD for children in families starting in high-poverty. Taken at face-value the estimates are slightly *smaller* for children who get a larger *reduction* in neighborhood poverty exposure suggesting that the change in neighborhood poverty alone isn't responsible for the positive effects we find. However, we cannot statistically distinguish the estimates, so we caution over-interpreting this pattern. The pattern for heterogeneity by baseline neighborhood crime mirrors that of neighborhood poverty: the point estimates are similar across families in high- and low-crime neighborhoods at baseline with slightly larger estimates for those families who experience smaller reductions in crime exposure.

The effects of public housing by baseline neighborhood conditions appear in Figure A.2b. While the sign of the effects of public housing differ across low-poverty and high-poverty, and low- and high-crime neighborhoods, it is unclear if this because the treatment effects are different or an artifact of sampling variance. Similar to vouchers, the patterns of heterogeneity do not seem to imply that households that experience the largest changes to neighborhood environment have fundamentally different treatment effects.

### E.0.2 Baseline Early Care Options

One potential pathway through which housing assistance may affect child development could be through moves to neighborhoods with more early childhood care and education programs, and thus greater access to and more opportunity for investment in skill development outside the home. We compiled data from public databases on the location and availability of Head Start — the federally funded preschool program for children from low-income households — centers, providers offering the state's voluntary, public pre-kindergarten (VPK) program, child care and preschool settings accepting the financial assistance provided through the state's School Readiness program, and child care and preschool providers with the state's high-quality "Gold Seal" designation. Using these data, we explore variation in housing assistance impact for children in households with less exposure to ECE opportunities prior to

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<sup>28</sup>We find an analogous pattern when we investigate heterogeneity in impacts on neighborhood crime in Appendix Figure A.5.

receipt of housing assistance.

We focus on baseline proximity to Head Start as results disaggregated by proximity to the other types of early care and education options did not differ. We define proximity to Head Start based on any Head Start center within  $3/4$  of a mile of the household’s address at time of housing assistance application. Results in Figure A.2b are suggestive of more pronounced effects for children in families living farther from any Head Start center prior to housing assistance. In the top panel showing heterogeneity in the impact of housing vouchers, the point estimates for those close to Head Start at baseline and those who were not are both positive and statistically indistinguishable though the effect for those who were farther away from Head Start is 0.28 SD, nearly three times larger than that of those who were close to Head Start at baseline. The effects for public housing, in the bottom panel, are also statistically indistinguishable from one another, but the point estimate for those close to Head Start prior to public housing assistance is large and negative.

## F Medicaid Coverage Data Appendix

Using Medicaid claims data to infer changes in health status is challenged by two common issues. First, most treatments could, in principle, impact Medicaid coverage leading to possible differential attrition patterns. Second, diagnosis of particular conditions requires some use of care which can conflate health conditions with health care utilization. In this appendix, we explore these concerns in more detail.

First, we evaluate whether the treatments we study significantly impacts subsequent Medicaid coverage in our sample. Table A.16 reports the results of regressing an indicator for any post-lottery Medicaid coverage (“Any Medicaid Coverage”) on the voucher and public housing instruments in Column (2). We find no statistically significant effect of either instrument on coverage of Medicaid coverage in the post-lottery period. In Column (4) we repeat the analysis, but the outcome is the share of years covered. Again, we find no significant effects on post-lottery coverage.

While the point estimates from Table A.16 are relatively small, they do not generally allow us to rule-out modest effects on coverage. Therefore, we also repeat our analysis in a sample of children covered in every post-lottery year. Appendix table A.14 panel A, reports the results for this balanced panel sample. The results are generally stronger in the sample with balanced coverage, suggesting that our results that vouchers reduce cognitive and behavioral health challenges are not driven by voucher holders being less likely to be covered in the post-lottery period. Since post-lottery coverage could be endogenous, we also predict post-lottery coverage using baseline covariates (waiting list application characteristics as

well as prior coverage patterns), then re-estimate our results in the sample of high-predicted coverage. The results appear in panel B of table A.14. The point estimates are nearly identical to the main results from Table 7. This lends additional support to the notion that coverage differences by treatment status do not drive our results.

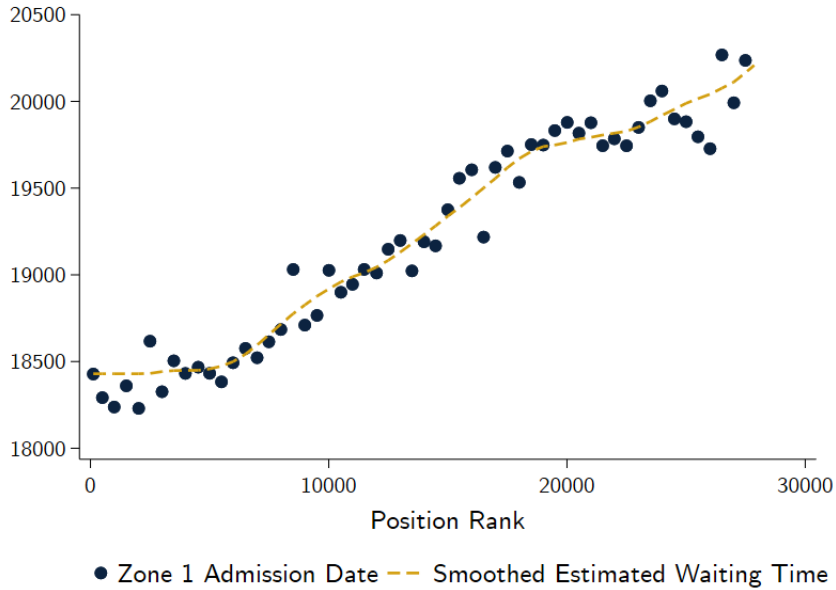
Determining whether differences in diagnosed conditions arise due to actual changes in health status or changes in healthcare utilization is a challenge. We explore this question in two ways. First, we directly estimate the effects on primary care provider visits as a proxy for health care utilization. The results appear in Appendix Table A.17. We find no evidence that voucher and public housing position increase the frequency of doctor's visits. We can rule-out increases over more than 1 visit over the entire post-lottery window. Second, since most children in the sample (93 percent) were covered by Medicaid at some point in the lead-up to the lottery, it is plausible that repeated diagnoses reflect increased use of health care services whereas first diagnosis for a given condition better reflect a change in health conditions. In panel B of Appendix Table A.15, we repeat our main analysis, but focus on children without a previous diagnosis for the listed conditions. The results are remarkably similar to those in panel A (which repeats the main analysis), which confirms that our results are not driven by repeated diagnoses, but rather by new instances.

## G Estimating Public Housing Waiting Time

For the analysis in section 6.2, we rely on zone-specific public housing waiting time estimates for constructing instruments for higher and lower quality public housing. In this appendix, we describe our approach to constructing these estimates.

To construct our waiting time estimates, we use our housing waiting list data linked to assistance records. The assistance records provide dates of admission for new public housing residents, as well as details on the development in which public housing residents move. Using the initial bedroom needs and initial randomized public housing waiting list positions, we estimate local cubic polynomial regressions of public housing admission date separately for each destination public housing zone and separately by bedroom needs using an epanechnikov kernel. We obtain smoothed estimates in bins of 250 rank positions. We use bins to smooth through sampling variance and minimize the influence of own-observations. We excluded households from estimation who selected an elderly preference, an assisted living facility preference, or a need for a handicap accessible unit, since they were eligible for different developments than families. Since the resulting estimates should be monotone in initial waiting list position, but may not be due to sampling variance or estimation error, we apply a simple algorithm which iterates from the first position bin to the last position bin (plausibly offered public housing) and detects any deviation from monotonicity and replaces the value with the latest position with a non-decreasing estimated waiting time. Appendix Figure A.10 plots binned mean admission times (navy circles) for public housing in zone 1 for 2-bedroom households against waiting list position rank, and the estimated waiting time function – using the procedure described above – is in gold-dashed lines. We generate these waiting time estimates for four-bedroom size needs: 1,2,3, or 4+ bedrooms, and for each of three public housing zones.

**Appendix Figure A.10: Estimated Admission Date for Zone 1 Public Housing (2 Bedroom)**



## H Public Housing Quality

In section 6.2, we explore whether moving into “higher quality” public housing yields different effects than moving into “lower quality” public housing. As described in 6.2, we use the leasing “zones” created by the housing authority for the purpose of making public housing apartment offers, to categorize public housing developments as either “Higher Quality” or “Lower Quality.” In this appendix, we provide evidence that our high/low-quality definition based on these zones is suitable for capturing some of the major differences across public housing developments.

Table A.19 reports the mean characteristics of the High- and Low-Quality public housing developments where families in our sample lease-up. The public housing developments categorized as “High Quality” appear to be significantly less distressed, on average, than those categorized as “Low Quality.” High quality developments have estimated hedonic quality that is roughly 20 percent higher than low quality developments. The typical poverty rate in the Census tract in which high-quality developments are located are 29 percent, compared to 44 percent for low-quality developments. Similarly, violent crime rates around high-quality developments are roughly half of those around low-quality developments. There is little difference in the average neighborhood test scores surrounding high- and low-quality developments. Finally, public housing developments categorized as high-quality are located in neighborhoods which have higher predicted income percentile ranks in adulthood according to the Opportunity Atlas (Chetty et al., 2018).

Appendix Figure A.11 provides examples of a higher- and lower-quality public housing developments. The higher-quality development is Arthur Mays Villas in panel A of Appendix Figure A.11. Built in 1976, Arthur Mays Villas is located on Miami's south side. The neighborhood surrounding Arthur Mays is considerably more middle class, with a median income of \$35,000. The poverty rate is modest at 18 percent, and violent crime is low-to-moderate at 9 violent crimes per 1,000. The housing is larger, a typical 2-bedroom unit was about 900 square-feet, and featured much newer building materials.

The lower-quality development, shown in panel B of Appendix Figure A.11, was Liberty Square.<sup>29</sup> Liberty square was built in 1937, making it the oldest public housing development in Miami. Located on Miami's north side, the development became notorious for the high concentration of poverty and crime. Liberty Square was located in a Census tract with a median income of just \$14,000, an average poverty rate of 49 percent, and incredibly high rate of violent crime with 44 violent crimes per 1,000. In addition to the age of the housing, the apartments were generally small, with the typical 2-bedroom apartment only about 700 square feet.

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<sup>29</sup>Partial demolition of Liberty Square began in 2020, reflecting the development's troubled history

**Appendix Figure A.11: Examples of High- and Low-Quality Public Housing**

**(a) High-Quality: Arthur Mays Villas**



**(b) Low-Quality: Liberty Square**



*Notes:* This figure presents photos of two of the most common public housing destinations in our sample. Panel A is a typical “High Quality” public housing location: Arthur Mays Villas. Panel B shows a typical “Low-Quality” public housing destination: Liberty Square. Photos are sourced from Google StreetView circa 2015.

# I Comparison with Chicago Housing Voucher Lottery

Our results imply that receipt of a housing voucher significantly improves developmental outcomes for young children. As we discuss in section 6.3, the closest study to ours is [Jacob et al. \(2015\)](#) (hereafter “JKL”), which uses a lottery of housing vouchers linked to data on 3-8th grade test scores, high school completion, crime and hospitalizations. In contrast to our results, JKL conclude that there are few, if any, meaningful effects of vouchers on children. This appendix explores the possible sources for differences between our work and JKL, which can be broadly categorized as differences in: treatment, outcome measures, population, and data.

While JKL also study housing choice vouchers, there are some differences in context, which may matter for understanding differences in the program’s effects. The vouchers studied by JKL were administered by the Chicago Housing Authority, which operates almost exclusively within the City of Chicago. Voucher holders seeking to lease-up outside of the city boundaries (say in Cook County suburbs) generally need to “port-out” - meaning there needs to be an agreement with neighboring housing authority to receive the family. This is regarded as administratively cumbersome, so CHA housing voucher recipients typically lease-up in central city neighborhoods. In contrast, we study a context where the housing authority serves an entire county. In theory, this could open-up a greater range of residential options ([Greenlee, 2011](#)). Consistent with these facts, we find that families in our context do use their vouchers to lease-up in observably “better” neighborhoods: Census tracts with lower crime and less poverty. Whereas JKL find no effects of voucher receipt on similar standard measures of neighborhood quality.

Another possibility is that the difference in effects of vouchers we observe is driven by differences in population served. There are two major observable dimensions across which our sample and that studied in JKL differ. The first is race and ethnicity. The sample studied in JKL is 94 percent Black and less than 4 percent Hispanic, while ours is 70 percent Black and 30 percent Hispanic. In Appendix Table A.22, we re-estimate our results reweighting the sample to match the race/ethnicity composition of JKL’s sample. If anything, our estimates appear *larger* when we match the race/ethnicity of JKL, which suggests this is not the source of differences. The more salient difference might be that of average age at time of treatment. By construction, our focus on early child development means that our sample is substantially younger at the time of random assignment. JKL *do* find marginally significant positive effects for boys ages 0-6 at time of voucher offer, however, their IV point estimate for boys is still quite a bit smaller (0.06 SD) than our main estimate for boys (0.55 SD). While age at treatment and time from randomization to measurement of outcomes may explain some

portion of the difference in results, those explanations are likely insufficient to fully account for the difference in estimates.

One subtle, but potentially important, difference relates to the data environment. There are several features of our data that differ from previous work. First, our waitlist data allow us to directly observe the children associated with an application and their identifying information necessary for linking to outcomes. In contrast, JKL do not observe the children attached to assistance applications, and must use public assistance records to identify the children likely to be in the waiting list household. As a result, our data should be less likely to suffer from measurement error than prior studies. Second, we define receipt of assistance *at the child level*, whereas prior work primarily measures assistance at the level of the household-head. While both approaches yield similar results, defining assistance at the household-head level will tend to overstate the fraction of children assisted, and hence understate the IV estimates of receipt of housing assistance. Finally, we leverage statewide education data which allows us to measure children’s outcomes even if they move across public school districts whereas prior work typically relied on data from just a single school district.

Finally, the outcomes we consider differ from previous work in two important ways: first, we examine effects upon entry into formal schooling which is proximal to children’s experiences of early-life conditions and resources, and secondly, our outcomes capture both cognitive and noncognitive skills. [Jacob et al. \(2015\)](#) study the medium-run impacts of vouchers on math and reading test scores in third through eighth grades.<sup>30</sup> Measuring test scores well into formal schooling will include not just the effects of changes in early-life resources and the home environment, but typically several years worth of inputs from the educational system. In contrast, our measures of skill development upon entering kindergarten are likely shaped almost entirely by the home experiences because the children are too young to have significant contact with the school system. Our measures are inherently shorter-run in nature, compared to the longer-run test score impacts studied by JKL.

Beyond this difference, the outcomes we consider are also different than standardized reading and math assessments, including both early literacy skills assessment during dedicated one-on-one test administration, and also classroom observations of children’s behavior and social-emotional development. These data contain direct measures of noncognitive skills such as responsible decision-making, social problem solving, and approaches to learning. Researchers have also demonstrated that cognitive assessments contain substantial noncognitive content, such as task persistence and attentiveness ([Balart et al., 2018](#); [Cornelissen and](#)

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<sup>30</sup>JKL also examine impacts on high school completion, but only for children older than six at the time of lottery, limiting comparability to our setting.

[Dustmann, 2019](#)), which could be particularly important for completing assessment tasks at young ages. Our measures thus capture both the noncognitive skills they are intended to measure and implicit noncognitive content associated with assessment task performance.